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Niemelä

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(54) **METHODS OF TARGETED DELIVERY TO A HOST USING A CARRIER**

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(57) **ABSTRACT**

According to some embodiments, a method of providing treatment of a host comprises binding a carrier to cell structures of cells of the host to reduce a likelihood of an agent binding to said cell structures to at least partially inhibits the agent from binding to said cell structures, wherein the carrier comprises a core and a surface functionalized on the core, wherein the functionalized surface bind to target areas of cell structures of the host's cells, and wherein the carrier is to be used as targeted treatment for one or more disease, infections or allergic reactions caused by a disease-causing agent or source.

19 Claims, 9 Drawing Sheets

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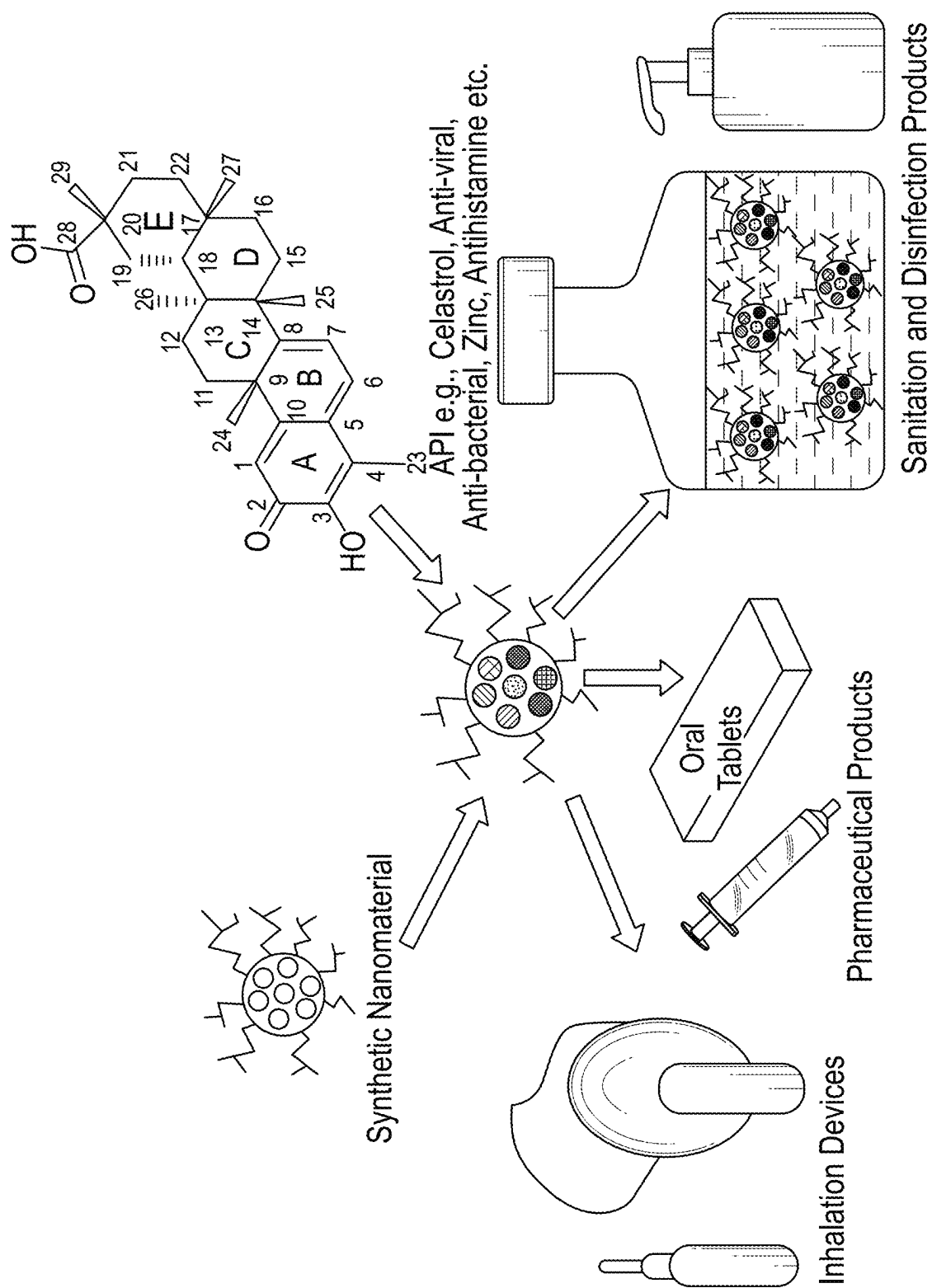


FIG. 1

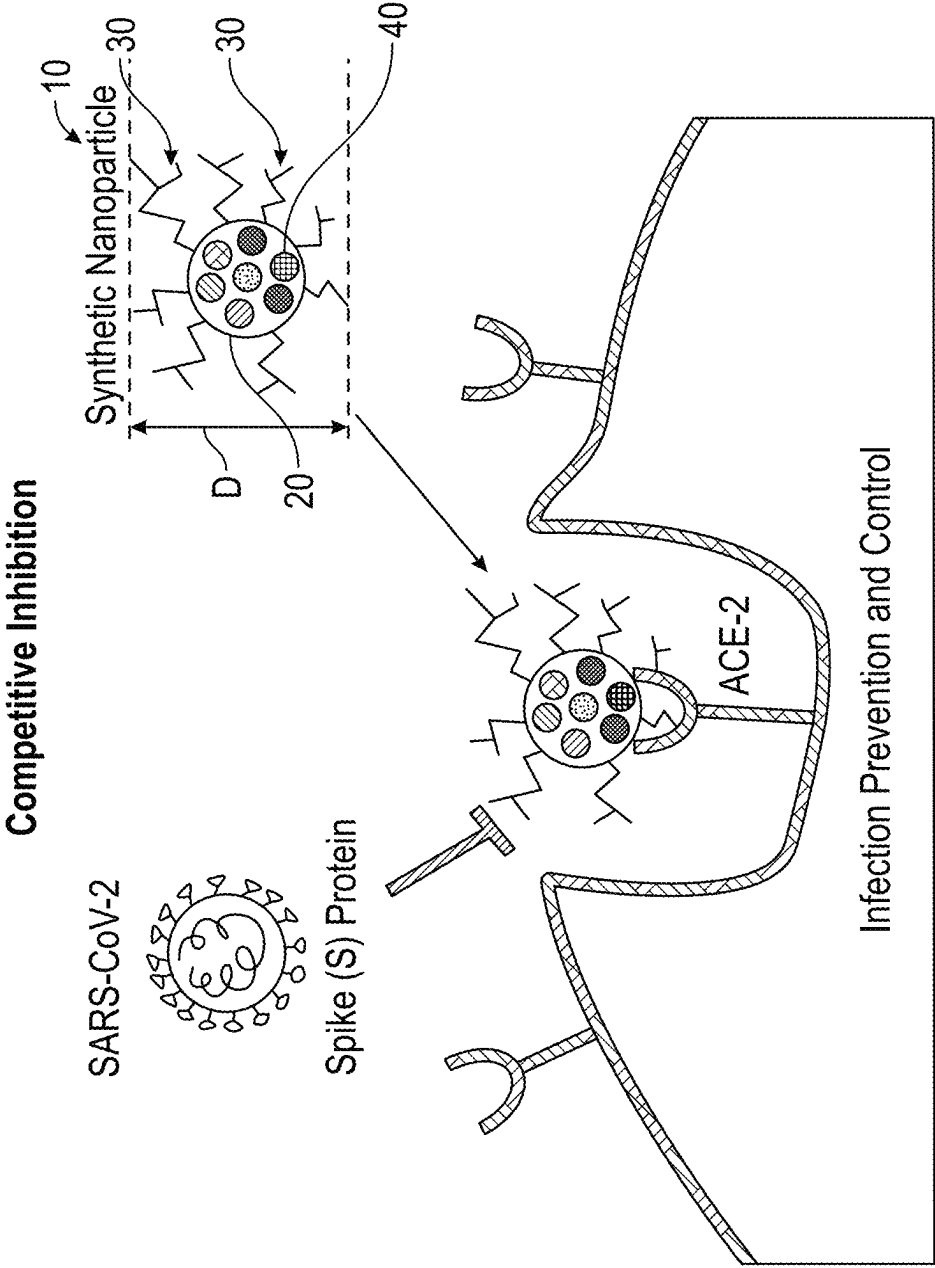
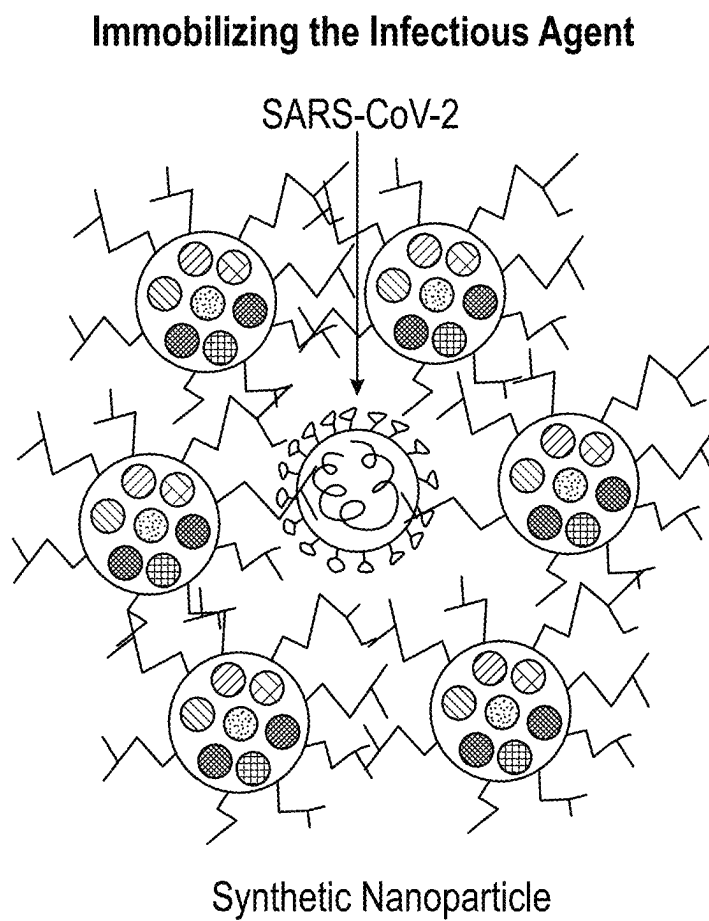


FIG. 2

**FIG. 3A**

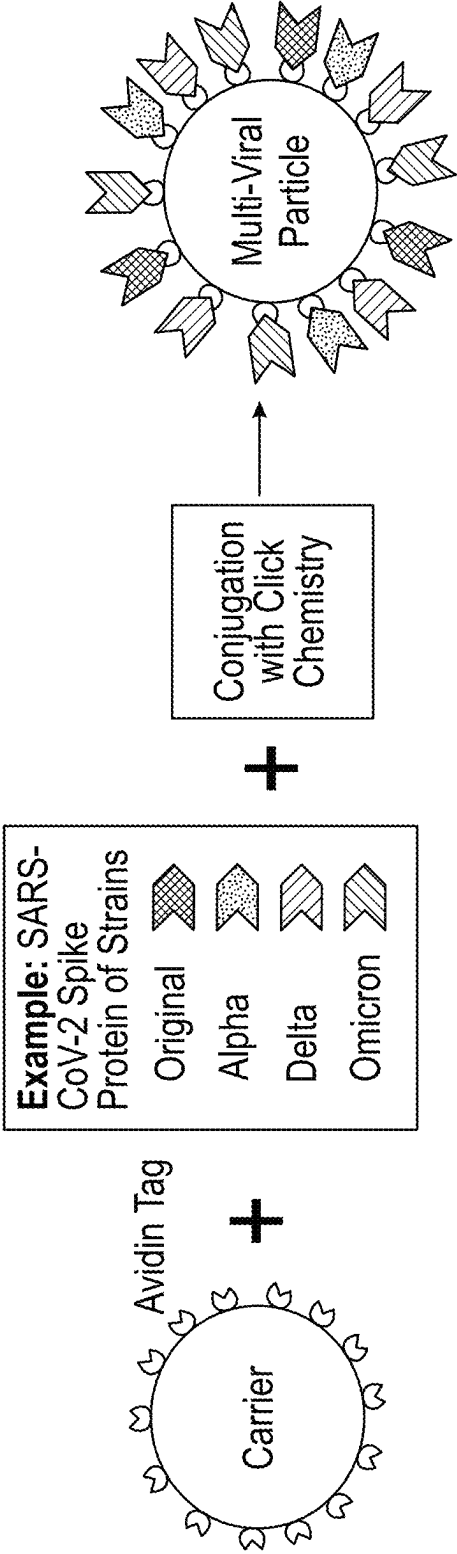
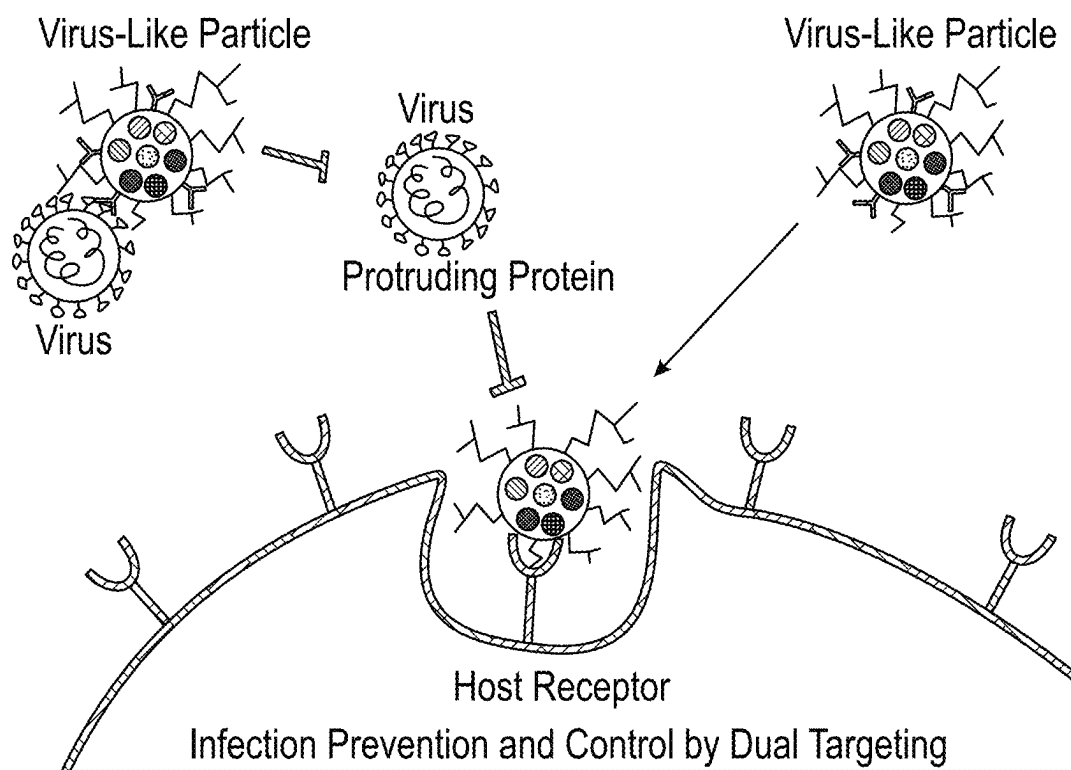


FIG. 3B

Competitive Inhibition & Immobilisation**FIG. 4A**

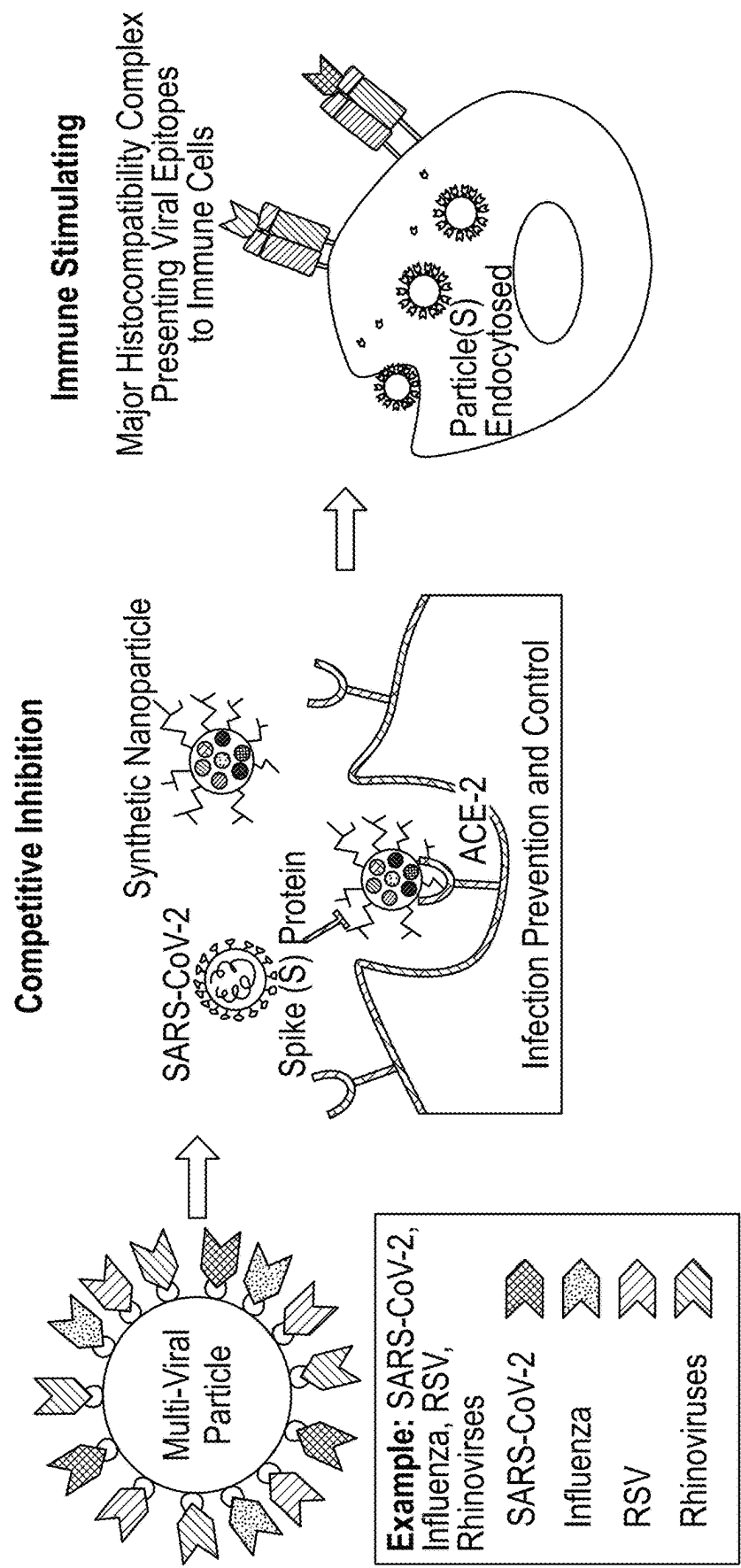


FIG. 4B

**Immobilizing the Infectious Agent by
Blocking Circulating Co-receptors**

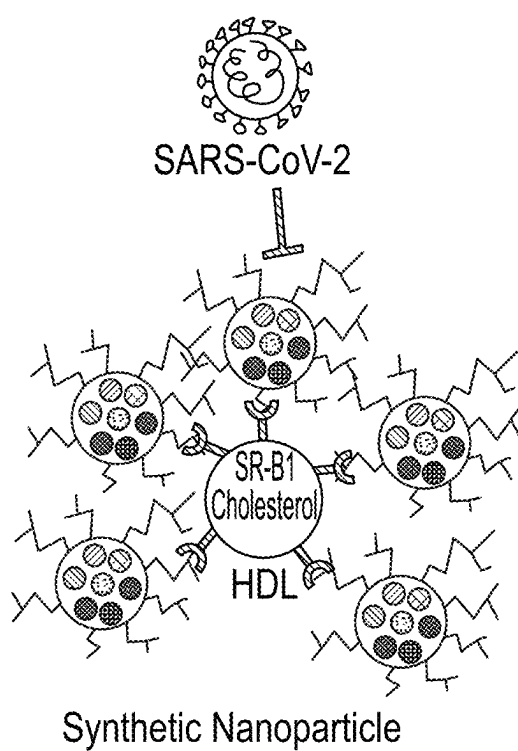


FIG. 5

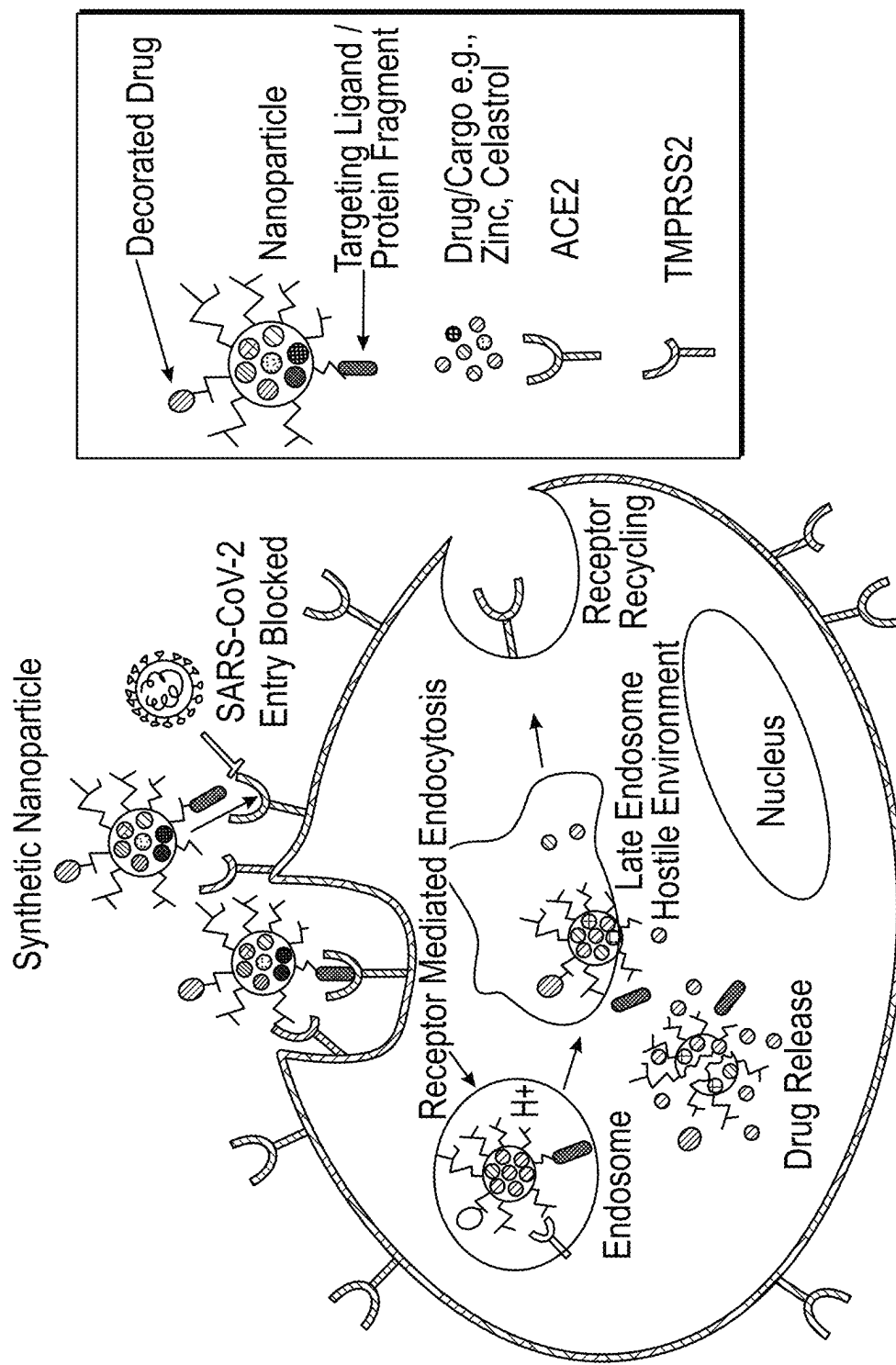


FIG. 6

XXXXXXXXXXXXXXXXXXXX	10	20	30	40	50	60	70	80	90	XXXXXXXXXXXXXXXXXXXXRVQPTESIVRFPNITNLCPFGEVFNATRFASVYAWNKRKRN
Signal Sequence			Tag		Spacer		Region: 319.....SARS-CoV-2 RBD.....			
CVADYSVLNSASFSTFKCYGVSPTKLNDLCFTINVYADSFVRGDEVQRQIAPGQTGIAD										
			100	110	120	130	140	150		
.....SARS-CoV-2 RBD										
YNYKL PDDFTGC V IAWNSNNLD SKVGG NYNYLYRLFRKS NLKPFERDISTEINYQAGSTPC										
			160	170	180	190	200	210		
.....SARS-CoV-2 RBD										
NGVEGFNCYFPLQSYGFQPTNGVGYPYRVVLSFELLHAPATVC GPKKSTNLVKNKCVNF										
			220	230	240	250	260	270		
.....SARS-CoV-2 RBD										
.....541										

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METHODS OF TARGETED DELIVERY TO A HOST USING A CARRIER

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/544,302, filed Dec. 18, 2023, which is a continuation-in-part (CIP) of U.S. patent application Ser. No. 18/152,930, filed Jan. 11, 2023, which is a continuation of U.S. patent application Ser. No. 17/684,341, filed Mar. 1, 2022 and issued as U.S. Pat. No. 11,564,892 on Jan. 31, 2023, which is a continuation-in-part (CIP) of PCT Application PCT/FI2021/050259, filed Apr. 9, 2021 and published on Oct. 14, 2021 as PCT Publ. WO 2021/205077, which claims priority to Finnish Application Nos. 20205382, filed Apr. 9, 2020 and 20215182, filed Feb. 19, 2021; and also claims priority to U.S. provisional application No. 63/433,755, which was filed Dec. 19, 2022. The contents of each of the aforementioned applications are incorporated by reference herein in their entireties and made part of the present application.

INCORPORATION BY REFERENCE OF MATERIAL IN SEQUENCE LISTING

This application incorporates by reference the material within the Sequence Listing contained in the following XML file being submitted concurrently herewith: File name: FICU.002P5C1_Replacement_SEQ_LIST.XML; created on Feb. 28, 2025 and is 10,456 bytes in size.

BACKGROUND

Field

The present application pertains to the field of pharmaceutical products, biologics, medical devices, over-the-counter drugs and consumer products preventing or reducing the spread of pathogens or other disease-causing agents, factors or sources, such as, for example, coronaviruses (e.g., the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2)), influenzas and viruses causing respiratory infection, diarrhea, common cold, cytokine storm, general discomfort and/or death, bacteria, other pathogens and the like. More specifically, the present application relates to nano- and/or micromaterials-based carriers, such as mimetic nano- and/or micromaterials-based carriers, synthesized to minimize the spread of pathogens and infectious agents (e.g., viruses (e.g., influenzas, rhinoviruses, noroviruses, respiratory syncytial virus (RSV), SARS-CoV-2, future strains and/or types of coronaviruses derived thereof, etc.), bacteria, parasites, antigens, prions, mold, fungi, toxins, poisons, and allergens), and/or other disease-causing agents, factors and/or sources.

The present application also relates to combinatory tailored treatments of an active pharmaceutical ingredient ("API") loaded inside a carrier system capable of delivering the drug specifically to target cells and/or tissues (e.g., targeted drug/material delivery). More specifically this application pertains to the fabrication and use of man-made materials in the nano- and/or microscale that would, at least partially, saturate and bind to receptors, proteins and/or macromolecules at the cellular level in order to prevent and/or minimize (or reduce the likelihood of) pathogen binding, in particular, for example, novel coronaviruses binding and entry to the host target cells and/or tissues by

competitive inhibition. In some embodiments, the targeted drug and/or other material delivery includes using a carrier to at least partially alter, inhibit and/or otherwise impact disease onset and/or progression. This can be accomplished by, in part, the carrier binding to a target site (e.g., a receptor, a protein, a macromolecule, etc.). Once carriers bind to host cells, they can enter said host cells (e.g., by endocytosis). Accordingly, the drug and/or other contents of the carrier can be advantageously provided to host cell (e.g., target host cells) with lower or less impact on other host cells. The carrier As a result, in some arrangements, an enhanced therapy can be provided by targeting particular cells of a host and being able to deliver specific drugs and/or other components to said cells. The carriers can increase the therapeutic window of drugs and/or other components loaded to (e.g., within) and/or otherwise provided with the carrier. In some embodiments, the drugs and/or other components provided with the carrier have low solubility and/or permeability and/or are prone to aggregate and be rendered non or less efficacious (e.g., when not using a carrier as a delivery vehicle to the subject's anatomy).

The present application further relates to a medical device or delivery device capable of releasing (e.g., on-demand, specific amounts of) the synthesized carrier system to targeted tissue or tissues susceptible to, for example, coronaviruses. In some embodiments, for example, such a device includes an inhalation device, e.g., meter dose, dry inhaler, nebulization, ultrasonication, nose or mouth drops, or a nasal spray for the respiratory tract. In some arrangements, the device includes tailored orally ingestible tablets or solutions for the gastrointestinal tract, a topically administrable cream or ointment intended to be applied to a subject's skin and/or an injectable substance intended to be applied subcutaneously, intravenously, intraperitoneally or otherwise injected and/or administered using, at least in part, for example, high pressure or laser.

Further, the present application relates to synthesized materials (e.g., carriers, nanoparticles, etc.) that are configured to bind to pathogen co-receptors and/or natural occurring carriers (e.g., high-density lipoprotein receptors in the blood and other secondary receptors, such as, for instance, the FcγR for, at least partially, preventing the infectious agent for entering the host cell). Therefore, according to some embodiments, the carrier system described herein is configured to reduce the likelihood of infection of host cells by blocking receptors and/or other binding sites/features of host cells to which viruses and/or other pathogens may bind (e.g., for entry into host cells), and by delivering immune stimulating properties to target cell populations, by having protrusions at the surface of the carrier that mimics-naturally occurring epitopes found at the pathogen.

Further, the present application relates to synthesized materials that have the capacity and ability of binding and encapsulating pathogens or pathogens co-receptors and natural occurring carriers for, at least partially, immobilizing and neutralizing the infectious agent for disrupting the infectious agent. Therefore, according to some embodiments, the carrier system described herein is configured to reduce the likelihood of infection of host cells by (1) blocking receptors and/or other binding sites/features of host cells to which viruses and/or other pathogens may bind (e.g., for entry into host cells), (2) delivering immune stimulating properties to target cell populations, and/or (3) immobilizing viruses and/or other pathogens by attracting virus and/or other pathogens (e.g., so such viruses and/or other pathogens are unable to bind to host cells).

Further, the present application relates to synthesized materials (e.g., carriers) that are configured to present and/or otherwise include epitopes capable of eliciting and/or configured to elicit an immunological response towards viruses and/or other pathogens. Therefore, according to some embodiments, the carriers disclosed herein are configured to be used as, for and/or in a vaccination (e.g., to have a vaccination-type effect) at target cell populations where the said epitopes can elicit an advantageous response. For example, in some embodiments, the epitopes can elicit a B cell response generating specific antibodies and/or T cell response priming and mounting a possible cytotoxic T-cell response against a virus and/or other pathogens. Accordingly, in some embodiments, one or more of the carrier systems described herein are configured to reduce the likelihood of infection of host cells by at least partially blocking receptors and/or other binding sites/features of host cells to which viruses and/or other pathogens may bind (e.g., for entry into host cells) and/or to provide immune stimulating properties to target cell populations (e.g., by mounting an efficient immunologic reaction).

Background

Different pathogens (e.g., viruses, bacterium, parasites, etc.) prefer environments typical of their specific niche inside the host tissues. For example, *Escherichia. Coli* prefers to colonize the intestine, whereas tuberculosis residues in the lungs of its host [1]. Malaria-bearing mosquitos may infect their human host by biting, allowing parasites to enter the blood stream and travel to the liver of the subject for maturation [2]. For the pathogens to colonize and replicate at their specific environments and tissues, they need to infect and/or inoculate their host [1-4]. At the cellular level, this mechanism of entry starts by the pathogen binding or getting in proximity of the host cell where specific receptors, macromolecules and/or proteins protruding at the cell membrane facilitates the endocytosis of the infectious agent. If the specific route of entry is known, such knowledge can be used for creating a man-made object that can allosterically hinder the specific pathogens entry by competitive inhibition [5]. For example, by creating a carrier (e.g., nanoparticle), such as a mimetic nanoparticle, of similar size, surface chemistry and charge as the pathogen of interest, it is possible to saturate and block the specific receptors at the host cells hindering the pathogens entry. Another possibility includes synthesizing man-made materials (e.g., carriers) that would efficiently bind to the pathogen of interest encapsulating and immobilizing the infectious agent, thereby minimizing the possible entry to the host. Another possibility includes synthesizing man-made materials (e.g., carriers) that would efficiently bind to the pathogen, such as the novel coronavirus and/or co-receptor of interest, thereby immobilizing the infectious agent and reducing the likelihood (e.g., minimizing) the possible spread of the disease further inside the body and/or reducing the likelihood (e.g., preventing) entry to the host.

In some embodiments, by creating a carrier (e.g., nanoparticle), such as a mimetic nanoparticle, of similar size, surface chemistry and/or charge as the pathogen of interest, it is possible to saturate and block the specific receptors at the host cells, hindering the pathogen's entry. Another possibility includes synthesizing man-made materials (e.g., carriers) that would efficiently bind to the pathogen co-receptor of interest, thereby at least partially hindering the infectious agent and reducing the likelihood (e.g., minimiz-

ing) the possible spread of the disease further inside the body and/or reducing the likelihood (e.g., preventing) entry to the host.

Viruses use components derived from their host for cell entry. For example, the SARS-CoV-2 virus that causes a respiratory infection called COVID-19 is decorated by glycoprotein spikes at the surface of the viral particle. These glycoproteins have high affinity for the human angiotensin converting enzyme 2 (ACE-2) allowing for specific internalization of the virus in the epithelial cells of the respiratory tract, possible intestinal tract and/or another tract or system of a subject where there is high expression of its target receptor [3,4]. Thus, potentially allowing for tailored molecules to be used for intervention of the SARS-CoV-2 virus enter to its human host. In bacterium it has been shown that surface topography together with surface charge greatly influences adhesion that modulates bacterial growth [6]. Using nanostructured surfaces, it could be possible to control bacterial adhesion and growth that could be used in medical applications for preventing infections. *Plasmodium falciparum*, which is the human Malaria parasites, uses dynamin like Eps15 homology domain-containing proteins for hijacking the endocytosis pathways important for infecting more erythrocytes in its host [2].

For example, the novel coronavirus SARS-CoV-2 that causes a respiratory infection called Coronavirus Disease 2019 (COVID-19), are decorated by glycoprotein spikes at the surface of the viral particle having high affinity for specific receptor(s). SARS-CoV-2, SARS-CoV and MERS-CoV belong to the betacoronavirus genus having a genome size of approximate 30 kilobases encoding both structural and non-structural proteins. Structural proteins include the envelope (E) protein, spike (S) glycoprotein, the nucleocapsid (N) protein and the membrane (M) protein, whereas to the non-structural proteins belong, for example, the RNA-dependent RNA polymerase. The spike (S) glycoproteins decorated on coronaviruses consist of a homotrimeric transmembrane protein, each 180 kDa monomer comprising two functional subunits S1 and S2, whereas the S1 unit consists of two domains: N-terminal domain (NTD) and C-terminal domain (CTD). Depending on the coronavirus type, either the NTD or CTD of S1 is used as the receptor binding domain (RBD) capable of binding to specific receptors at the host cell surface. Each of SARS-CoV-2 and SARS-CoV utilizes the CTD as its RBD for the human angiotensin converting enzyme 2 (ACE2) allowing for specific internalization of the virus in the epithelial cells of the respiratory tract and possible intestine where there is high expression of its target receptor [3,4,6]. However, due to the novel amino acid sequence and structure of SARS-CoV-2, the affinity for the ACE2 receptor is significantly higher compared to SARS-CoV [6,7]. In both of the aforementioned coronaviruses, the RBD forms a concave surface that contains a ridge loop that has the ability to bind to the receptor binding motif, which is the outer surface of the human ACE2 receptor at its N-terminal helix. In SARS-CoV, this loop contains a three-residue motif proline-proline-alanine, and these prolines repeats makes the ridge loop to make a sharp and short turn. SARS-CoV-2 has a four-residue motif of glycine-valine/glutamine-glutamate/threonine-glycine that give rise to two bulkier residues and two flexible glycine residues that creates a different and compact conformation, allowing the viral loop to be closer to the ACE2 receptors N-terminal helix forming additional hydrogen bonds between the loop and the human receptor resulting in a stronger binding [6]. The S2 subunit from the S protein is necessary for viral fusion with the host cellular

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membrane mediated by proteolytic cleavage by the human transmembrane serine protease 2 (TMPRSS2), leading to the internalization of SARS-CoV-2, enabling viral replication inside its host cell [7]. The inhibitory effect of RBD spike fragment hexapeptide 438YKYRYL443 (SEQ ID No: 1) of the SARS-CoV-2 has been estimated to have the highest affinity for ACE2 when compared to another known coronavirus derived hexapeptides. The specific hexapeptide YKYRYL (SEQ ID No: 1) carries the dominant binding amino acid sequence that binds to ACE2 with a high affinity of $KD=46 \mu M$. However, the simulation gives rise to potential alternative synthetic hexapeptide variants YKYNYY (SEQ ID No: 2) and YKYNLY (SEQ ID No: 3) with even stronger binding affinity towards the ACE2 receptor which is highly conserved among different mammalian organisms allowing transmission from animals to humans and vice versa [8]. Furthermore, some amino acid sequences from viruses such as SARS-CoV-2, have been shown to induce predominantly B cell response whereas other have been shown to induce predominantly T cell response. For example, but not limited to epitopes from SARS-CoV-2 spike protein: 1220FIAGLIAIV1228, 17VLLFLAFVV25, 20FLAFVVFLL28, 204VLAWLYAAV212 and 184VLWAHGFEL192 that stimulates at least partly the T cell response. Whereas for example but not limited to epitopes from SARS-CoV-2 spike protein: 287DAVD-CALDPLSETKCTLKSFTVEKGHYQTSN317, 524VCGPKKSTNLVKNKCVNFNGLTGTGVLTESN-KKFLPFQQFGRDIADTTDAVRDPQTLEILDITPC SFGGVSV1598, 601GTNTSNQVAVLYQDVNCTE-VPVAIHADQLTPTWRVYSTGS640, 802FSQIL-PDPSKPSKRSFIE819, 888FGAGAALQIPFAMQMAY-RFNGI909, that stimulates at least partly the B cell response. There are variations among human populations and the animal kingdom regarding the ACE2 receptor in terms of expression levels and polymorphisms that could influence the susceptibility of SARS-CoV-2 and outcome of COVID-19 disease [6-10]. Thus, potentially allowing for tailored molecules to be used for intervention of the SARS-CoV-2 virus enter to its human and animal hosts.

In the context of this specification, both ACE2 and ACE-2 may be understood as referring to human angiotensin converting enzyme 2.

Influenza, rhinoviruses, coronaviruses, respiratory syncytial virus (RSV) and noroviruses are non-limiting examples of viruses causing respiratory infection, diarrhea, common cold, cytokine storm, general discomfort, death and/or other symptoms or ailments. For example, influenza virus is a negative-sense, single-stranded RNA that causes a respiratory infection commonly called the “flu” which affects millions of individuals annually and causes thousands of deaths and millions of hospitalizations. The flu viral envelope is decorated with the fusion protein hemagglutinin (HA) that binds to the host sialic acid receptors and neuraminidase (NA), an enzyme located at the viral surface that cleaves the glycosidic bonds of the monosaccharide sialic acid, aiding in penetrating the host mucus and enabling the escape of newly formed viral particles. The size of influenza virus particles is around 80-120 nm, which is quite close to the size of coronaviruses. However, influenzas differ by having two main proteins on the surface, i.e., HA and NA, whereas coronaviruses only have the spike proteins protruding on the surface. To further complicate matters, there are currently 17 different HA proteins and 10 different NA proteins that have been characterized. It is these different combinations of proteins that give influenzas their subtype

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names. For example, the sequence homology of the HA2 subunit compared to the other HA subtypes is around 51-80%, whereas the HA1 subunit has an 34% to 59% sequence homology, rendering different genetic and protein varieties between the influenza subtypes. Combining the genetic variation with the potential protein combination of these viruses contributes to the immune evasive properties of influenzas, rendering efficacious vaccination development to a difficult task. This limited protection against influenzas, induced by vaccination, is listed by the Center for Disease Control (CDC) as follows, “flu vaccination reduces the risk of flu illness by between 40% and 60% among the overall population.” Therefore, there exists a significant need for developing anti-viral compounds and broad immune stimulating influenza vaccines in order to mitigate the spreading of the flu.

Rhinoviruses is one category of other major causative agents for the common cold, and there is currently no efficient vaccination against these types of viruses. Human Rhinoviruses (HRV) belong to the picornavirus family and are positive-sense, single-stranded ribonucleic acid (RNA) viruses that have an icosahedral symmetry with a particle size of around 30 nm. The viral capsid is composed of four main proteins: VP1, VP2, VP3, and VP4, whereas the VP4 protein is located inside of the virus anchoring the genetic information to the capsid structure. There are over 150 different serotypes of HRVs with the two most common types being HRV-A and HRV-B that uses the intercellular adhesion molecule-1 (ICAM-1) as the cell receptor for entering the host. However, some of the HRV serotypes use heparin sulfate proteoglycan as an additional receptor, and there are around 10 serotypes that use low-density lipoprotein as the cell receptor. Additionally, a new serotype of HRVs that arose in 2002 was given the name HRV-C, having a route of cell entry still remains elusive rendering rhinoviruses to a difficult task to mitigate.

Respiratory syncytial virus (RSV) belongs to the family of paramyxoviridae viruses and are negative-sense, single-stranded RNA viruses that usually cause a mild cold in most healthy humans. However, for infants, the elderly and/or other humans that are immunocompromised or otherwise susceptible to disease, the RSV can cause a more serious disease such as bronchiolitis and pneumonia, oftentimes leading to hospitalization. The RSVs have an average size of around 200 nm and contain three membrane proteins: 1) the host receptor attachment protruding glycoprotein (G), 2) the fusion protein (F), and 3) a short hydrophobic (SH) protein that forms a ion channel. RSVs can be further divided into two groups, A and B, depending on the reaction with monoclonal antibodies directed against the F and G proteins. The A group is the most prevalent circulating virus, and the largest genetic divergence is associated with the gene encoding for the G protein, rendering this protein to the most variable protein of the virus. This diverse variation of proteins explains, at least partially, why no effective vaccination against RSV currently exists on the market. Noroviruses (NoV) belong to the family Caliciviridae, which are genetically a diverse group of single-stranded positive-sense RNA that are non-enveloped viruses that cause an infection commonly called gastroenteritis or “stomach flu,” giving sudden onset of vomiting, diarrhea and other symptoms, which are often relatively severe. The most common symptoms for norovirus include nausea, vomiting, stomach pain or cramps, diarrhea, fever and/or muscle pain, with early symptoms usually beginning about 12 to 48 hours after exposure to the virus. Such symptoms can last up to several days. Infected individuals may continue to shed noroviruses

in their feces for several weeks after recovery, thereby transmitting the disease, without knowing, to other individuals. The norovirus consists of an ~7.7-kb RNA genome with three open reading frames (ORFs), where ORF1 encodes a polyprotein precursor which is processed into several non-structural proteins, and where the two other ORFs encode the major (VP1) and minor (VP2) capsid proteins. The viral particles are around 27-30 nm in diameter having an icosahedral symmetry where the viral capsid is built of 90 dimers consisting of VP1, each protein having a shell (S) domain and a protruding (P) domain connected by a flexible linker. The S domain is responsible for the assembly of the virus capsid shell encapsulating the viral genome and is highly conserved domain of the VP1 protein. The P domain, on the other hand, is more variable and includes a P1 and a P2 subdomain. The P1 subdomain binds the S domain with the P2 domain, and the P2 subdomain contains the host receptor binding site which is also a target for neutralizing antibodies. The norovirus enters its host by binding to cell-associated glycans located on the cell membranes, including sialic acid and histo-blood group antigens. Then soluble cofactors facilitate viral binding to its host receptor. For murine norovirus (MNoVv), the receptor is a CD300If an immunoglobulin (Ig) domain-containing membrane protein, whereas for the feline calicivirus (FCV), the receptor is a feline junctional adhesion molecule A (fJAM-A). However, the receptor for the human norovirus (HNoV) remains elusive. Taking together the highly variable P2 domain in combination of the VP1 protein's ability to inhibit cytokine induction and VP2 protein's capability of regulating antigen presentation and the complex transmission routes makes noroviruses challenging pathogens to combat.

Although vaccines can be effective in protecting against infectious agents, they often take significant time and resources to develop. For example, effective vaccines that can safely be administered to patients require many clinical tests that need to be performed before approval. Secondly, vaccinations only work if the correct antigens for the specific pathogen are being administered with sufficient immunological reaction, creating an immunity for the specific disease [11]. For example, seasonal influenza strains normally vary from one year to the next, and the vaccines usually contain only a few epitopes of different influenza strains, thereby rendering the creation of some vaccinations to an educated guess work.

A vaccination's efficacy is based on, at least in part, administering to the subject a composition comprising epitopes that mimic specific antigens of the said pathogen of interest. Once inside the host, these compositions containing epitopes can be internalized by cells of the host and then at least partly presented via the viral antigen presentation pathway of the immune system. In humans, such an antigen presenting pathway is named human leukocyte antigen (HLA) system and comprises of major histocompatibility complex (MHC) I and II. MHC I pathways can be crucial in stimulating the immune responses in which CD8+ cytotoxic T cells are primed for eliminating virus-infected cells, while the MHC II class is responsible for presenting antigens from the virus or epitopes from an vaccination composition that primes and stimulates B cells in producing antibodies against that specific antigen.

Anti-viral medicine can also be effective against viral infections if treated correctly. However, these medications often interrupt viral DNA or RNA replication machinery, and thus, it is not always plausible to use such medicines as a proactive drug. Unfortunately, these compounds can be harmful for the patient if used under prolonged periods [12].

Antibiotics are effective against the spread of bacteria by disrupting their cell division and/or the synthesis of the proteoglycan-based cell wall [1]. Formulating the most efficient antibiotic depends on if the target bacterium is gram positive (having a cell wall) or gram negative (lacking a cell wall). Recently, there have been numerous cases where multi-resistant bacteria have emerged that are not responding to traditional antibiotics. In these cases, broad spectrum antibiotics have been used to combat infection. However, such strong cocktails of antibiotics can take their toll on the patient and potentially can give rise to more antibiotic resistance bacteria [1]. Therefore, there is an unmet need of developing medications, such as over-the-counter (OTC) drugs and consumer products, that can be used for preventing or reducing the likelihood of the spread of pathogens, including mutating novel coronaviruses, using proactive purposes and having minimal or minor side effects. Also, the use of tailored medicine using carriers (e.g., nanomaterials) loaded with an active pharmaceutical ingredient (API) for both inhibiting the endocytosis of the target pathogen, in particular viruses, as well as stopping the replication of already infected cells and/or tissues.

In addition, according to some embodiments, the immune response includes cytokines produced by the host such as Interferons (IFNs) that have potent or substantially potent antiviral effects. These cytokines are part of the first line of defense against viral infections and have important roles in immunosurveillance. There are two different classes of interferons: type I IFNs and type II IFN. There are many type I interferons, including for example interferon alpha (IFN- α), interferon beta (IFN- β) and interferon epsilon (IFN- ϵ) whereas there is only one type II interferon named interferon gamma (IFN- γ). Interferon (IFN)-gamma is a critical antiviral mediator central to the elimination of viruses from the host and could be used as an immune modulating drug.

In addition, according to some embodiments, part of the immune response includes activation of innate immune system by the particle (e.g., carrier). Where the particle contains component or components capable at least partly to target dendritic cells (DCs) to aid in the antigen presenting pathway to inform the immune response what to fight against such as invasive pathogens and/or cancerous cells while. The component loaded into or onto the particle can at least partly activate macrophages by engaging for example, pattern-recognition receptors such as Toll-like receptor or receptors (TRL) that is part of the interferon type I secretion pathway.

Protein and proteasome inhibitors show great potential as these compounds can specifically bind and allosterically hinder the enzymatic reaction by binding to the active site blocking the target molecules interaction with the enzyme [13]. However, one of the major drawbacks of proteasome inhibitors is their instability and possible low solubility. Further, due to their high specificity, such molecules often only show efficacy to only a few or one specific enzyme per drug molecule.

Monoclonal antibodies have been used since 1986. The first such drug approved by the FDA was Orthoclone OKT3, which is used for reducing kidney rejection after transplantation. Monoclonal antibodies that are used in cancer therapeutics include trastuzumab (Herceptin), which is a drug that binds to the human epidermal growth factor receptor 2 (HER2) slowing down the growth of malignant HER2 positive breast cancer cells [14]. The major limitation of antibody-based therapeutics is that these proteins are foreign. For example, such therapeutics are produced in mice or

other animals so when they are introduced to patients, they can invoke an immunologic reaction, potentially giving adverse reaction of the treatment.

Cells of multicellular organisms, such as humans and animals, need to maintain their cellular homeostasis, which can be a delicate balance or equilibrium that is constantly challenged by disease-causing agents. Disease is more likely to occur when this balance or equilibrium is disrupted or otherwise impacted by one or more disease-causing agents.

In some embodiments, disease causing agents, factors and/or sources include, without limitation, include a pathogen, an infectious agent and/or any other type of disease-causing agent or source, such as, for example, and without limitation, viruses (e.g., SARS-CoV-2 viruses, other coronaviruses, influenza viruses, rhinoviruses, noroviruses, respiratory syncytial viruses (RSV), other viruses that impact the respiratory system, any other type of viruses, etc.), bacteria, parasites (e.g., worms), protozoa, microorganisms, prions, fungi (e.g., molds), allergens, antigens, toxins, poisons (e.g., venoms, pollutants, etc.), carcinogens, compounds or molecules (e.g., organic, inorganic, etc.), chemical agents (e.g., naturally-occurring, synthetic or man-made, etc.), radiation (e.g., electromagnetic radiation, radioactive compounds/materials, other radiation, etc.), enzymes or proteins, regulatory elements and/or substances (e.g., hormones, enzymes, other proteins, DNA/RNA repair elements or agents, replication machinery, chaperone machinery, other host-producing elements or substances, etc.), inflammatory agents, elements or substances and/or the like. Pathogens, infectious agents or other types of disease-causing agents or sources can be internal and/or external to the host's cells.

In some embodiments, internal disease-causing agents, factors and/or sources include, by way of example, enzymes and proteins responsible for correcting or otherwise impacting the cellular homeostasis within the cells. For instance, the disease-causing agents can include DNA-repair machinery enzymes, which are proteins that correct or adjust genetic material, and/or protein folding machinery, which can refold or otherwise adjust incorrectly folded or aggregated proteins. However, in some cases, unwanted changes can occur in such repair machinery, which can result in one or more diseases. For example, in some arrangements, chemical agents, carcinogens and/or radiation can result in mutations to genetic code. This can lead to accumulation of cells and other unwanted results because, for example, repair machinery of cells has become faulty or otherwise impaired, causing diseases or other health issues, such as cancer, immune disorders, etc. In some embodiments, mutations arising from internal or external agents can cause other changes in the cellular homeostasis, leading to, by way of example, excessive cell removal that may cause neurodegenerative diseases, such as, for instance, Huntington's disease and Alzheimer's disease, degenerative disorders of organs (e.g., liver cirrhosis, muscular dystrophies, etc.) and/or any other disease, disorder or other health issue.

Nanomedicine shows great potential in the field of targeted drug delivery, where nanotechnology and medicine are combined for the development of personalized diagnostics, as well as the treatment and prevention of different diseases. In some arrangements, nanomaterials or other carriers include man-made and/or naturally occurring objects with dimensions between 0.2 nm to 100 nm. The physical properties of such materials can be drastically different compared to their bulk counterpart. For example, nanomaterials can be more reactive on both biological and chemical substances due to higher surface area to volume

ratio. Functionalized nanoparticles have shown to be able to target specific cell types opening the possibility of targeted drug delivery lowering the off-target effects [15]. Combining these different fields, it would be possible to develop a carrier (e.g., a synthetic particle or object) that mimics (e.g., partially, completely, etc.) the pathogen or pathogens of interest (e.g., viruses, bacteria, other pathogens, etc.) in order to hinder the spread of the disease by competitive inhibition. Further, such carriers can be used to advantageously deliver the appropriate API, drug, molecule and/or other substance or material (e.g., as provided in one or more portions of this specification) to the target tissues with increased efficacy and minimal or reduced side effects. For example, by utilizing such targeting strategies, it is possible to create a carrier that can deliver one or more materials to specific cell populations. In some embodiments, carriers can be loaded with or otherwise include different active molecules used in treating diseases caused by disruptions in the cellular homeostasis.

SUMMARY

According to some embodiments, a synthetic carrier comprises a particle forming a core and having a maximum size in at least one dimension in a nanometer or a micrometer range, and a functionalized surface configured to bind to target areas of cell surfaces of the host, wherein the functionalized surface comprises protruding features that include targeting moieties, wherein the protruding features at least partially mimic protrusions of a disease-causing agent or source, wherein at least one component is loaded into or onto or otherwise included with the carrier, wherein the carrier is configured to bind to specific areas of the host cells for targeted delivery of the at least one component to the host, and wherein the carrier is configured to assist with a treatment of a disease or adverse impact caused by the pathogen or other disease-causing agent or source.

According to some embodiments, the target areas comprise at least one of a receptor, a target molecule, an amino acid and a nucleotide.

According to some embodiments, the disease-causing agent or source comprises a pathogen. In one embodiment, the pathogen comprises a virus (e.g., a coronavirus, a SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV) and a virus that impacts the respiratory system, another type of virus, etc.).

According to some embodiments, the carrier is made, at least in part, using a bottom-up method or a top-down method of fabrication.

According to some embodiments, the disease causing agent or source comprises one or more of the following: an infectious agent, a pathogen, a virus, a bacterium, a parasite, an antigen, a protein, a prion, a mold, a fungus, a toxin, a poison and an allergen.

According to some embodiments, the carrier is used for personalized treatment or personalized diagnostics related to a disease, infection or allergic reaction or other condition caused by the disease-causing agent or source.

According to some embodiments, the at least one component comprises at least one of the following: an API, a drug, a protein, an amino acid, RNA, DNA and a molecule.

According to some embodiments, the carrier is configured to be delivered to the host using an inhalation device, an aerosol, a spray, a drop, a tablet, a cream, an ointment, an injection or an intravenous administration.

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According to some embodiments, the carrier is configured to be delivered to the host using a dispensing device, the dispensing device comprising an inhalation device configured to deliver the carrier to at least a portion of the host's respiratory tract.

According to some embodiments, a method of providing treatment of a host comprises binding a carrier to cell structures of the host to reduce a likelihood of a disease-causing agent or source binding to said cell structures, thereby to at least partially inhibiting the disease-causing agent or source from binding to said cell structures, wherein the carrier comprises a core and a surface functionalized on the core, wherein the functionalized surface is configured to bind to target areas of the cell structures of the host, wherein at least one component is loaded into or onto or otherwise included with the carrier, the carrier being configured for targeted delivery of the at least one component to the host, and wherein the carrier is configured to assist with a treatment of a disease or adverse impact caused by the pathogen or other disease-causing agent or source.

According to some embodiments, a size of the carrier is in the nanometer to micrometer range.

According to some embodiments, the disease-causing agent or source comprises a pathogen. In one embodiment, the pathogen comprises a virus (e.g., a coronavirus, a SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV) and a virus that impacts the respiratory system, another type of virus, etc.).

According to some embodiments, the carrier is made, at least in part, using a bottom-up method or a top-down method of fabrication.

According to some embodiments, the disease causing agent or source comprises one or more of the following: an infectious agent, a pathogen, a virus, a bacterium, a parasite, an antigen, a protein, a prion, a mold, a fungus, a toxin, a poison and an allergen.

According to some embodiments, the carrier is used for personalized treatment or personalized diagnostics related to a disease, infection or allergic reaction or other condition caused by the disease-causing agent or source.

According to some embodiments, the at least one component comprises at least one of the following: an API, a drug, a protein, an amino acid, RNA, DNA and a molecule.

According to some embodiments, the carrier is configured to be delivered to the host using an inhalation device, an aerosol, a spray, a drop, a tablet, a cream, an ointment, an injection or an intravenous administration.

According to some embodiments, the carrier is configured to be delivered to the host using a dispensing device, the dispensing device comprising an inhalation device configured to deliver the carrier to at least a portion of the host's respiratory tract.

According to some embodiments, a method of providing treatment of a host comprises binding a carrier to cell structures of cells of the host to reduce a likelihood of an agent binding to said cell structures to at least partially inhibits the agent from binding to said cell structures, wherein the carrier comprises a core and a surface functionalized on the core, wherein the functionalized surface bind to target areas of cell structures of the host's cells, and wherein the carrier is to be used as targeted treatment for one or more disease, infections or allergic reactions caused by a disease-causing agent or source.

According to some embodiments, a synthetic carrier comprises a particle forming a core and having a maximum size in at least one dimension in a nanometer or a micrometer

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range and a functionalized surface capable of binding to target areas of cell surfaces of the host, wherein the functionalized surface features contain targeting moieties protruding from the surface, wherein the protruding surface features at least partially physically mimic naturally-occurring protrusions of the pathogen, wherein the carrier is loaded into or onto with an additive, wherein the carrier is configured to bind to specific areas of the host cells for targeted delivery; the one or more diseases, infections or allergic reactions, and wherein the carrier is to be used as targeted treatment for one or more disease, infections or allergic reactions caused by a disease-causing agent or source.

According to some embodiments, the target areas of the host's cells comprise a receptor, a target molecule, an amino acid and/or a nucleotide.

According to some embodiments, the disease-causing agent or source comprises a pathogen. In some embodiments, the pathogen comprises a virus (e.g., a coronavirus, a SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV), another virus that impacts the respiratory system, any other type of virus, etc.).

According to some embodiments, the carrier is made, at least in part, using a bottom-up method or a top-down method of fabrication.

According to some embodiments, the disease causing agent or source comprises one or more of the following: an infectious agent, a chemical agent, a pathogen, a virus, a bacterium, a parasite, an antigen, a protein, a prion, a mold, a fungus, a toxin, a poison or an allergen, enzyme, electromagnetic radiation, carcinogens, molecules, hormones, inflammation, other agents, etc.

According to some embodiments, the carrier is used for personalized diagnostics, treatment and/or prevention of one or more diseases, infections or allergic reactions.

According to some embodiments, the one or more diseases, infections or allergic reactions comprises at least one of the following: autoimmunity, inflammation, cancer, diseases caused by imbalance of hormones, neurological disorders, muscle dystrophies, cystic fibrosis, diabetes, rheumatoid arthritis, systemic lupus erythematosus, etc.

According to some embodiments, wherein the carrier further comprises at least one additional component, the at least one additional component comprises at least one of: an API, a drug, a protein, an amino acid, RNA, DNA and a molecule.

According to some embodiments, the carrier is configured to be delivered to the host using an inhalation device, an aerosol, a spray, an eye drop, a tablet, a cream, an ointment or an injection. In some embodiments, the carrier is configured to be delivered to the host using a dispensing device, the dispensing device comprising an inhalation device configured to deliver the carrier to at least a portion of the host's respiratory tract.

According to some embodiments, a method of reducing the spread of infection by a target pathogen in a host comprises binding a carrier to a target areas of the host's cells to reduce a likelihood that the target pathogen binds to the target areas, thereby blocking at least some of the target areas, and providing a gained advantage to the host's immune system to fight a disease caused by the target pathogen, wherein providing the gained advantage comprises decorating the carrier with epitopes to be used as a vaccination at target cell populations.

According to some embodiments, a method of reducing the onset and/or spread of infection by a target pathogen in

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a host comprises binding a carrier to a target areas of the host's cells to reduce a likelihood that the target pathogen binds to the target areas, and creating an advantage to the host's immune system to fight a disease caused by the target pathogen, wherein creating the advantage comprises deco-

rating the carrier with epitopes or targeting ligands. According to some embodiments, the target areas of the host's cells comprise a receptor, a target molecule, an amino acid, a nucleotide and/or the like.

According to some embodiments, the carrier comprises a core and a functionalized surface. In some arrangements, the functionalized surface comprises features that mimic surface properties of the target pathogen. In some embodiments, the features comprise protruding structures (e.g., peptide structures in the form of protein spikes, other protruding structures, etc.).

According to some embodiments, the carrier further comprises at least one of an API, a drug, a protein, an amino acid, RNA, DNA, a molecule and/or the like, wherein the at least one of an API, a drug, a protein, an amino acid, RNA, DNA, a molecule and/or the like is configured to reduce the spread of the target pathogen or is configured to hinder replication of the target pathogen inside the host.

According to some embodiments, the carrier further comprises a compound configured to hinder replication of the target pathogen. In some embodiments, the compound comprises an anti-pathogenic compound, an anti-viral compound, an anti-microbial compound and/or any other compound or substance.

According to some embodiments, the target pathogen comprises a virus (e.g., SARS-CoV-2, a coronavirus, influenza, rhinoviruses, noroviruses, respiratory syncytial virus (RSV), etc.).

According to some embodiments, the carrier is configured to resemble the target pathogen.

According to some embodiments, the carrier can be loaded to a medical device capable of releasing an amount of the carrier to specific tissue of the host. In some embodiments, the specific tissue of the host comprises the host's upper or lower respiratory tract.

According to some embodiments, a carrier for reducing a likelihood of a pathogen binding to cell structures of a host comprises a core having an exterior surface, a plurality of surface features extending from the exterior surface of the core, wherein the surface features are configured to bind to target areas of cell structures of the host, wherein binding of the carrier to at least one of the target areas of cell structures of the host is configured to at least partially block the pathogen from binding to said target areas as a result of competitive inhibition, and wherein the surface features at least partially physically mimic naturally-occurring protrusions of the pathogen, and wherein the surface features are configured to comprise immune stimulating properties, wherein a size of the carrier is in the nanometer to micrometer range (e.g., in the nanometer or micrometer range).

According to some embodiments, a carrier for reducing a likelihood of a pathogen or other disease-causing agent binding to cell structures of a host comprises a core having an exterior surface, a plurality of surface features extending from the exterior surface of the core, wherein the surface features are configured to bind to target areas of cell structures of the host, wherein binding of the carrier to at least one of the target areas of cell structures of the host is configured to at least partially block the pathogen from binding to said target areas as a result of competitive inhibition, and wherein the surface features at least partially physically mimic naturally-occurring protrusions of the

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pathogen, and wherein the surface features are configured to comprise immune stimulating properties. The carrier further comprises a plurality of binding sites along the exterior surface, wherein the binding sites are configured to attract at least one portion of the pathogen, wherein the binding sites are configured to at least partially mimic binding sites of the cell structures of the host, wherein the binding sites are recognizable by the pathogen or other disease-causing agent and are able to be bound by the pathogen or agent, thereby at least partially immobilizing the pathogen and reducing the likelihood of the pathogen binding to target areas of cell structures of the host, and wherein a size of the carrier is in the nanometer to micrometer range (e.g., in the nanometer or micrometer range).

According to some embodiments, the pathogen comprises a virus. In some embodiments, the virus includes one or more of the following: a coronavirus, SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV), another virus that impacts the respiratory system and any other type of virus.

According to some embodiments, the pathogen comprises a bacterium, a parasite, an antigen, a prion, a mold, a fungus or an allergen.

According to some embodiments, the naturally-occurring protrusions of the pathogen comprise proteins at the surface of the viral exterior.

According to some embodiments, the carrier is sized, shaped or otherwise configured to reach targeted portions of the host that are susceptible to infection by the pathogen. In some embodiments, the targeted portions of the host that are susceptible to infection by the pathogen comprise the lungs or other area of the host's upper or lower respiratory tract. In some embodiments, the carrier is configured to be delivered via the respiratory tract of the host to the targeted portions of the host that are susceptible to infection by the pathogen.

According to some embodiments, the carrier comprises at least one coating that improves a binding affinity of the carrier to the pathogen relative to a binding affinity of the cell structures of the host to the pathogen.

According to some embodiments, the carrier further comprises at least one component positioned at least partially on and/or within carrier. In some embodiments, the at least one component comprises a pharmaceutical agent (e.g., API). In some embodiments, the pharmaceutical agent comprises at least one of an anti-viral compound, an adjuvant, a nucleic acid and an RNA or DNA sequence. In several embodiments, the adjuvant is selected from the group consisting of potassium alum, aluminum hydroxide, aluminum phosphate, calcium phosphate hydroxide, paraffin oil, peanut oil, inactivated bacteria and/or bacterial products from *Bordetella pertussis*, inactivated bacteria and/or bacterial products from *Mycobacterium bovis*, inactivated bacterial toxoids, monophosphoryl lipid A, saponins (e.g., QS-21), soybean derivatives, senegaroot derivatives, interleukin (IL) 1, IL2, IL12, squalene or squalene-containing compounds, MF59, AS03, and combinations thereof.

According to some embodiments, a carrier for reducing a likelihood of a pathogen or other disease-causing agent binding to cell structures of a host comprises a core, surface features extending from an exterior surface of the core, wherein the surface features are configured to bind to target areas of cell structures of the host to at least partially block the pathogen or other agent from binding to said target areas as a result of competitive inhibition, and wherein the surface features comprise immune stimulating properties.

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According to some embodiments, a carrier for reducing a likelihood of a pathogen or other disease-causing agent binding to cell structures of a host comprises a core, surface features extending from an exterior surface of the core, wherein the surface features are configured to bind to target areas of cell structures of the host to at least partially block the pathogen or other agent from binding to said target areas as a result of competitive inhibition, and wherein the surface features comprise immune stimulating properties. The carrier further comprises a plurality of binding sites along the exterior surface, wherein the binding sites are configured to attract at least one portion of the pathogen or other agent, wherein the binding sites are configured to at least partially mimic binding sites of the host, and wherein the binding sites are recognizable by the pathogen or other agent and are able to be bound by the pathogen or other agent, thereby at least partially immobilizing the pathogen or other agent and reducing the likelihood of the pathogen or other agent binding to target areas of cell structures of the host.

According to some embodiments, the pathogen or other disease-causing agent comprises a virus. In some embodiments, the virus includes one or more of the following: a coronavirus, a SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV), another virus that impacts the respiratory system and any other type of virus.

According to some embodiments, the pathogen comprises a bacterium, a parasite, an antigen, a prion, a mold, a fungus or an allergen.

According to some embodiments, the carrier is sized, shaped and otherwise configured to reach targeted portions of the host that are susceptible to infection by the pathogen, the targeted portions of the host that are susceptible to infection by the pathogen comprise the lungs or other area of the host's respiratory tract.

According to some embodiments, the carrier is configured to be delivered via a respiratory tract of the host to the targeted portions of the host that are susceptible to infection by the pathogen or other disease-causing agent.

According to some embodiments, the carrier further comprises at least one component positioned at least partially on and/or within carrier (e.g., an anti-viral compound, a nucleic acid, an adjuvant, an RNA or DNA sequence and another pharmaceutical agent, etc.).

In some embodiments, additional adjuvants, other additives and/or modifications to the carriers, adjuvants, particles and/or objects can be used. Examples of adjuvants include, by way of example and without limitation, mineral salts (e.g., aluminum hydroxide, magnesium hydroxide, etc.), cytokines and/or chemokines (e.g., GM-CSF, Type I IFN- α , IL-12, etc.), modified microbial antigens (e.g., LT and CT derivatives, monophosphoryl lipid A (MPL), etc.), immunostimulatory oligonucleotides (e.g., CpG motifs), oil emulsions (e.g., MF59, 25 AS02, AS03, Mantanide ISA-51, ISA-720, AF03, etc.), liposomes (e.g., AS01, AS015, etc.), particulate adjuvants (e.g., virosomes, AS04, ISCOMS) and/or any other materials. In some embodiments, such additives and/or modification can help enhance the immunogenic reaction to said carriers, adjuvants, particles and/or objects.

In one embodiment, the additives of the carriers, adjuvants, particles and/or objects include one or more of the following, for example and without limitation, a preservative (e.g., thimerosal, 2-phenoxyethanol, etc.), a stabilizer (e.g., sorbitol, sucrose, glycerol, trehalose, gelatin, urea, etc.), a surfactant (e.g., polysorbate 80, polysorbate 20, etc.) and/or any other enhancing component or other additive.

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According to some embodiments, a carrier for reducing a likelihood of a pathogen binding to cell structures of a host comprises a core, surface features extending from an exterior surface of the core, wherein the surface features are configured to bind to target areas of cell structures of the host to at least partially block the pathogen from binding to said target areas as a result of competitive inhibition, and a plurality of binding sites along the exterior surface, thereby at least partially reducing the likelihood of the pathogen binding to target areas of cell structures of the host. According to some embodiments, the pathogen comprises a virus. In some embodiments, the virus includes one or more of the following: a coronavirus, a SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV), another virus that impacts the respiratory system and any other type of virus.

According to some embodiments, a carrier for reducing a likelihood of a pathogen or other disease-causing agent binding to cell structures of a host comprises a core, surface features extending from an exterior surface of the core, wherein the surface features are configured to bind to target areas of cell structures of the host to at least partially block the pathogen or other agent from binding to said target areas as a result of competitive inhibition, and a plurality of binding sites along the exterior surface, wherein the binding sites are configured to attract at least one portion of the pathogen or other agent, wherein the binding sites are recognizable by the pathogen or other agent and are able to be bound by the pathogen or other agent, thereby at least partially immobilizing the pathogen or other agent and reducing the likelihood of the pathogen or other agent binding to target areas of cell structures of the host. According to some embodiments, the pathogen comprises a virus. In some embodiments, the virus includes one or more of the following: a coronavirus, a SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV), another virus that impacts the respiratory system and any other type of virus.

According to some embodiments, the pathogen or other disease-causing agent comprises a bacterium, a parasite, an antigen, a prion, a mold, a fungus or an allergen. According to some embodiments, the carrier is sized, shaped and otherwise configured to reach targeted portions of the host that are susceptible to infection and/or other impact by the pathogen or other disease-causing agent, the targeted portions of the host that are susceptible to infection and/or other impact by the pathogen or other agent comprise the lungs or other area of the host's respiratory tract. According to some embodiments, the carrier is configured to be delivered via a respiratory tract of the host to the targeted portions of the host that are susceptible to infection or other impact by the pathogen or other agent. According to some embodiments, the carrier further comprises or is accompanied by (e.g., delivered before, with, or after the carrier) at least one component positioned at least partially on and/or within carrier (e.g., an anti-viral compound, a nucleic acid, an RNA or DNA sequence and another pharmaceutical agent (such as an adjuvant), etc.).

According to some embodiments, a method of reducing a spread of pathogens or other disease-causing agents within a host comprises at least partially blocking pathogens or other agents from binding to said target areas as a result of competitive inhibition by delivering a carrier to the host, wherein the carrier comprises a core, surface features extending from an exterior surface of the core, and a plurality of binding sites along the exterior surface, and at least partially immobilizing pathogens or other agents and

reducing the likelihood of pathogens or other agents binding to target areas of cell structures of the host by binding the carrier to at least one of the pathogens or agents, wherein the surface features are configured to bind to target areas of cell structures of the host, wherein the surface features at least partially physically mimic naturally-occurring protrusions of the pathogen or other agent, wherein the surface features are configured to comprise immune stimulating properties.

According to some embodiments, a method of reducing a spread of pathogens within a host comprises at least partially blocking pathogens from binding to said target areas as a result of competitive inhibition by delivering a carrier to the host, wherein the carrier comprises a core, surface features extending from an exterior surface of the core, and a plurality of binding sites along the exterior surface, and reducing the likelihood of pathogens binding to target areas of cell structures of the host by competitive inhibition, wherein the surface features at least partially physically mimic naturally-occurring protrusions of the pathogen, wherein the surface features are configured to comprise immune stimulating properties to be delivered to target cell populations and subsequently mounting an efficient immunologic reaction, wherein the binding sites are configured to attract at least one portion of the pathogen, wherein the binding sites are configured to at least partially mimic binding sites of the cell structures of the host, and wherein, and wherein the binding sites are recognizable by the pathogen and are able to be bound by the pathogen.

According to some embodiments, a method of reducing a spread of pathogens within a host comprises at least partially blocking pathogens from binding to said target areas as a result of competitive inhibition by delivering a carrier to the host, wherein the carrier comprises a core, surface features extending from an exterior surface of the core, and a plurality of binding sites along the exterior surface, where the exterior surface at least partially includes specific set of epitopes capable of eliciting an immunological response towards viruses and/or other pathogens.

According to some embodiments, the carrier system described herein is configured to be used as a vaccination at target cell populations where the said epitopes can elicit a B cell response generating specific antibodies and/or T cell response priming and mounting a possible cytotoxic T-cell response against virus and/or other pathogens. Therefore, according to some embodiments, the carrier system described herein is configured to reduce the likelihood of infection of host cells by (1) blocking receptors and/or other binding sites/features of host cells to which viruses and/or other pathogens may bind (e.g., for entry into host cells), (2) delivering immune stimulating properties to target cell populations and by mounting an efficient. In several embodiments, the carrier system is utilized with an adjuvant to enhance the development of an immune response to the virus and/or other pathogens. The adjuvant may be organic or inorganic, depending on the embodiment. In several embodiments, the adjuvant is selected from the group consisting of potassium alum, aluminum hydroxide, aluminum phosphate, calcium phosphate hydroxide, and combinations thereof. An oil, such as paraffin oil or peanut oil may be used in some embodiments. Inactivated bacteria and/or bacterial products may be used in some embodiments, such as *Bordetella pertussis*, *Mycobacterium bovis*, or inactivated bacterial toxoids, including monophosphoryl lipid A. Plant-derived adjuvants can also be used in some embodiments, such as saponins (e.g., QS-21), soybean derivatives, senegalo root derivatives and the like. Cytokines, such as interleukin (IL) 1, IL2 and/or IL12 can be used in some embodiments.

Squalene or squalene-containing adjuvants may also be used, such as MF59 and/or AS03, depending on the embodiment. In some embodiments, additives, preservatives and/or other materials can be used, such as, for example, thimerosal and 2-phenoxyethanol, stabilizers (e.g., sorbitol, sucrose, trehalose, glycerol, purified gelatin, urea, etc.), surfactants (e.g., polysorbate 80, polysorbate 20, etc.) and/or the like.

According to some embodiments, the pathogen comprises a virus. In some embodiments, the virus includes one or more of the following: a coronavirus, a SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV), another virus that impacts the respiratory system and any other type of virus.

According to some embodiments, the pathogen comprises a bacterium, a parasite, an antigen, a prion, a mold, a fungus or an allergen.

According to some embodiments, the carrier is sized, shaped and otherwise configured to reach targeted portions of the host that are susceptible to infection by the pathogen, the targeted portions of the host that are susceptible to infection by the pathogen comprise the lungs or other area of the host's respiratory tract.

According to some embodiments, the carrier is configured to be delivered via a respiratory tract of the host to the targeted portions of the host that are susceptible to infection by the pathogen.

According to some embodiments, the carrier further comprises at least one component positioned at least partially on and/or within carrier (e.g., an anti-viral compound, a nucleic acid, an RNA or DNA sequence and another pharmaceutical agent, such as an adjuvant, etc.).

According to some embodiments, a carrier for reducing a likelihood of a pathogen or other disease-causing agent binding to cell structures of a host, the carrier comprising a core having an exterior surface, a plurality of surface features extending from the exterior surface of the core, wherein the surface features are configured to bind to target areas of cell structures of the host, wherein binding of the carrier to at least one of the target areas of cell structures of the host is configured to at least partially block the pathogen or other agent from binding to said target areas as a result of competitive inhibition, wherein the surface features are configured to at least partially physically mimic naturally-occurring protrusions of the pathogen or other agent, and wherein the surface features are configured to comprise immune stimulating properties, and a plurality of binding sites along the exterior surface, thereby at least partially reducing the likelihood of the pathogen binding to target areas of cell structures of the host, and wherein a maximum cross-sectional dimension of the carrier in at least one dimension is in the nanometer to micrometer range (e.g., in the nanometer or micrometer range).

According to some embodiments, a carrier for reducing a likelihood of a pathogen or other disease-causing agent binding to cell structures of a host, the carrier comprising a core having an exterior surface, a plurality of surface features extending from the exterior surface of the core, wherein the surface features are configured to bind to target areas of cell structures of the host, wherein binding of the carrier to at least one of the target areas of cell structures of the host is configured to at least partially block the pathogen or other agent from binding to said target areas as a result of competitive inhibition, wherein the surface features are configured to at least partially physically mimic naturally-occurring protrusions of the pathogen or other agent, and wherein the surface features are configured to comprise immune stimulating properties, and a plurality of binding

sites along the exterior surface, wherein the binding sites are configured to attract at least one portion of the pathogen or other agent, wherein the binding sites are configured to at least partially mimic binding sites of the cell structures of the host, wherein the binding sites are recognizable by the pathogen or other agent and are able to be bound by the pathogen or other agent, thereby at least partially immobilizing the pathogen or other agent and reducing the likelihood of the pathogen or other agent binding to target areas of cell structures of the host, and wherein a maximum cross-sectional dimension of the carrier in at least one dimension is in the nanometer to micrometer range (e.g., in the nanometer or micrometer range).

It is an aim of the present application to control and hinder (e.g., slow or prevent) the spread of pathogens and other infectious agents, e.g., viruses, bacteria, parasites, antigens, proteins, prions, toxins, allergens, other substances that are foreign and/or potentially harmful and the like. Specifically, the application provides ways of targeting viruses, such as, for example, influenzas, rhinoviruses, noroviruses, respiratory syncytial virus (RSV), coronaviruses (e.g., SARS-CoV-2, future mutated strains derived from a coronavirus, etc.) and the like, that could otherwise give rise to disease, infections or allergic reactions in the host.

It is an aim of the inventions disclosed herein to decrease the risk of infection (or at least decrease the spread of infection if a host has been infected) by a pathogen or pathogens. As such, various embodiments disclosed herein can be helpful combatting infection and any resulting symptoms and other consequences (e.g., respiratory infection, diarrhea, common cold, cytokine storm, other inflammatory reactions, general discomfort, intubation, other symptoms, death, etc.). Accordingly, various embodiments disclosed herein are configured to, at least partially, resist and otherwise combat the effects of contracting the COVID-19 disease caused by SARS-CoV-2 (e.g., via entry of the virus into a host for a temporary or prolonged duration), to give a targeted treatment for the specific disease caused by the infectious agent, to provide one or more additional benefits or advantages, etc.

Further, it is an aim of the inventions disclosed in the present application to provide a method for preventing the spreading and/or for lowering the infection rate of pathogens, such as SARS-CoV-2, influenzas, rhinoviruses, noroviruses, respiratory syncytial viruses (RSVs) and/or the like, by, at least in part, competitive inhibition using synthesized carriers (e.g., nanomaterials, particles, objects, etc.).

Thus, in one aspect, the inventions disclosed herein include carriers (e.g., synthesized nano- or micro sized materials) that mimic, at least partially, the pathogen or pathogens of interest, such as coronaviruses (e.g., SARS-CoV-2), influenzas, rhinoviruses noroviruses, respiratory syncytial viruses (RSVs) and other viruses (e.g., viruses capable of causing respiratory infection) using surface functionalization to hinder or otherwise lower the likelihood of the infectious agent entering the host. In some embodiments, carriers can be used to target pathogens other than viruses, including without limitation, bacteria, parasites, antigens, prions, mold, fungi, toxins, poisons, and allergens. In some embodiments, this is accomplished, at least in part, by competitive inhibition (e.g., at the cellular level).

It is another aim of the inventions disclosed in the present application to create a carrier (e.g., man-made particle, object, material, etc.) that efficiently binds to the pathogen(s) of interest (e.g., coronavirus) and/or circulating co-receptors (e.g., high-density lipoprotein (HDL) receptor in the blood and other secondary receptors such as the FcγR that affects

SARS-CoV-2 infection dynamics by antibody-mediated enhancement (ADE)). Thus, in some embodiments, the carrier at least partially binds to the pathogen or other infectious agent co-receptor of the host and at least partly hinders the receptor mediated viral entry. In some embodiments, the carrier is configured to at least partially inhibit the infectious agent's reproduction capabilities by including appropriate API, thereby reducing the spread of said host organism. In some embodiments, the carriers immune stimulating protrusions make it easier for the host body to identify, engulf and/or eliminate the pathogen and/or pathogen infected cell, resulting in elimination or neutralization of the infectious agent.

It is another aim of the inventions disclosed in the present application to create a carrier (e.g., man-made particle, object, material, etc.) that efficiently binds to the pathogen(s) of interest (e.g., coronavirus) and/or circulating co-receptors (e.g., high-density lipoprotein (HDL) receptor in the blood and other secondary receptors such as the FcγR that affects SARS-CoV-2 infection dynamics by antibody-mediated enhancement (ADE)). Thus, in some embodiments, the carrier at least partially encapsulates and/or immobilizes the pathogen or other infectious agent and at least partly hinders the receptor mediated viral entry. In some embodiments, the carrier is configured to at least partially inhibit the infectious agent's reproduction capabilities, thereby reducing the spread of said host organism. In some embodiments, the carrier makes it easier for the host body to identify, engulf and/or filter the macromolecule holding the pathogen, resulting in elimination or neutralization of the infectious agent.

A synthetic carrier can be used to at least partially prevent or reduce infection of a host by one or more pathogens, such as, for example, influenzas, rhinoviruses, noroviruses, respiratory syncytial viruses (RSVs), corona viruses (e.g., SARS-CoV-2), other viruses or pathogens. In some embodiments, the carrier is configured to bind to target areas of cell surfaces of a host. For example, the carrier can bind to ACE2 receptors, TMPRSS2 receptors, sialic acid, histo-blood group antigens, ICAM-1, IGF1R receptors, intracellular or extracellular receptors, other receptors, or combination of receptors and/or any other portion of the cell structures of the host that may be susceptible to the pathogen. According to some embodiments, a carrier is formed by biocompatible particles having a maximum size in at least one dimension in the nanometer or micrometer range (e.g., a core of the carrier includes a maximum size in at least one dimension in the nanometer or micrometer range). Further, a functionalized surface can be formed on or along a core of the carrier that is capable of binding to said target areas of the cell surfaces of the host to at least temporarily and/or at least partially block the target areas, thereby, at least partially, preventing or minimizing pathogen binding and internalization. Accordingly, the risk of the host being infected or contracting a disease caused by said pathogen, such as a virus, can be beneficially reduced.

In an aspect, the present application provides for loading of synthetic particles (e.g., carriers) with API or molecules, such as for example and without limitation, Celastrol, antiviral drugs, Zinc and/or immune stimulating molecules such as, for example, Interferon-Gamma modulators alternatively administering RNA vectors encoding Interferon-Gamma producing said protein in the host, anti-viral compounds that prevent the spread of (and/or hinder the replication of) the targeted pathogen or pathogens (e.g., coronavirus, influenzas, rhinovirus, noroviruses, respiratory syncytial virus, etc.) inside the host body and/or the like.

In some embodiments, particles (e.g., carriers) having prophylactic properties include interfering and/or modulating nucleotides, such as for example and without limitation, RNA interference (RNAi), microRNA (miRNA), short hairpin RNA or small hairpin RNA (shRNA), small interfering RNA (siRNA), long non-coding RNA (lncRNA), small nucleolar RNA (snoRNA), Piwi-interacting RNA (piRNA). Such carrier embodiments can help prevent or reduce the spread of (and/or hinder the replication of) the targeted pathogen or pathogens (e.g., coronavirus, influenzas, rhinovirus, noroviruses, respiratory syncytial virus, etc.) inside the host body and/or the like.

It is another aim of the inventions disclosed in the present application to provide a medical device capable of delivering the synthesized carrier, particle, object or other material (e.g., on-demand by the patient or other user). Therefore, in some embodiments, the carrier can be administered to the anatomy by the subject or patient, making the carrier capable of self-administration. For example, in some embodiments, such a medical device comprises an inhalation device, an aerosol, a spray, eye or oral drops, an intravenous injection, a tablet, a topically applicable cream, an ointment or other material.

It is another aim of the inventions disclosed in the present application to provide a medical countermeasure similar to that of chelating agents used in toxications of metal complexes (e.g., arsenic poisoning, snake venoms, mold toxins, etc.) [16]. In some embodiments, the present inventions provide a functionalized nanomaterial or carrier, which, in some configurations, is effectively an antidote capable of binding toxic components of a specific virus to larger entities that can be metabolized, degraded or secreted from the body and/or capable of binding to viral co-receptors inside the host to minimize or otherwise reduce potential spreading inside the organism [15]. The antidote (e.g., nanomaterial or other carrier) can be inhaled, orally ingested or administered through intravenous injection and/or any other delivery method or technology (e.g., inhalation, ingestion, topical application, etc.), as desired or required.

It is another aim of the inventions in the present application to provide a medical countermeasure similar to that of chelating agents used in toxications of metal complexes (e.g., arsenic poisoning, snake venoms, mold toxins and the like). In some embodiments, the present inventions provide a carrier (e.g., a functionalized nanomaterial) that acts as an antidote, and is advantageously capable of binding toxic metal complexes, toxins, poisons and/or the like to larger entities that can be metabolized, degraded and/or secreted/otherwise removed from the body [15]. The antidote (e.g., carrier or nanomaterial) could be inhaled, orally ingested, administered to the host using intravenous injection and/or any other delivery method or technology.

In some embodiments, a carrier (e.g., a functionalized nanoparticle) is loaded, coated and/or decorated with an API and/or RNA/DNA and/or other molecules or materials capable of binding to cell structures of the host (e.g., receptors of the host's cell structures) and delivering its cargo or contents to targeted cell population. Beneficially, this can at least partially block or otherwise limit entry of a specific pathogen, such as, for example, influenzas, rhinoviruses, noroviruses, respiratory syncytial viruses, coronaviruses and other viruses causing infection (e.g., in the respiratory tract or other anatomical location). Further the use of such carriers can advantageously provide the capability of releasing certain materials (e.g., APIs, therapeutics, other molecules, etc.) to the host thereby minimizing or at least reducing the spread of the infectious agent.

According to some embodiments, the present application discloses a carrier (e.g., a polymeric or protein/peptide functionalized nano and/or micro particle) that is loaded with one or more anti-viral molecules capable of binding to one or more receptors and/or other portions of a host cell structure (e.g., ACE2 and/or TMPRSS2 receptors, sialic acid, histo-blood group antigens, ICAM-1, IGF1R receptors, other receptors in humans that at least partially hinder (e.g., allosterically hinder) the targeted virus or other pathogen (e.g., influenza, rhinovirus, norovirus, respiratory syncytial virus, SARS-CoV-2 or another corona virus) from binding to its target receptor). As a result, the risk of infecting the host can be reduced, e.g., for a limited or prolonged duration.

Certain advantages are obtainable as a result of the present inventions, as carriers (e.g., nanoparticles, other particles or objects, etc.) can be synthesized using different materials and/or functionalized (or otherwise configured) with virtually endless combinations of features and/or functionality. In some embodiments, the carriers include mesoporous silica nanoparticles (MSNs) or other inorganic silica-based materials which have shown great potential for targeted drug delivery. For example, MSNs can have tunable ordered repetitive mesostructures of pores that can be loaded with different drugs and/or other components or materials. In some arrangements, such carriers or particles can be synthesized in various sizes, shapes and/or other configurations, as desired or required for its particular purpose or application. Furthermore, inorganic silica materials, such as MSNs, are safe, biocompatible, stable, customizable and versatile. For instance, inorganic silica materials have been given a Generally Recognized As Safe (GRAS) designation by the FDA as silica degrades in aqueous solution to silicic acid and gets excreted or otherwise removed (e.g., via the urine), and is, therefore, considered biocompatible [15]. In some embodiments, since MSNs and other inorganic silica-based materials have been proved to be a versatile delivery vehicle with beneficial features and properties (e.g., improved stability, large surface area, tunable pore sizes and volumes, capable easy encapsulation of drugs, proteins, biogenic molecules, etc.), they are well suited to be used in carriers.

In some embodiments, the carriers comprise lipid-based micelles (e.g., forming the cores of the carriers). Such carriers can be provided by synthesizing, for example, cholesterol based lipid particles decorated with SARS-CoV-2 spike protein fragments that bind both to ACE2 and TMPRSS2 as well as to cholesterol and its high-density lipoprotein (HDL) scavenger receptor B type 1 (SR-B1) and/or FcγR receptor. These co-receptors can facilitate ACE2-dependent entry of the carrier (e.g., the envisioned nanoparticle) loaded with one or more selected APIs for combating a targeted disease (e.g., COVID-19 disease). In some embodiments, advantageously, by creating a carrier (e.g., a nanoparticle and/or microparticle) that competes with the spike protein-HDL interaction, the ability of SARS-CoV-2 for ACE2-mediated internalization is lowered and viral entry to host cells is blocked and replication hindered.

In some embodiments, the carriers comprise solid lipid particles synthesized by microfluidics and/or protein-based particles such as ferritin-based particles that self-assemble decorated or conjugated with viral mimicking protrusion capable of both binding to the target receptor and stimulating the host immune system against the said virus.

In some embodiments, carriers comprise quantum nanoparticles (e.g., as a core). Such quantum particles or carriers once decorated or otherwise provided with the desired surface features (e.g., SARS-CoV-2 or other viral spike protein receptor binding domain (RBD)) can be

capable of binding to ACE2 receptors, other receptors and/or other binding sites of the host cell structure. Accordingly, such carriers can be internalized (e.g., by ACE2-GFP HEK293T cells after a certain time period, for example, approximately 3 hours), thereby validating that it is possible to produce man-made virus-like particles that efficiently bind and are internalized by target cells. Furthermore, protein-lipid entities, such as Dalbavancin, an antibiotic, can bind to the ACE2 receptor (or another receptor or binding site) to prevent or reduce the likelihood that the targeted pathogen (e.g., SARS-CoV-2) is able to enter its host via intervening viral-receptor interactions.

In some arrangements, carriers (e.g., nano-sized and/or micro-sized materials) are synthesized that mimic (e.g., accurately, approximately) the targeted pathogen by using, for example, the known size, morphology, surface properties and/or other properties of the infectious agent. Thus, a man-made carrier (e.g., particle, object) can be produced that at least partially hinders or otherwise mitigates the spread of the disease caused by the target infectious agent (e.g., virus, bacterium, other pathogen) by competitive inhibition. In some embodiments, such man-made carriers (e.g., particles, materials, etc.) are configured to (i) bind to receptors or other binding sites of host cell structures, thereby blocking the attachment of pathogens to such receptors or binding sites and preventing (or lowering the likelihood of) the infectious agent from entry into the cell structure, and (ii) the mimicking protrusion of the carrier elicits and immune response against said infectious agent. Accordingly, the carrier embodiments disclosed herein and equivalents thereof provide two possible approaches to reducing infection of a host's cells, and as such, have immense potential in different applications in medicine, drug development, medical devices, consumer, sanitation products and/or the like.

In some arrangements, carriers (e.g., nano-sized and/or micro-sized materials) are synthesized that mimic (e.g., accurately, approximately) the targeted pathogen by using, for example, the known size, morphology, surface properties and/or other properties of the infectious agent. Thus, a man-made carrier (e.g., particle, object) can be produced that at least partially hinders or otherwise mitigates the spread of the disease caused by the target infectious agent (e.g., virus, bacterium, other pathogen) by competitive inhibition. In some embodiments, such man-made carriers (e.g., particles, materials, etc.) are configured to (i) bind to receptors or other binding sites of host cell structures, thereby blocking the attachment of pathogens to such receptors or binding sites and preventing (or lowering the likelihood of) the infectious agent from entry into the cell structure and/or (ii) bind to the infectious agent itself thus immobilizing the threat caused by the infection agent. Accordingly, the carrier embodiments disclosed herein and equivalents thereof provide two possible approaches to reducing infection of a host's cells, and as such, have immense potential in different applications in medicine, drug development, medical devices, consumer, sanitation products and/or the like.

According to some embodiments, it is possible to synthesize nano-sized and/or micro-sized materials (e.g., carriers, particles, objects) that mimic, at least partially, targeted viruses or other pathogens (e.g., SARS-CoV-2 virus, the spread of which resulted in a global epidemic starting in 2020, pandemic strain/type of the influenzas, rhinoviruses, noroviruses, respiratory syncytial viruses, coronaviruses (including mutated forms thereof) derived from SARS-CoV-2, other viruses, other pathogens, etc.) by using the known size, morphology, surface properties and/or other characteristics or properties of the targeted virus or other pathogen. Thus,

carriers (e.g., man-made particles or objects) can beneficially hinder or at least slow the spread of the disease by competitive inhibition.

In some embodiments, such carriers (e.g., man-made materials) are designed and otherwise configured to bind to surface receptors, co-receptors and/or other binding sites of a host cell structure, and thereby blocking, at least partially, the entry of the virus or other pathogen. In some arrangements, in addition to binding to host cell structures (and thus at least partially blocking or preventing the attachment of a pathogen to said host cell structures) and by immune stimulating the host of said pathogen, carriers are configured to simultaneously administer or otherwise deliver APIs and/or other materials. Such APIs and/or other materials delivered to target cells and tissues can be configured to create an environment that is hostile to viral replication and that provides a synergistic approach to the host. Such carriers and the associated methods of treatment can have immense potential in different applications in medicine, drug development, medical devices, consumer products and the like. In some embodiments, for instance, adding Zinc ions, Celastrol Cannabinols, anti-viral molecules, other APIs and/or other substances or materials to the nanoparticle or other carrier, it would be possible to create an environment for at least partially arresting the viral replication cycle inside host cells (e.g., cells that are expressing ACE2 receptors or other targeted receptors on their cell surface).

In some embodiments, such carriers (e.g., man-made materials) are designed and otherwise configured to bind to surface receptors, co-receptors and/or other binding sites of a host cell structure, and thereby blocking, at least partially, the entry of the virus, other pathogen or other disease-causing agent. In some arrangements, in addition to binding to host cell structures (and thus at least partially blocking or preventing the attachment of a pathogen to said host cell structures) and/or immobilizing the pathogen or other agent by binding to the pathogen or other agent itself, carriers are configured to simultaneously administer or otherwise deliver APIs and/or other materials. Such APIs and/or other materials delivered to target cells and tissues can be configured to create an environment that is hostile to replication (e.g., viral replication) and that provides a synergistic approach to the host. Such carriers and the associated methods of treatment can have immense potential in different applications in medicine, drug development, medical devices, consumer products and the like. In some embodiments, for instance, adding Zinc ions, Celastrol Cannabinols, anti-viral molecules, other APIs and/or other substances or materials to the nanoparticle or other carrier, it would be possible to create an environment for at least partially arresting the viral replication cycle inside host cells (e.g., cells that are expressing ACE2 receptors or other targeted receptors on their cell surface).

In some embodiments, a carrier (e.g., a synthetic nanoparticle and/or microparticle) can be used to reduce the spread of influenzas, rhinoviruses, noroviruses, respiratory syncytial viruses (RSVs), coronaviruses (e.g., SARS-CoV-2, other derived coronaviruses), other viruses that cause symptoms such as respiratory infection, diarrhea, common cold, cytokine storm, general discomfort, serious illness, death and/or any other viruses or other pathogens. To that end, in some embodiments, the synthetic particles or other carriers can be manufactured to match, imitate, emulate or substantially match, imitate or emulate one or more characteristics or other properties of one or more targeted pathogens (such as, for example, influenzas, rhinoviruses, noroviruses, respiratory syncytial viruses (RSVs), coronaviruses (e.g., SARS-

CoV-2 virus) and/or any other viruses or other pathogens). More specifically, according to some arrangements, the particle or carrier is preferably fabricated to a size of around 10 to 200 nm, for example 50 to 150 nm (such as around 100 to 120 nm), 10 to 200 nm, 10 to 100 nm, 100 to 200 nm, 50 to 200 nm, 10 to 150 nm, values between the foregoing values and ranges).

Further, in some embodiments, the carriers can be coated with similar amino acids and peptides as the targeted virus and/or other pathogen. For example, in some embodiments, the carrier can contain glycoprotein spike proteins, other protrusions, similar molecules and/or other surface features. In some embodiments, such features of the carrier can be configured to mimic or substantially mimic certain surface features of the viral envelope. More specifically, in some embodiments, the particle is preferably fabricated to a size of around 100-120 nm and coated with similar amino acids and peptides as the virus contains, for example, glycoprotein spikes, protrusions and/or other features at the viral surface or similar molecules that mimic the surface of the viral envelope. In some embodiments, the carrier (e.g., particle or object) is fabricated to a size of around 100 nm (e.g., 80 to 120 nm, 90 to 110 nm, 100 nm, values between the foregoing, etc.) and is coated with similar amino acids and peptides as the targeted virus or other pathogen (e.g., glycoprotein spikes and/or protrusions at the viral surface or similar molecules that mimic the surface of the viral envelope). In more specific embodiments, the carrier (e.g., object or particle) is fabricated to a size of around 0.2 to 100 nm and coated with similar amino acids and peptides as the virus contains e.g., protrusions at the viral surface.

Further, in some embodiments, carriers are coated and/or otherwise functionalized with amino acids and/or peptides similar to those of a targeted virus and/or other pathogen for targeted drug delivery. The carrier can be loaded with one or more materials, such as, for example, an API, a drug, a protein, an amino acid, RNA, DNA, a molecule and/or the like (e.g., any other materials disclosed in this specification). In some embodiments, the carrier (loaded with one or more materials) is configured to provide a targeted treatment for a specific disease or other health issue caused by the disease causing agent or other agent. In some arrangements, the carrier is configured to release one or more materials loaded on and/or within the carrier to target cells with reduced off-target effects or impacts.

In some embodiments, carriers (e.g., man-made materials) that mimic, at least partially, certain features and/or other aspects of a targeted virus and/or other pathogen for targeted drug delivery can increase the drug accumulation at target tissue with minimal or reduced off-target effects or other impacts. This can be due to, at least in part, the carrier having an enhanced or "high" uptake in specific cells that have the specific receptors to which the carrier binds. In some embodiments, the binding of the carrier is partly due to the functionalization properties enabling binding to the target receptor that initiates the endocytosis of the carrier that can be loaded with specific APIs. In some embodiments, such a specific endocytosis at targeted tissues increases the therapeutic window and improves the therapeutic index of the specific API. Many APIs and/or other components have relatively poor solubility and/or permeability that makes such compounds difficult to administer to the host. Using a carrier that can be loaded with APIs and/or other components that mimic or substantially mimic, at least partially, a pathogen to be used as an "trojan horse" could increase the therapeutic window of such compounds. The targeting embodiments disclosed herein or variation thereof can be

classified in to either active or passive targeting. Passive targeting commonly refers to non-functionalized carriers (e.g., particles, objects, etc.) that usually accumulate in the liver and spleen for clearance. Whereas active targeting commonly refers to functionalized carriers (e.g., particles, objects, etc.) that enable active uptake of carriers at target site. Functionalizing carriers surfaces with an active targeting ligand can, in some embodiments, improve or otherwise enhance drug accumulation at the target site and lower drug accumulation at unwanted off-target sites. In some embodiments, this can minimize or reduce possible side effects.

In some embodiments, carriers (e.g., man-made materials) that mimic, at least partially, certain features and/or other aspects of a targeted virus and/or other pathogen for targeted drug delivery can be loaded into or onto with additional components, API, drug, chemicals, protein, peptides or any combination thereof. The loading degree depends on the material used in the carrier for example silica based carriers with mesopores can be loaded inside the pores with hydrophobic drugs. Whereas polymer based carriers are capable of being loaded with hydrophilic drugs such as Poly(Lactic Acid) based micelles. Whereas lipid based particles such as solid lipid particles can be loaded with either hydrophobic components in the lipid components or hydrophilic components inside the particle hollow core.

In some embodiments, the amount of one or more additional components (e.g., APIs, drugs, chemicals, proteins, peptides, other materials and/or any combination thereof, etc.) loaded into or onto the carrier is 0.1 to 40 percent by weight (% weight or % wt) of the carrier (e.g., 0.1 to 40, 0.1 to 5, 5 to 10, 0.1 to 10, 10 to 20, 20 to 30, 30 to 40, 10 to 40, 10 to 30, 15 to 40, 20 to 40, 30 to 40, 5 to 30%, values between the foregoing ranges and values, etc.). In some embodiments, the core comprises a lipid core, which in some embodiments, allows for a greater "loading" of a carrier with an additional component. In some embodiments, the carrier is configured to be loaded with more than 5% by weight of one or more component relative to the carrier (e.g., at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 40%, at least 50%, more than at least 50%, values between the foregoing, etc.).

Next, certain embodiments will be described in more detail.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features, aspects and advantages of the present application are described with reference to drawings of certain embodiments, which are intended to illustrate, but not to limit, the present inventions. It is to be understood that these drawings are for the purpose of illustrating the various concepts disclosed herein and may not be to scale.

FIG. 1 is a schematic depiction of different applications of synthesized nanomaterials according to some embodiments of the present technology;

FIG. 2 is a schematic depiction of infection prevention and control by competitive inhibition using synthetic nanoparticles according to some embodiments of the present technology;

FIG. 3A is a schematic depiction of immobilization of an infectious agent by functionalized nanomaterials according to some embodiments of the present technology;

FIG. 3B is a schematic depiction of the carrier having protrusions mimicking several amino acids sequences of different viruses or viral strains capable of both binding to

targeted receptor and/or receptors, blocking viral entry, and/or functioning as an vaccination at targeted cell population;

FIG. 4A is a schematic depiction of dual targeting strategy compromising both immobilization of an infectious agent and by infection prevention and control by competitive inhibition functionalized nanomaterials according to some embodiments of the present technology;

FIG. 4B is a schematic depiction of multi-viral targeting strategy compromising multiple epitopes of different pathogens capable of electing a broad immunologic reaction and by infection prevention and control by competitive inhibition functionalized nanomaterials according to some embodiments of the present technology;

FIG. 5 schematic depiction of utilization of nanoparticles coated with peptides resembling the binding motif of the spike protein from the SARS-CoV-2 virus for binding and blocking co-receptor and as a consequence decrease the virus mobility, thus minimizing the risk of the virus's infecting the host cells or infecting the host to other organs;

FIG. 6 is a schematic depiction of functionalized nanoparticles mimicking SARS-CoV-2 to be used as targeted intervention and therapy against COVID-19 and other respiratory diseases according to some embodiments of the present technology; and

FIG. 7 shows an example of a SARS-CoV-2 spike RBD expression construct. Different expression cassettes (shown as the XXX region) can be used for expressing the desired construct e.g., influenza H7 haemagglutinin (indicated as "Signal sequence"), Tag, spacer, and SARS-CoV-2 RBD.

DETAILED DESCRIPTION

In the present context, the term "around" means, when used in connection with numerical values, that a variation of $\pm 25\%$, in particular $\pm 20\%$, for example $\pm 10\%$, or $\pm 5\%$, of the exact value is included by a literal reading of that value.

In the present context, the term "about" means, when used in connection with numerical values, that a variation of $\pm 25\%$, in particular $\pm 20\%$, for example $\pm 10\%$, or $\pm 5\%$, of the exact value is included by a literal reading of that value.

The term "polymer" is used herein in a broad sense and refers to materials, compounds, amino acids and proteins characterized by repeating moieties or units.

The term "functionalization" is used herein in a broad sense and refers to conjugating, coating, covalently and/or otherwise adding (e.g., allosterically adding) materials, compounds, drugs, amino acids and/or proteins to the synthesized particle or object.

The term "biocompatible" refers herein to "the ability of a material to perform with an appropriate host response in a specific application" (e.g., William's definition) [19].

The term "therapeutic window" as used herein, refers to the dosage range between a minimum effective therapeutic concentration and the minimum toxic concentration.

The term "therapeutic index" (also referred to as therapeutic ratio) as used herein, is a quantitative measurement of the relative safety of a drug. The ratio of the dose that exerts toxicity in 50% of the population (TD50) divided by the dose that exerts a therapeutic response (ED50) in 50% of the population (TI=TD50/ED50). For clarification, a higher therapeutic index is preferable to a lower one for API, drugs and components to be given to the host, as it means, for example, a greater likelihood of successful outcomes (e.g., with respect to combating an infection, a disease, etc.).

Nanomaterials and nanomedicine can be classified according to the targeting strategies used, which can include,

for instance, active or passive targeting. In some embodiments, passive targeting utilizes non-functionalized particles for accumulation in organs and tissues that are responsible for clearance of foreign objects such as macrophages, e.g., in the liver, spleen, etc. In some arrangements, tumor microenvironments typically show an enhanced permeability and retention effect (EPR), which can be a consequence of leaky and fenestrated blood vessels around tumors. Active targeting, on the other hand, uses a targeting ligand or functionalization that enhances the accumulation of the carrier at target site [15].

There are virtually endless functionalization possibilities by covalently attaching, adhering, saturating or binding (e.g., allosterically binding) molecules, polymers, proteins, amino acids, compounds and/or drugs onto the nanomaterial for achieving active targeting. One of the major advantages of functionalizing a smaller molecule to a larger entity, e.g., proteins, epitopes, amino acid sequences or hydrophobic molecules to a nanomaterial, is to increase the combined objects stability and/or solubility and/or possible minimize the unwanted immunologic reaction [15].

Described herein are carriers (e.g., fabricated nanomaterials or other carriers) used for inhibiting or improving the ability to inhibit, at least partially, pathogen entry of certain pathogens or other unwanted organisms, in particular coronaviruses (e.g., SARS-CoV-2), influenzas, rhinoviruses, noroviruses, respiratory syncytial viruses and other viruses causing respiratory infection to the host organism. Accordingly, such nanomaterials or other carriers can be advantageously used to limit or reduce the replication and spread of a disease or virus.

Embodiments disclosed herein have capabilities of carrying or otherwise delivering or providing anti-pathogenic pharmaceuticals or other materials, such as anti-viral drugs, in the carrier (e.g., nanomaterial) to reduce the replication and growth of the infectious agent.

As discussed, any of the carrier embodiments disclosed herein can be configured to deliver or otherwise provide one or more materials to cells of a subject, such as, for example, drugs, pharmaceuticals, APIs, viscosity modulators, antihistamines, immunosuppressors other molecules, compounds, substances, materials, etc. to give targeted treatment for one or more specific diseases, disorders and/or other health issues (e.g., caused by a pathogen, other disease causing agent, etc.).

Embodiments disclosed herein pertain to the fabrication of man-made (e.g., fabricated) carriers or materials (e.g., in the nano- and/or microscale) that are configured to at least partially saturate and bind to receptors, proteins and/or macromolecules at the cellular level in order to reduce the likelihood (e.g., prevent) and reduce (e.g., minimize) pathogens or other disease-causing agents (e.g., coronavirus) binding to and/or entering into receptors and/or target tissues of the host. In some embodiments, the synthesized carrier (e.g., nanomaterial) can be stored and loaded onto a medical device capable of releasing (e.g., on-demand, specific amounts) the synthesized carrier system to specific tissue. Such medical devices or other devices or systems include, without limitation, inhalation devices, oral tablets, injectable substances, lotions or creams and/or any other device, system or component, as desired or required.

In one aspect or embodiment, a synthetic carrier, particle or object is configured to at least partially hinder or impede the spread of the COVID-19 disease by competitive inhibition and to deliver an API, drug or molecule to targeted cells and/or tissues with increased or improved efficacy. In some

embodiments, the use of such carriers, particles or objects is configured to have few or minimal side effects for creating a hostile environment for the virus or other targeted pathogen. According to some arrangements, the carrier or particle (e.g., the mimetic nanoparticle) is functionalized with, in one non-limiting example, hexapeptide resembling that of the RBD from SARS-CoV-2. This can, according to some embodiments, allow high binding affinity to the ACE2 receptor at the lining of the respiratory system, thus blocking, at least partially, a route (e.g., a primary route) of infection. In some embodiments, this approach virtually eliminates the problem associated with mutations of the viral strain, because the specific target is the human receptor and not the constantly evolving coronavirus. This may be especially significant, for instance, in light of the impactful SARS-CoV-2 mutations that have appeared starting in 2021 and beyond, which have and will have a significant impact on the state of world health. In some embodiments, alternatively or simultaneously, ACE2 receptor binding moieties or antibodies designed to bind and immobilize the virus at specific sites can be used.

Therefore, in some embodiments, as noted above and discussed in greater detail herein, carriers (e.g., particles, obstacles, etc.) are configured to prevent or reduce the likelihood of infection by pathogens or other disease-causing agents using one or more principles or mechanisms. For example, in some arrangements, the carriers are sized, shaped and otherwise configured to prevent or reduce the likelihood of negative health impacts (e.g., pathogen infection) by competitive inhibition (e.g., blocking receptors).

In another aspect or embodiment, a carrier (e.g., a synthetic particle or object) is configured to deliver its “cargo” or content to targeted cell population (e.g., for targeted treatment of a disease, disorder, ailment or other health issue, to target specific pathogens and/or disease-causing agents or sources, etc.). For example, in some arrangements, the carrier comprises a core material that can be “loaded” or otherwise provided with (e.g., into or onto) drugs, API and/or other molecules or materials. Such substances provided in and/or on the carrier can be targeted with higher efficacy to specific cells and tissues using, for example, functionalization (e.g., protrusions that are capable of binding to host cell structures such as receptors that facilitates carrier uptake at said cells enabling targeted therapeutics). Thus, potentially, the therapeutic effect of the drug can be improved, increased or otherwise enhanced, e.g., by accumulating the local dosage in specific cells, reducing at least some side-effects of the drug (e.g., by reducing off-target effect in unwanted cells and/or the like).

Therefore, in some embodiments, as noted above and discussed in greater detail herein, carriers can be “loaded” or otherwise provided with (e.g., into or onto) one or more of the following: a drug, an API, a mineral, another molecule or material used to at least partially reduce or alleviate symptoms, the progression, discomfort and/or other issues caused by an disease causing agent or source. In some embodiments, disease causing agents or sources include, but are not limited to, viruses (e.g., SARS-CoV-2 viruses, other coronaviruses, influenza viruses, rhinoviruses, noroviruses, respiratory syncytial viruses (RSV), other viruses that impact the respiratory system, any other type of viruses, etc.), bacteria, parasites (e.g., worms), protozoa, microorganisms, prions, fungi (e.g., molds), allergens, antigens,

toxins, poisons (e.g., venoms, pollutants, etc.), carcinogens, compounds or molecules (e.g., organic, inorganic, etc.), chemical agents (e.g., naturally-occurring, synthetic or man-made, etc.), radiation (e.g., electromagnetic radiation, radioactive compounds/materials, other radiation, etc.), enzymes or proteins, regulatory elements and/or substances (e.g., hormones, enzymes, other proteins, DNA/RNA repair elements or agents, replication machinery, chaperone machinery, other host-producing elements or substances, etc.), inflammatory agents, elements or substances and/or the like. In some embodiments, the disease caused by the disease-causing agents or sources include, by way of example, inflammation local or systemic or both, neurodegenerative diseases, autoimmune diseases (e.g., psoriatic arthritis, rheumatoid arthritis (RA), Sjögren’s syndrome, Systemic lupus erythematosus (e.g., Lupus, SLE), cancer (e.g., bladder cancer, breast cancer, colon cancer, rectal cancer, endometrial cancer, kidney cancer, leukemia, liver cancer, lung cancer, melanoma, Non-Hodgkin lymphoma, pancreatic cancer, prostate cancer, thyroid cancer, any other type of cancer, etc.), muscle dystrophies, cystic fibrosis, diabetes, rheumatoid arthritis, diseases caused by imbalance of hormones (e.g., hormone imbalance, infertility, acne, metabolism, thyroid disease, etc.) and/or any other type of disease, ailment, syndrome and/or condition.

In some embodiments, the carrier is loaded with or otherwise comprises one or more materials (as noted in the preceding paragraph and/or other portions of this specification) that the carrier can advantageously deliver to target tissues, cells and/or areas or portions of the subject’s anatomy with increased efficacy and with reduced side effects. This can help create an environment in which disease progression or other negative health issue is at least partially halted or otherwise attenuated. As noted herein, in some arrangements, the size of the carrier (e.g., the particle or object) is similar or substantially similar to the size of the virus or other pathogen or other disease-causing agent being targeted. For example, a diameter or other cross-sectional dimension of the carrier can be 50% to 200% (e.g., 50-200, 50-100, 50-150, 50-200, 100-150, 150-200%, values and ranges between the foregoing, etc.) of the diameter or other cross-sectional dimension of a targeted pathogen and/or agent (e.g., influenza, other virus or pathogen, other disease-causing agent or source, etc.).

The following tables provide certain non-limiting examples of materials that can be delivered to portions of a subject’s anatomy using a carrier for targeted treatment (e.g., for combatting the effects or other impacts of certain pathogens or disease-causing agents or sources, to help with the symptoms and/or other impacts of a disease, disorder or other health-related issue, etc.). The information included in the tables below (and/or information included in other portions of the application) is not intended to be inclusive of all possible uses of the embodiments disclosed herein. Accordingly, there may be additional and/or different particles or carriers, anatomical locations, diseases and/or conditions, purposes, advantages and/or benefits, modes of delivery and/or the like that exist which are not explicitly listed herein. Further, it should be understood that a particular carrier or set of carriers can be used to target more than one disease and/or condition, to provide more than one purpose, benefit and/or advantage and/or the like. Relatedly, a carrier can be delivered (e.g., simultaneously, non-simultaneously, etc.) to more than one anatomical location.

		Disease/condition that is at least partially prevented, treated, combatted and/or otherwise impacted to benefit a recipient (not intended to be exhaustive of all diseases/conditions)	Other purposes/benefits/advantages (not intended to be exhaustive of all purposes/benefits/advantages)
Target delivery location of carrier (at least in part)			
Lungs/respiratory tract (e.g., upper respiratory tract, lower respiratory tract, etc.)	Infection by pathogens	Therapeutic, preventative/prophylactic, vaccine	
Intravascular system	Infection by pathogens	Therapeutic, preventative/prophylactic, vaccine	
Tissues (e.g., muscle, fat, connective tissues, bones, skin, etc.)	Infection by pathogens	Therapeutic, preventative/prophylactic, vaccine	
Organs (e.g., solid organs (e.g., liver, pancreas, spleen, kidneys, adrenal glands, etc.), hollow organs (e.g., stomach, intestines, gallbladder, bladder, rectum, etc.), etc.)	Infection by pathogens	Therapeutic, preventative/prophylactic, vaccine	
Lungs/respiratory tract (e.g., upper respiratory tract, lower respiratory tract, etc.)	Cancer (e.g., caused by mutating agents and/or mutations in the germline)	Therapeutic, preventative/prophylactic, vaccine	
Intravascular system	Cancer (e.g., caused by mutating agents and/or mutations in the germline)	Therapeutic, preventative/prophylactic, vaccine	
Tissues and connective organs	Cancer (e.g., caused by mutating agents and/or mutations in the germline)	Therapeutic, preventative/prophylactic, vaccine	
Organs (e.g., solid organs (e.g., liver, pancreas, spleen, kidneys, adrenal glands, etc.), hollow organs (e.g., stomach, intestines, gallbladder, bladder, rectum, etc.), etc.)	Cancer (e.g., caused by mutating agents and/or mutations in the germline)	Therapeutic, preventative/prophylactic, vaccine	
Lungs/respiratory tract (e.g., upper respiratory tract, lower respiratory tract, etc.)	Cellular imbalances and/or deficiencies	Therapeutic, preventative/prophylactic, vaccine	
Intravascular system	Cellular imbalances and/or deficiencies	Therapeutic, preventative/prophylactic, vaccine	
Tissues: muscle, fat, connective tissues, bones, skin, etc.	Cellular imbalances and/or deficiencies	Therapeutic, preventative/prophylactic, vaccine	
Organs (e.g., solid organs (e.g., liver, pancreas, spleen, kidneys, adrenal glands, etc.), hollow organs (e.g., stomach, intestines, gallbladder, bladder, rectum, etc.), etc.)	Cellular imbalances and/or deficiencies	Therapeutic, preventative/prophylactic, vaccine	
Lungs/respiratory tract (e.g., upper respiratory tract, lower respiratory tract, etc.)	Fibrosis, other scars and aggregates, etc.	Therapeutic, preventative/prophylactic, vaccine	
Intravascular system	Fibrosis, other scars and aggregates, etc.	Therapeutic, preventative/prophylactic, vaccine	
Tissues (e.g., muscle, fat, connective tissues, bones, skin, etc.)	Fibrosis, and other scars and aggregates, etc.	Therapeutic, preventative/prophylactic, vaccine	
Organs (e.g., solid organs (e.g., liver, pancreas, spleen, kidneys, adrenal glands, etc.), hollow organs (e.g., stomach, intestines, gallbladder, bladder, rectum, etc.), etc.)	Fibrosis, other scars and aggregates, etc.	Therapeutic, preventative/prophylactic, vaccine	

Lungs/respiratory tract (e.g., upper respiratory tract, lower respiratory tract, etc.)	Metabolic disorders or diseases	Therapeutic, preventative/prophylactic, vaccine
Intravascular system	Metabolic disorders or diseases	Therapeutic, preventative/prophylactic, vaccine
Tissues (muscle, fat, connective tissues, bones, skin, etc.)	Metabolic disorders or diseases	Therapeutic, preventative/prophylactic, vaccine
Organs (e.g., solid organs (e.g., liver, pancreas, spleen, kidneys, adrenal glands, etc.), hollow organs (e.g., stomach, intestines, gallbladder, bladder, rectum, etc.), etc.)	Metabolic disorders or diseases	Therapeutic, preventative/prophylactic, vaccine
Material (either alone or with one or more other material(s))		
Anti-pathogenic	Infection (e.g., viral infection, bacterial infection, infection caused by other pathogens, microorganisms or sources, etc.)	Therapeutic, preventative/prophylactic, vaccine
Anti-Viral	Infection (e.g., viral infection)	Therapeutic, preventative/prophylactic, vaccine
Anti-bacterial	Infection (e.g., bacterial infection)	Therapeutic, preventative/prophylactic, vaccine
Anti-parasitic	Infection (e.g., parasitic infection)	Therapeutic, preventative/prophylactic, vaccine
Anti-fungal	Infection (e.g., fungal infection)	Therapeutic, preventative/prophylactic, vaccine
Minerals, vitamins and/or supplements	Infection (e.g., viral infection, bacterial infection, infection caused by other pathogens, microorganisms or sources, etc.)	Therapeutic, preventative/prophylactic, vaccine
Minerals, vitamins and/or supplements	Cellular imbalances	Therapeutic, preventative/prophylactic, vaccine
Minerals, vitamins and/or supplements, etc.	Deficiencies	Therapeutic, preventative/prophylactic, vaccine
Dilating drugs, nitro glycerin, vasodilating beta blockers, etc.	Hypoxemia-caused conditions and/or diseases	Therapeutic, preventative/prophylactic, vaccine
Immune modulating drugs, viscosity modulators, bronchodilators, etc.	Fibrosis, other scars, obstructions and aggregates, etc.	Therapeutic, preventative/prophylactic, vaccine
Cancer drugs and medications, chemotherapy, hormone therapy, immunotherapy, pain killers, immune modulating drugs, etc.	Cancer	Therapeutic, preventative/prophylactic, vaccine
Neurodegenerative drugs, protein aggregation modulating drugs, chaperons, etc.	Neurodegenerative disorders and diseases	Therapeutic, preventative/prophylactic, vaccine
Metabolic steroids, metabolic drugs, insulin, etc.	Help prevent, cure and/or otherwise treat or target metabolic disorders	Therapeutic, preventative/prophylactic, vaccine

In another aspect or embodiment, a carrier (e.g., a synthetic particle or object) is configured to hinder, at least partially, the spread of influenza or the "flu" by competitive inhibition of virus entry. As noted above, in some arrangements, the size of the carrier (e.g., the particle or object) is similar or substantially similar to the size of the virus or other pathogen being targeted. For example, a diameter or other cross-sectional dimension of the carrier can be 50% to 200% (e.g., 50-200, 50-100, 50-150, 50-200, 100-150, 150-200%, values and ranges between the foregoing, etc.) of the diameter or other cross-sectional dimension of influenza or other targeted virus or pathogen.

In some embodiments, the carrier is loaded with or otherwise comprises an API, drug, molecule and/or other materials to be delivered to target cells and tissues with increased efficacy and with minimal or reduced side effects, while creating a hostile environment for the virus. However, in other arrangements, the carrier (e.g., particle or object) does not contain any API, drug or other molecule that is intended to be delivered to targeted cells and tissues. Even in such embodiments, the carriers or particles can be configured to reduce the likelihood of infection (e.g., by preventing the actual virus from binding to and infecting targeted cells of the host). This can be accomplished by, at least in part, blocking receptors (or other binding sites or

portions) of host cells. The mimetic carrier or particle can be functionalized with, for example and without limitation, protein fragments resembling HA and NA binding moiety. In such embodiments, high binding affinity to the host sialic acid receptors at the lining of the respiratory system if facilitated, thereby at least partially blocking a route (e.g., the primary route) of infection. Additionally, in some embodiments, the protrusions are at least partially capable of electing or configured to elect an immune response against said pathogen.

In another aspect, a carrier (e.g., a synthetic particle or object) mimics (or is configured or adapted to mimic), for example, Rhinoviruses in order to hinder, at least partially, the spread of the major causative agent of the common cold by competitive inhibition. In some embodiments, the appropriate API, drug or molecule is delivered to target cells and tissues with increased efficacy and with minimal or reduced side effects while creating a hostile environment for the virus. The mimetic particle (e.g., carrier) can be functionalized with, for example, but not limited to, VP1 and VP2 capsid protein allowing high binding affinity to the ICAM-1 and other related receptors at the lining of the respiratory system. Thus, the primary route of infection can be, at least partially, blocked. Additionally, in some embodiments, the protrusions are at least partially capable of electing or configured to elect an immune response against said pathogen.

In another aspect, a carrier (e.g., a synthetic particle or object) mimics (or is configured to mimic), at least approximately or substantially, respiratory syncytial virus (RSV) in order to hinder, at least partially, the spread of influenza or the “flu,” a respiratory disease, by competitive inhibition. In some embodiments, with the use of such carriers, the appropriate (e.g., desired, required, etc.) API, drug or molecule is delivered to target cells and tissues with increased efficacy and with minimal or reduced side effects while creating a hostile environment for the virus. The envisioned carrier (e.g., mimetic particle or object) functionalized with, for example and without limitation, receptor attachment protruding glycoprotein (G) allowing high binding affinity to the IGF1R receptor at the lining of the respiratory system. Accordingly, such a carrier can be configured to block, at least partially, the primary route of infection. Additionally, in some embodiments, the protrusions are at least partially capable of electing or configured to elect an immune response against said pathogen.

In another aspect, a carrier (e.g., a synthetic particle or object) mimics or is configured or adapted to mimic) Noroviruses. In such embodiments, the carrier can hinder, at least partially, the spread of “stomach flu” a gastroenteritis disease by competitive inhibition. In some embodiments, with the use of such carriers, the appropriate (e.g., desired, required, etc.) API, drug or molecule is delivered to the target cells and tissues with increased efficacy and with minimal or reduced side effects while creating a hostile environment for the virus. The carrier (e.g., the envisioned mimetic particle) functionalized with, for example and without limitation, VP1 containing the P2 subdomain allowing high binding affinity to the including sialic acid and histo-blood group antigens at the lining of the respiratory system. Thus, the carriers can block, at least partially, the primary route of infection. Additionally, in some embodiments, the protrusions are at least partially capable of electing or configured to elect an immune response against said pathogen.

FIG. 1 schematically shows some non-limiting applications of synthesized carriers (e.g., nanomaterials, objects,

particles, etc.) according to some embodiments of the present technology. Thus, by way of an example, the carrier (e.g., nanomaterial) can be loaded with one or more active pharmaceutical ingredients (API), such as, e.g., Celestrol, Zinc, anti-viral compounds, Interferon-Gamma modulators, etc., and then used in inhalation devices, oral tablets or injectables, or other devices or tools of administering the carriers, to name just a few. Such nanomaterials and/or micromaterials can be used to hinder, at least partially, the entry of novel coronaviruses within host cells, thereby reducing or minimizing the spreading of the disease.

Embodiments disclosed herein allow for decreasing the risk of a pathogen or pathogens, such as coronaviruses, influenzas, rhinoviruses, other viruses causing respiratory infection (e.g., SARS-CoV-2), entering its host for a temporary or prolonged duration. Accordingly, such embodiments can advantageously give a targeted treatment for the specific disease caused by the infectious agent.

In a first embodiment, a synthetic carrier is provided, which comprises biocompatible particles having a maximum size in at least one dimension in the nanometer or micrometer range. In some embodiments, said maximum size in at least one dimension is 10 to 500 nanometers (e.g., 10 to 500, 10 to 50, 10 to 100, 50 to 100, 1 to 100, 100 to 200, 200 to 300, 300 to 400, 400 to 500, 200 to 400 nanometers, values and ranges between the foregoing, etc.). In some embodiments, such carriers form or include a core and include a functionalized surface capable of binding to target areas of cell surfaces of a host. Advantageously, such binding can at least temporarily block the target areas to prevent or minimize pathogens (e.g., influenzas, rhinoviruses, coronaviruses including but not limited to SARS-CoV-2, other viruses causing respiratory infection, thereby reducing the risk of the host contracting the disease caused by the pathogen (e.g., the COVID-19 disease, diarrhea, respiratory infections, common cold, etc.).

In one embodiment, a carrier (e.g., a synthetic carrier) comprises biocompatible particles having a maximum size in at least one dimension in the nanometer or micrometer range. In some embodiments, said maximum size in at least one dimension is 10 to 500 nanometers (e.g., 10 to 500, 10 to 50, 10 to 100, 50 to 100, 1 to 100, 100 to 200, 200 to 300, 300 to 400, 400 to 500, 200 to 400 nanometers, values and ranges between the foregoing, etc.). In some embodiments, such carriers comprise (e.g., include or form) a core and a functionalized surface configured to at least partially bind to target areas of cell surfaces of a host. Advantageously, such carriers can be utilized for targeted drug delivery.

As used herein, the term “host” means, but is not necessarily limited to, an individual mammal, in particular a human or an animal.

As schematically illustrated in FIG. 2, in some embodiments, the carrier 10 comprises a core 20 and a plurality of surface features 30 related to the core. As disclosed herein, the surface features can include protrusions that resemble or mimic, at least partially, spike proteins or other protrusions or features of a target virus or other pathogen. With continued reference to FIG. 2, the carrier 10 can be “loaded” or otherwise provided with one or more materials or other substances (e.g., APIs, other pharmaceuticals or agents, etc.) 40. As disclosed herein, such materials 40 can be delivered by the carrier to or near the site of a targeted virus or other pathogen for improved treatment (e.g., therapeutic treatment, infection prevention or mitigation, etc.).

In some embodiments, the synthetic carrier comprises a “nano” material which can be of nano- or micrometer or larger size. In some arrangements, the synthetic carrier has

a size in at least one dimension which is in the nanometer scale. In some arrangements, the synthetic carrier has a size in at least one dimension which is in the micrometer scale. For example, such a size in at least one dimension is schematically depicted in FIG. 2 as dimension D. In other embodiments, the carrier has a size in at least one dimension that is outside the nanometer or micrometer range, as desired or required. For instance, the carrier can have a size in at least one dimension which is smaller than one nanometer (e.g., in the picometer range or smaller) or greater than one millimeter, depending on the targeted pathogen or other factors. The nanomaterial or other synthetic carrier can be formed as a particle, spheroid, cubical, cigar-shaped, elongated, triangle, sharp and pointy, a sheet and film and/or any configuration or shape.

According to some embodiments, a maximum cross-sectional dimension of the core 20 of the carrier 10 is 10% to 1000% (e.g., 10 to 1000, 500 to 1000, 10 to 500, 50 to 150, 10 to 300, 100 to 500, 10 to 100, 75 to 125%, values between the foregoing, etc.) of a maximum cross-sectional dimension of the pathogen (e.g., virus, bacterium, other pathogen, etc.).

In some embodiments, the synthetic carrier has a maximum size in at least one dimension which is smaller than 2500 μm (e.g., less than 2500 μm , less than 2000 μm , less than 1500 μm , less than 1000 μm , less than 500 μm , less than 100 μm , less than 50 μm , less than values between the foregoing, etc.). In one embodiment, the material has a maximum size in at least one dimension which is smaller than 10 μm (e.g., less than 10 μm , less than 5 μm , less than 1 μm , less than values between the foregoing, etc.). In one embodiment, the material in particular nanomaterial has a maximum size in at least one dimension which is smaller than 1000 nm, in particular smaller than around 500 nm or around 100 nm or smaller than around 10 nm or smaller than around 0.2 nm.

In one embodiment, the synthetic carrier is biocompatible. For example, according to some arrangements, such a material is configured to cause no reaction or only a minor unwanted reaction in the end-user (e.g., toxicity, off-target effects, etc.).

Generally, in some embodiments, the carriers disclosed herein are synthetic, which is used interchangeably with "synthesized" to denote that they are man-made or non-natural.

Embodiments of the carriers comprise organic or inorganic materials, protein based, ferritin protein particles, lipid droplets, micelles, solid lipids, or any combination of these.

The synthetic material can be selected from inorganic and organic, monomeric and polymeric materials capable of forming biocompatible nano- or micro-sized particles as explained herein.

Examples of materials for the carriers comprise one or more of the following: synthetic polymers (e.g., thermoplastic or thermosetting materials, such as polyolefins, polyesters, including biodegradable polyesters (e.g., polylactides, polycaprolactones, etc.), polyamides, polyimides, polynitriles, etc.). Further non-limiting examples of possible materials include, for example and without limitation, silica, polysiloxanes, silicone materials which optionally may contain organic and metal residues, and/or the like. In some embodiments, silica particles are preferred, but not in all embodiments.

According to some embodiments, the carrier comprises amino acids, proteins, salts and minerals and/or similar molecules or materials, as desired or required.

In one embodiment, the material, which forms the core structure of the carrier, is manufactured or otherwise obtained using one or more of the following: 3D printing, microfluidics, sol-gel methods (e.g., bottom-up methods or top-down methods of fabrication), genetically engineered organism producing specific proteins or amino acids that can either self-assemble such as ferritin protein particles or conjugate to larger entities any other method or technique, and/or combinations thereof.

In one embodiment, the core material comprises one or more materials, such as, for example, mesoporous silica nanoparticles with ordered mesostructures of pores. Such pores can be loaded with different drugs. The most common methods for drug loading to particles is either by physical adsorption using a highly saturated drug solution (e.g., a hydrophobic solvent such as cyclohexane with a hydrophobic drug) or an aqueous solution for water-soluble drugs. In some embodiments, loading further includes covalently conjugating the molecule to the particle surface using, for example, thiol chemistry and/or attracting the cargo molecule by having a different charge than the particle (e.g., particles having a positive charge which will allow loading of negatively charged drug, RNA/DNA molecules).

The carriers (e.g., particles or objects) disclosed herein can be synthesized in various sizes and shapes. In one embodiment, the material forming the core of the carrier contains pores with diameters between 1 and 75 nm, such as, for example, 2 to 50 nm, 2.5 to 30 nm, 2 to 5.5 nm, other values or ranges within the foregoing. In some embodiments, determining the hydrodynamic size using dynamic light scattering (DLS) makes it possible to confirm redispersibility of particle. In some arrangements, the morphology and particle diameter can be measured by either scanning electron microscope (SEM) or transmission electron microscopy (TEM). In order to determine surface area, pore size and pore volume, N₂-sorption measurements can be used. The size and volume of the of the mesopores can be detected using small angle X-ray (SAXRD), according to some embodiments. The drug loading is, in some embodiments, measured by Thermogravimetric analysis (TGA) and alternatively or additionally measured by UV/vis spectroscopy measurements at a wavelength of 425 nm. Any alternative method or technology of forming the carriers and/or determining measurements can be used, either in addition to or in lieu of those disclosed herein, as desired or required.

In one embodiment, the core material comprises mesoporous silica nanoparticles (MSNs).

In one embodiment, the material comprises a nanoparticle core with coated targeting ligands with a possibility of (or configured to allow for) loading the particle with API, drugs, molecules, proteins and amino acids, RNA or DNA and compounds of interest.

In one embodiment, the material comprises a nanoparticle core and/or microparticle core with coated and/or functionalized targeting ligands with a possibility of (or configured to allow for) loading the particle into or onto with API, drugs, molecules, proteins and amino acids, RNA or DNA and compounds of interest.

Thus, in one embodiment, a nano and/or micro sized particle for example solid lipid particle (e.g., palmitate-based or stearylamine and the matrix lipid Compritol) having a net positive charge can be decorated/coated with negative molecules, such as RNA or DNA encoding for example interferon gamma for targeted delivery.

Thus, in one embodiment, the nanomaterial comprises a core particle or object functionalized with targeting moieties, drugs, amino acids, protein or any combination

thereof, such as hybrid materials containing but not limited to 1,2-Dioleoyl-3-trimethylammonium propane (DOTAP), Cholesterol (Chol), Dioleoylphosphatidylethanolamine (DOPE) and/or 1,2-Distearoyl-sn-glycero-3-phosphorylethanolamine (DSPE), polyethylene glycol (PEG) (e.g., DOTAP:Chol:DOPE:PEG or DOTAP:Chol:DSPE:PEG) loaded with, for example and without limitation, RNA and/or DNA. In some embodiments, the object is preferably loaded with an active substance, drug or API.

In some embodiments of the present application and the technologies disclosed herein, two ways of synthesizing nanomaterials or other carriers are in particular employed. These include the top-down and the bottom-up approach or hybrid approach where some of the particle components are done with one approach and another component with another approach. In other embodiments, however, carriers can be synthesized or otherwise manufactured using other methods or approaches, as desired or required.

In the top-down approach, for example, the building materials have larger dimensions than the final product, which means that the materials undergo physical stresses, such as, e.g., grinding, milling etc., in order to be reduced in size. This process can lead to surface imperfections that could give rise to some variations in the final product that affect particle distribution in the host and binding kinetics.

In some embodiments, the bottom-up method starts by using smaller building blocks in solution transforming gradually to the final product, which can provide a more cost-efficient way of producing nanomaterials and/or micro-materials. Common bottom-up methods include, for example, co-precipitation, template synthesis and sol-gel method where the building blocks are often copolymers, colloids and liquid crystals and self-assembling components such as ferritin protein particles.

The carrier or particle system comprising of a core and functionalization can be characterized, in some arrangements, using Scanning electron microscopy and/or electron microscopy to confirm the size, monodispersity, morphology and non-agglomerated state of the particles. In some embodiments, to find (e.g., accurately, approximately) the amount of drug loading in the particle if the drug is fluorescent, particles can be dispersed in ethanol for complete drug elution. The concentration of drug can be determined by UV/vis spectroscopy measurements at a wavelength of 425 nm, for example with Celestrol. In some embodiments, from such measurements, the drug loading amount can be calculated or approximated. The mesoscopic ordering of the particles can be detected by powder-XRD using a Kratky compact small-angle system or similar X-ray diffraction (XRD) methods. In some embodiments, the hydrodynamic size of the particles can be determined by dynamic light scattering, and the mesoporosity by nitrogen sorption measurements. Thermogravimetric analysis can be used in order to estimate the amount of Polyethyleneimines (PEI), sugar motifs, folic acid (FA) or methotrexate (MTX) or other organic content functionalized to the particle. In some embodiments, thermogravimetric analysis can be used to estimate the amount of organic contact or other molecule and/or drug content functionalized to the particle.

In one embodiment, inhibiting the spread of the virus SARS-CoV-2, influenza, rhinovirus, other viruses causing respiratory infection and/or any other virus includes using a carrier (e.g., a mesoporous silica nanoparticle, lipid nanoparticle, protein-based nanoparticle or any combination thereof with similar size as the virus). In some embodiments, such nanoparticles or other carriers are configured to be strategically provided or otherwise administered to a host in

one or more ways (e.g., via inhalation, oral ingestion, intravenous injection, topical application, etc.), as desired or required. In some arrangements, the carriers (e.g., nanoparticles) include a size of 1 to 200 nm (e.g., 1 to 200, 10 to 120, 50 to 100, 90 to 110, 100 nm, values between the foregoing ranges, etc.). In some embodiments, the carriers include a size of 0.01 to 1000 nm (e.g., 0.01 to 1000, 10 to 1000, 50 to 1000, 100 to 1000, 1 to 500, 500 to 1000, 200 to 800, 400 to 600 nm, values between the foregoing ranges, etc.). In some embodiments, the carriers include a size of 0.2 to 100 nm (e.g., 0.2 to 100, 1 to 10, 2 to 20, 5 to 50, 10 to 100 nm, values between the foregoing ranges, etc.). Further, the nanoparticles can be fabricated using the bottom-up sol-gel method or top-down method.

A carrier in accordance with any of the embodiments disclosed herein can be configured to mitigate or otherwise reduce the progression of one or more diseases, disorders and/or other health-related symptoms, complications or other issues caused by one or more disease-causing agents or sources. The carrier can include a nanoparticle, a mesoporous silica nanoparticle, lipid nanoparticle, protein-based nanoparticle or any combination thereof. The carrier can include a similar or substantially similar size as the target pathogen or other disease-causing agent or source. The carrier can further include protrusions and/or other surface features, which can mimic, at least in part, protrusions and/or surface features of the targeted pathogen or disease-causing agent or source. In some embodiments, as discussed throughout this application, the carrier is configured to be strategically provided or otherwise administered to a host in one or more ways (e.g., via inhalation, oral ingestion, intravenous injection, topical application, etc.), as desired or required. In some arrangements, a diameter or other cross-sectional dimension of the carriers (e.g., nanoparticles) is 1 to 200 nm (e.g., 1 to 200, 10 to 120, 50 to 100, 90 to 110, 100 nm, values between the foregoing ranges, etc.). In some embodiments, the diameter or other cross-sectional dimension of the carriers is 0.01 to 1000 nm (e.g., 0.01 to 1000, 10 to 1000, 50 to 1000, 100 to 1000, 1 to 500, 500 to 1000, 200 to 800, 400 to 600 nm, values between the foregoing ranges, etc.). In some embodiments, the diameter or other cross-sectional dimension of the carrier is 0.2 to 100 nm (e.g., 0.2 to 100, 1 to 10, 2 to 20, 5 to 50, 10 to 100 nm, values between the foregoing ranges, etc.). The diameters or cross-sectional dimensions provided above can be the maximum diameter or cross-sectional dimension of the carrier. Alternatively, the diameters or other cross-sectional dimensions of the carrier can be the average or mean diameter or cross-sectional dimension of the carrier. Further, the carrier can be fabricated using the bottom-up (e.g., sol-gel) method or top-down method.

In some embodiments, by using known viral genetic information, such as known viral (e.g., coronaviral, influenza viral, rhinoviral and/or other viral, etc.) genetic information, it is possible to produce similar peptides present in targeted viruses. For example, peptides or other structures can be similar or substantially similar to those found in viral glycoprotein spikes and/or protein protrusions, thus, in some arrangements, mimicking (e.g., at least substantially or approximately) at least some of the viral surface properties that assist with the binding of the carrier to certain receptors (e.g., ACE2 N-terminal helix or sialic acid, histo-blood group antigens, ICAM-1, IGF1R, other target receptors ACE2, etc.). In some arrangements, the carrier can include amino acid sequences found in the viral receptor binding domain (RBD) or the viral receptor binding motif (RBM) in the S protein, HA or NA or VP other decorated proteins that

could be used or functionalizing the particle with similar (e.g., substantially similar) or identical peptides. Alternatively or additionally, the carrier's ability to at least partially inhibit entry of viruses can be enhanced by including organic polymers as part of the protrusion (e.g., of cationic polyamidoamine dendrimer (PAMAM)) or by predicting an amino acid sequence or polymer for producing a surface coating which is similar in surface charge as the viral surface or by attaching targeting motifs which are known to bind to the target receptor allowing selective internalization in target cells [6-9,20].

In one embodiment, the carriers disclosed in the present application or variations thereof comprise mesoporous silica particles. In some embodiments, such carriers preferably include a spherical or substantially spherical form or shape. In some arrangements, the particles or other carriers are provided with a plurality of protruding (e.g., relative to a spherical or substantially spherical core) peptide structures in the form of protein spikes or protein fragments/protrusions on their surfaces. In some embodiments, each of the particles include 5 to 500 protruding peptide structures (e.g., 5 to 500, 0 to 100, 100 to 200, 200 to 300, 300 to 400, 400 to 500, 100 to 500, 200 to 500, 300 to 500, 0 to 200, 0 to 300, 0 to 400, 0 to 500, values between the foregoing ranges and values, etc.). In some embodiments, each of the particles include 1 to 1000 protruding peptide structures (e.g., 1 to 1000, 0 to 100, 100 to 200, 200 to 300, 300 to 400, 400 to 500, 100 to 500, 200 to 500, 300 to 500, 100 to 600, 200 to 600, 300 to 600, 400 to 600, 500 to 600, 100 to 700, 200 to 700, 300 to 700, 400 to 700, 500 to 700, 600 to 700, 100 to 800, 200 to 800, 300 to 800, 400 to 800, 500 to 800, 600 to 800, 700 to 800, 100 to 900, 200 to 900, 300 to 900, 400 to 900, 500 to 900, 600 to 900, 700 to 900, 800 to 900, 100 to 1000, 200 to 1000, 300 to 1000, 400 to 1000, 500 to 1000, 600 to 1000, 700 to 1000, 800 to 1000, 900 to 1000, 0 to 200, 0 to 300, 0 to 400, 0 to 500, 0 to 600, 0 to 700, 0 to 800, 0 to 900, 0 to 1000, values between the foregoing ranges and values, etc.).

In one embodiment, the surface features or other members that protrude from a core of the carrier (e.g., spikes) have a length of about 1 to 200 nm (e.g., 1 to 200, 1 to 100, 2 to 80, 5 to 50, 20 to 100, 50 to 100, 100 to 200 nm, values between the foregoing, etc.). In some embodiments, the surface features or other members that protrude from a core of the carrier (e.g., spikes) have a length of 0.2 to 100 nm (e.g., 0.2 to 100, 1 to 10, 2 to 20, 5 to 50, 10 to 100 nm, values between the foregoing ranges, etc.). In some embodiments, the length includes the actual length of a spike or other protrusion is the total liner length of such a spike or protrusion. However, in other embodiments, the length includes the distance from the spherical or other core of the carrier to the outermost radial distance of the protrusion.

In some embodiments, allowing the carrier (e.g., synthetic particle) to compete with viral particles, such as coronaviruses (e.g., the SARS-CoV-2 virus, variants thereof, etc.), influenzas, rhinoviruses, Respiratory Syncytial Viruses (RSVs), noroviruses, other viruses, etc.) for the same receptor and/or other binding site or portion of a host cell (e.g., ACE2, sialic acid, histo-blood group antigens, ICAM-1, IGF1R receptor, etc.) can function as a hindrance and/or other obstacle (e.g., allosteric regulation or hinder, other competitive or non-competitive inhibition, etc.) for the viral particle to bind to the receptor or other site or portion. This can advantageously minimize or reduce the likelihood of endocytosis of the virus or other pathogen, thereby lowering the risk of infecting the host cell.

One embodiment of the principle of competitive inhibition is schematically illustrated in FIG. 2. As shown, by way of an example, in some embodiments, a host receptor (e.g., ACE2) is responsible for mediating the SARS-CoV-2 infection responsible for coronavirus disease 19 (e.g., COVID-19). In some configurations, by binding carriers (e.g., the novel synthetic nanoparticles, other particles, objects, etc.) to that reception site (e.g., receptor), to the specific host receptors motifs and/or other any other site or portions of the host cell, infection (e.g., caused by the SARS-CoV-2 viruses, other viruses, etc.) can be advantageously prevented, controlled and/or otherwise mitigated.

With continued reference to FIG. 2, by way of an example, a host receptor (e.g., ACE2) responsible for mediating the infection resulting in a specific disease is generally depicted (schematically). In some embodiments, by binding a carrier (e.g., a novel synthetic particle, object, etc.) to that specific area or to the specific host receptors motifs, the infection caused by the specific virus, viruses and/or other pathogen can be prevented and controlled (e.g., the likelihood of infection can be reduced or otherwise mitigated, etc.). The competitive inhibition can be utilized against different viruses and/or other pathogens (such as, for example and without limitation, influenzas, rhinoviruses, RSVs, noroviruses, other respiratory and gastrointestinal viruses, other viruses or pathogens, etc.).

Based on, for example, the foregoing, in an embodiment, carriers (e.g., synthetic nanoparticles, other particles, etc.) are selected such that they resemble, at least partially, coronaviruses (e.g., SARS-CoV-2), influenzas, rhinoviruses, noroviruses, other common cold viruses and/or any other viruses or pathogens, as desired or required. In some embodiments, preferably, synthetic nanoparticles are enhanced or otherwise optimized, at least partially, for competitive inhibition. For example, the particle morphology, size, surface properties and/or any other properties or features of such particles can be modified to achieve higher (or otherwise improve) affinity for the target receptor angiotensin converting enzyme 2 (ACE2) and/or TMPRSS2, sialic acid, histo-blood group antigens, ICAM-1, IGF1R or other target receptors. Thus, the binding affinity for the specific receptor can be advantageously increased, thereby blocking the internalization of the viral envelope more efficiently and potentially prolonging the gained viral protection [8-10].

A carrier system as described herein, wherein the carrier (e.g., synthetic nanoparticle, other particle or object, etc.) resembling a targeted virus (e.g., the SARS-CoV-2, other corona or spiked viruses, influenza, rhinoviruses, noroviruses, other common cold viruses, etc.) can be enhanced or optimized for personalized medicine as variations and mutations in individuals might give rise to slightly different target receptors. Thus, the surface properties and functionalization of the carrier can be changed to match or substantially match the individual properties (e.g., mutations or variations) in target receptors and/or other binding sites or locations of a host cell for tailored therapies.

One embodiment of a targeted and/or personalized medicine is schematically illustrated in FIGS. 4A and 4B. As depicted, by way of an example, by designing a carrier (e.g., synthetic nanoparticle, other particle or object, etc.) that has features that resemble the selected or targeted virus or other pathogen (e.g., SARS-CoV-2, other corona or spiked viruses, influenzas, rhinoviruses, noroviruses, other common cold viruses, etc.). For example, the synthetic particles or other carriers can include spike protein fragments, protein protrusions, other protrusions, other surface features and/or

any other feature or property. Accordingly, it is possible for targeted drug delivery at (e.g., at, to, near, etc.) host cells that are susceptible for the virus and/or other pathogen. In some embodiments, as discussed herein, the carrier can include (e.g., can be "loaded" or otherwise provided with) one or more drugs and/or other compounds, substances and/or materials (for example, anti-viral compounds, zinc, immune modulating drugs (e.g., Celastrol, other interferon-gamma or stimulating molecules, penicillium, Dalbavancin or other anti-bacterial compounds, drugs intended to combat virus-related pneumonia, voriconazole, isavuconazole, drugs intended to combat viral-associated pulmonary aspergillosis, anti-fungal compounds, etc.) and/or the like, as desired or required by a particular application or use.

In some embodiments, a synthetic particle or other carrier is configured to be used for targeted and/or personalized medicine. For example, in some embodiments, the carrier's targeted therapy possibilities improve the therapeutic index of the API, drug, component and/or other material loaded into or onto the carrier. In some embodiments, the carrier improves the therapeutic index of the API, drug, component and/or other material by 1 to 1000 (e.g., 1 to 1000, 0 to 100, 100 to 200, 200 to 300, 300 to 400, 400 to 500, 100 to 500, 200 to 500, 300 to 500, 100 to 600, 200 to 600, 300 to 600, 400 to 600, 500 to 600, 100 to 700, 200 to 700, 300 to 700, 400 to 700, 500 to 700, 600 to 700, 100 to 800, 200 to 800, 300 to 800, 400 to 800, 500 to 800, 600 to 800, 700 to 800, 100 to 900, 200 to 900, 300 to 900, 400 to 900, 500 to 900, 600 to 900, 700 to 900, 800 to 900, 100 to 1000, 200 to 1000, 300 to 1000, 400 to 1000, 500 to 1000, 600 to 1000, 700 to 1000, 800 to 1000, 900 to 1000, 0 to 200, 0 to 300, 0 to 400, 0 to 500, 0 to 600, 0 to 700, 0 to 800, 0 to 900, 0 to 1000, values between the foregoing ranges and values, etc.), vis-à-vis, for example, delivering the same or equivalent amount of a material or component (e.g., API, drug, etc.) to the host without using the carrier as a delivery tool (e.g., a targeted delivery tool or mechanism). The therapeutic index (and the likelihood of successfully combatting a pathogen or other agent) is improved relative to no carrier or same API/other component that is delivered without a carrier (e.g., ingested, injected, crème, etc.). In some embodiments, delivery using a carrier in accordance with any of the embodiments disclosed herein or equivalents thereof increases or otherwise improves or enhances solubility and/or permeability of the drug. For example, the solubility and/or permeability of the component being delivered to the host using the carrier can be improved by 0.1 to 50% (e.g., 0.1 to 5, 5 to 10, 10 to 20, 20 to 30, 30 to 40, 40 to 50, 0.1 to 50, 1 to 20, 1 to 50, 20 to 50, 10 to 40%, values between the foregoing, etc.), more than 50% (e.g., 50 to 60, 60 to 70, 70 to 80, 80 to 100, 50 to 100%, values between the foregoing, greater than 100%, greater than 200%, etc.), relative to, for example, the same or similar component delivered by other means (e.g., via ingestion, injection, droplets, lotions, etc.).

In some embodiments, the synthetic particle or other carrier comprises (e.g., is provided with) a coating and/or functionalization that has higher affinity towards the receptor favoring the binding of the synthetic particle or other carrier than the viral one (e.g., the virus, other pathogenic or infectious agent or member, etc.).

In one embodiment, for example, the synthetic particle or other carrier comprises an amino acid sequence that is similar to that of the said viral protrusion having affinity for the same target receptor as the pathogen thus having competition for the same receptor.

In one embodiment, for example, the synthetic particle or other carrier is further optimized for improved binding to said host receptor in order to achieve improved blocking effect by competitive inhibition to the said pathogen.

In one embodiment, for example, the synthetic particle or other carrier having coating and/or functionalization of epitopes similar to that of the pathogen of interest in order to give a vaccination at target cell population.

According to some embodiments, the synthetic particle or other carrier is decorated and/or loaded with immunogenic epitopes of the pathogen or pathogens of interest for the targeted cell populations to present the said epitopes to B and/or T cells in order to elicit an immunological response towards said pathogen or pathogens. In some embodiments, this advantageously results in the infected cells to be hindered and/or eliminated by the host's immune system.

In some embodiments, by way of example, the synthetic particle (e.g., carrier) is decorated and/or loaded with polypeptide protrusions containing epitopes from several different pathogens of interest to be used in the composition of a particle-based vaccine composition. The carrier protrusions, which can include both B cell stimulating and/or T cell stimulating epitopes, can be configured to comprise a specific sequence obtained from the amino-acid sequence of the protein of interest (e.g., derived from a coronavirus, SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV), another virus that impacts the respiratory system, any other type of virus, etc.).

In some embodiments, for example, the synthetic particle or other carrier is decorated and/or loaded with a selected and isolated nucleic acid molecules (e.g., RNA or DNA) having a nucleic acid sequence that encodes a specific polypeptide sequence. Such a sequence can include both B cell stimulating and/or T cell stimulating epitopes that include specific sequences. In some embodiments, the sequences are obtained from the amino-acid sequence (e.g., complete sequence) of the protein of interest (e.g., derived from a coronavirus, SARS-CoV-2 virus, an influenza virus, a rhinovirus, a norovirus, a respiratory syncytial virus (RSV), another virus that impacts the respiratory system, any other type of virus, etc.).

In one embodiment, the synthetic particle or other carrier is configured to prevent or at least partially hinder the ability of the SARS-CoV-2 pathogen to infect and/or replicate in the host (e.g., human, animal, etc.).

In some arrangements, the synthetic particle or carrier is synthesized using different materials and functionalization in order to match or substantially match the optimal or beneficial properties to be administered to target cell populations (e.g., hydrophobic or hydrophilic properties, depending on the intended use).

In one embodiment, for example, silica (e.g., stable organic silica) is used as the core material that could exhibit a blocking effect that, optionally after modification of the particle, could be prolonged for hours, days or longer as it takes time for silica nanoparticles to degrade in aqueous conditions similar to the environment of the human body.

In one embodiment, for example, solid lipid particles (e.g., fabricated by a bottom-up method using microfluidics) are used as the core material for the carrier to be further coated, functionalized and/or loaded into or onto with API, epitopes, proteins, RNA/DNA, anti-virals and immune stimulating compounds such as Celastrol, interferon gamma.

In one embodiment, for example, self-assembling protein particles produced by genetically-engineered bacterial or mammalian cells producing proteins or protein fragments, such as ferritin heavy or light chain, are used as the core

material for the carrier. Such particles can be further functionalized and/or loaded into or onto other molecules, epitopes, API, epitopes, proteins, RNA/DNA, anti-virals and immune stimulating compounds such as Celastrol, interferon gamma.

According to some embodiments, the administration route of a carrier depends on the tissue that the virus has invaded. For example, if the virus or other targeted pathogen resides in the upper or lower respiratory tract, it may be preferred to use an inhalation device for administering the carriers (e.g., synthetic particles) with a desired dosage. In some arrangements, such an inhalation device can allow a desired (e.g., optimal, effective, etc.) dosage of a carrier to be provided to a targeted anatomical location on demand.

In one embodiment, there is provided an inhalation device which compromises a container (e.g., a small plastic container) with dried carriers (e.g., synthetic particles, objects, etc.) like that of a dry powder inhaler or as a meter dose inhaler where the carriers (e.g., particles) are sprayed from the inhaler as an aerosol, as an vaporizer creating a fine mist of particles and solution, as an nasal spray dispersed in an aqueous solution and/or in any other form or configuration or hybrid form, as desired or required.

In some embodiments, for improving or enhancing (e.g., maximizing) the coverage of the upper respiratory tract, an inhalation mask is used. As a result, the entry of carriers (e.g., particles) into the nasal cavity and lower respiratory tract (where epithelial cells expressing ACE2, sialic acid, histo-blood group antigens, ICAM-1, IGF1R or other receptors that may also reside) can be enhanced or otherwise improved, thereby lowering (e.g., minimizing) the risk of being infected by the virus or other pathogen, at least temporarily.

In embodiments where the viral infection is (or would be) in the gastrointestinal tract, a tablet, an orally ingestible liquid and/or any other ingestible material is the preferred route of administration of the carrier to the host or subject. The synthetic particles or other carriers of such orally administered compositions can advantageously temporarily protect, at least partially, the end-user from infection by the virus (e.g., orally, via fecal-oral transmission, etc.).

In some embodiments, the carrier (e.g., nanomaterial, other particle or object, etc.) can also be fabricated and configured to have a high or a favorable affinity for the pathogen, thus, at least partially, encapsulating and immobilizing the threat of infection e.g., coating or functionalizing the particle with molecules that has high binding affinity towards the pathogen. Accordingly, such carriers could be used in disinfecting products (e.g., cleaning solution, hand sanitizer products, disinfecting wipes, etc.).

FIG. 3A shows, schematically and by way of an example, utilization of carriers (e.g., nanoparticles, other particles or objects, etc.) coated and/or otherwise provided with peptides resembling the binding motif of the target receptor, such as, e.g., ACE-2, sialic acid, histo-blood group antigens, ICAM-1, IGF1R or other receptors that the specific or targeted virus or other pathogen uses. For example, such targeted viruses or other pathogens include, without limitation or restrictions, coronaviruses (e.g., SARS-CoV-2), influenzas, rhinoviruses, noroviruses and other common cold viruses and/or the like. In some embodiments, as discussed throughout this application, the carriers are configured to encapsulate and/or immobilize the virus and/or other pathogen, thus minimizing or otherwise reducing the risk of the virus and/or other pathogen infecting the host.

FIG. 3B shows, schematically and by way of an example, utilization of carriers (e.g., nanoparticles, other particles or

objects, etc.) coated and/or otherwise provided with peptides (epitopes) resembling the naturally occurring protrusions of the pathogen or pathogens capable of binding to targeted receptor and/or receptors, blocking viral entry and functioning as a vaccine for targeted cell populations. For example, such targeted viruses or other pathogens can include, without limitation or restrictions, coronaviruses (e.g., SARS-CoV-2), influenzas, rhinoviruses, noroviruses and other common cold viruses and/or the like. In some embodiments, as discussed throughout this application, the carriers are configured to at least partially hinder viral or pathogen spread and elicit a protective immune response against said virus and/or other pathogen, thus minimizing or otherwise reducing the risk of viral or other pathogenic infection within the host.

In some embodiments, the carrier (e.g., nanomaterial, other particle or object, etc.) can also be fabricated to have high or favorable affinity for the receptor and co-receptor. Thus, the carrier can be configured to bind to multiple receptors (e.g., receptors that various pathogens use for cell entry). Further, as noted herein, the synthetic particle or other carrier can be provided with a coating or similar layering or component that has higher or otherwise favorable or improved affinity towards the receptor favoring the binding of the carrier (e.g., synthetic particle). This can allow or otherwise facilitate the carrier to be used in a multiple targeting approach (e.g., further reducing (e.g., minimizing) the risk of contracting said disease or diseases (e.g., viral or pathogenic infection and the diseases originating therefrom)).

FIGS. 4A and 4B schematically shows, by way of an example only, utilization of nanoparticles or other carriers coated (or otherwise provided) with peptides resembling the binding motif of a viral protrusion protein, such as, for example, the spike protein from the SARS-CoV-2 or other coronaviruses, Hemagglutinin (HA) and Neuraminidase (NA) proteins from influenza A virus, etc. combined with peptides resembling the binding motif of host receptors (e.g., ACE2, sialic acid, histo-blood group antigens, ICAM-1, IGF1R or other receptors) of the subject. As a result, the carriers can advantageously be provided with dual targeting strategies, thereby minimizing or reducing the risk of viruses or other pathogens infecting the host.

According to some embodiments, the carrier (e.g., nanomaterial, particle or object, etc.) can be fabricated or otherwise configured to have high or favorable affinity for the targeted pathogen(s) (e.g., virus(es) circulating co-receptors e.g., high-density lipoprotein (HDL) scavenger receptor B type 1 (SR-B1), thus immobilizing the treat which could be used as an antidote preventing further spreading of the virus in the said host.

FIG. 5 schematically shows, by way of an example, utilization of nanoparticles or other carriers coated (and/or otherwise provided) with peptides resembling the binding motif of the spike protein from a coronavirus (e.g., the SARS-CoV-2 virus) for at least partially binding to the co-receptor. As a consequence of such carriers within the host, the mobility of the virus can be advantageously decreased, thus minimizing or otherwise reducing the risk of the targeted virus or other pathogen infecting the host or spreading the viral infection inside the said host (e.g., spread to other organs or locations of the host).

Based on, for example, the above, the following represents non-limiting embodiments of the present technology:

A carrier (e.g., synthesized carrier in the nano- or microscale or any other object that has the capacity of saturating and binding to target receptors, proteins and/or

macromolecules for example but not limited to ACE2, sialic acid, histo-blood group antigens, ICAM-1, IGF1R or other receptors at the surface of cells that prevents and minimize pathogen, such as influenzas, rhinoviruses, RSVs, noroviruses, coronaviruses (e.g., SARS-CoV-2), other viruses causing respiratory infection, binding and entry to the host lowering the risk of contracting the specific disease, such as COVID-19 disease, diarrhea, common cold, cytokine storm, death or generally discomfort or a combination thereof.

A carrier (e.g., a synthesized carrier in the nano- or microscale or any other object that has the capacity of binding and encapsulating the pathogen of interest, thus immobilizing, at least partially, the pathogens' ability to bind and entry to the host, thereby lowering the risk of contracting the specific infectious agent).

A carrier (e.g., a carrier as above), wherein the core structure of the carrier is obtained (e.g., manufactured, fabricated, etc.), at least in part, by 3D printing, microfluidics, supercritical solution method, sol-gel method, other bottom-up and/or top-down method of fabrication self-assembling components and/or any other method or technology.

A carrier (e.g., as provided above and/or herein), where the core material is made of or comprises, however not limited to, organic or inorganic components, lipid droplets, micelles, cholesterol, amino acids, proteins, salts and minerals or other molecules.

One embodiment comprises lipid-based micelles made by, for example, cholesterol decorated with SARS-CoV-2 spike protein fragments and/or other protrusions that bind both to host receptor sites or other portions of the host cell (e.g., ACE2, TMPRSS2, etc.) and to cholesterol and its high-density lipoprotein (HDL) scavenger receptor B type 1 (SR-B1) that would facilitate ACE2-dependent entry of the nanoparticle and/or microparticle loaded with selected API for combating COVID-19 disease or other disease resulting from infection by a virus or other pathogen. In some embodiments, the cholesterol recognition amino acid consensus (CRAC) motifs near the inverted cholesterol recognition motif (CARC) have been proven to bind with SARS-CoV-2 S1 subunit and this HDL complex enhances viral entry to host cells facilitating replication [17]. Therefore, by creating a carrier (e.g., nanoparticle, other particle or object, etc.) that would compete with this spike protein-HDL interaction would potentially lower the ability of SARS-CoV-2 (or the targeted pathogen for ACE2-mediated (or other receptor-mediated) internalization, at least partially blocking viral entry to host cells and at least partially hindering replication. In one arrangements, this co-receptor incarceration could be blocked by decorating the nanoparticle with spike protein fragments from CARC-CRAC region of SARS-CoV-2 preferably but not limited to 129KKKKVCEFCNDPFLGVYYHKNNKKKK150 (SEQ ID No: 4) together with other amino acids for example the RBD spike fragment hexapeptide 438YKYRYL443 (SEQ ID No: 1) that binds to the ACE2 receptor creating a nanoparticle capable of blocking viral-host interaction on multiple positions loaded with selected API for targeted therapeutics (e.g., Celastrol, Zinc, ITX 5601, etc.). [8,15-18].

One embodiment comprises the use of simultaneous inhibiting of multiple receptors by multiple targeting approaches, where the carrier (e.g., mimetic particle) includes protrusions (e.g., located on or along an outer surface of the carrier, extending from the outer surface, etc.) that are similar or substantially similar to those of the target virus or viruses. Such protrusions can include, for example,

spike proteins, HA and NA or VP that would bind to the specific host receptor for inhibiting viral entry by competitive inhibition, etc. In some embodiments, the carrier (e.g., virus-like particle) also includes surface protrusions that mimic epitopes selected from said viruses for eliciting an immune response against said viruses.

According to one embodiment comprises the use of simultaneous targeting of multiple receptors by multiple targeting approaches, where the carrier (e.g., mimetic particle) includes protrusions (e.g., located on or along an outer surface of the carrier, extending from the outer surface, etc.) that are similar or substantially similar to those of the target virus or viruses. Such protrusions can include, for example, spike proteins, HA and NA or VP that would bind to the specific host receptor for inhibiting viral entry by competitive inhibition, etc. In some embodiments, the carrier (e.g., virus-like particle) also includes drugs, API or other molecules for targeted delivery to the host.

Another embodiment comprises the use of simultaneous inhibiting and immobilizing by dual targeting approaches, where the carrier (e.g., mimetic particle) has protrusions on the outer surface that are similar to those of the virus, for example, spike protein, HA and NA or VP that would bind to the specific host receptor for inhibiting viral entry by competitive inhibition. In some embodiments, the carrier (e.g., virus-like particle) also includes surface protrusions that mimic the host component (e.g., ACE2, sialic acid, histo-blood group antigens, ICAM-1, IGF1R receptors, and/or antibodies such as the monoclonal antibody bebtelovimab, etc.).

Another embodiment comprises using self-assembling recombinant protein-based nanoparticle constructs, such as, for example, SpyTag/SpyCatcher system and ferritin-based constructs [23]. Where the constructs are expressed in *E. Coli*; the proteins are purified and then assembled like a two-component "super glue" into virus-like particles (VLPs) conjugated with the selected antigens, viral epitopes or fragments [24]. The carrier could be assembled using the SpyTag/SpyCatcher system or ferritin (heavy or light chain) based particle core and then conjugated, coated and/or functionalized with the selected SARS-CoV-2 spike protein or selected hexapeptide 438YKYRYL443 derived thereof or peptides from the CARC-CRAC region or other proteins of interest. Several studies show that it is possible to construct such a VLP using SARS-CoV-2 spike protein (RBD) candidate combined with SpyCatcher technology and ferritin based particle systems [24, 25]. The selected studies using RBD-SpyVLP demonstrate that the construct is easily producible and scalable, and that the final product is thermally stable even at room temperature for several weeks [25]. The SARS-CoV-2 RBD conjugated to SpyCatcher-mi3 nanoparticle (abbreviated: RBD-mi3 NP) shows higher binding affinity for the ACE2 receptor than viral RBD monomers detected using Biolayer interferometry (BLI) kinetic assays [24]. Therefore, it appears possible to develop mimetic nanoparticles or other carriers for preventing the spreading and lowering the infection rate of novel coronaviruses with higher affinity than the RBD monomer.

The synthetic carrier or nanoparticle may comprise or be decorated with a polypeptide or protein having an amino acid sequence of the ACE2 binding sequence and/or the SARS-CoV-2 spike protein RBD or a fragment thereof. In an embodiment, the amino acid sequence of the ACE2 binding sequence and/or the SARS-CoV-2 spike protein RBD or a fragment thereof is optimized, for example such that it has a higher binding affinity for the ACE2 receptor and enhanced blocking properties that of the spike protein of the corona-

virus interaction compared to the corresponding, unmodified spike protein sequence (SEQ ID No: 5). By optimizing the amino acid sequence of the ACE2 binding sequence and/or the SARS-CoV-2 spike protein RBD or a fragment thereof it is possible to achieve even higher binding affinity for example with combining hexapeptides 438YKYRYL443 (SEQ ID No: 1) or 438YKYNYL443 (SEQ ID No: 3) with the optimized spike protein sequence.

In a carrier as above, according to some embodiments, the core or core material may be made of, for example, self-assembling virus-like protein nanoparticles that can be saturated with different drugs. These particles can be synthesized in various sizes and shapes.

A carrier as above, where the core material is made of, for example, mesoporous silica nanoparticles with ordered mesostructures of pores that can be loaded with different drugs and that these particles can be synthesized in various sizes and shapes.

A carrier as above, wherein the core material is functionalized with one or several of the following: amino acids, epitopes, peptides or proteins and/or protein fragments, chemical agents, active pharmaceutical ingredients (API), organic or inorganic polymers or molecules.

A functionalized carrier as above, wherein the carrier with its functionalization provides a method of specifically binding to receptors, proteins and macromolecules at the cellular level in order to prevent and minimize pathogen entry to the host target tissues by competitive inhibition.

A functionalized carrier as above, wherein the carrier with its functionalization provides a method of specifically binding to receptors, proteins and macromolecules at the cellular level in order to prevent and minimize SARS-CoV-2, influenza, rhinoviruses, respiratory syncytial virus, norovirus and other viruses causing respiratory infection entry to the host target receptors by competitive inhibition.

A carrier system as above, wherein the carrier with its functionalization provides a method of loading drugs, API, molecules, peptides inside or onto the carrier system.

A carrier system as above, where the functionalized and drug loaded carrier system can be used for targeted drug delivery of anti-pathogenic, anti-viral or anti-microbial compounds in order to decrease the growth of the infectious agent.

A carrier system as above, where the functionalized and drug loaded carrier system can be used for targeted drug delivery of, anti-viral compounds in order to decrease the replication rate of the coronavirus.

A carrier system as above, wherein the synthetic nanoparticle resembling the SARS-CoV-2 virus is loaded into or onto the nanoparticle for further enhancing the anti-viral properties of the invention. For example, Zinc which has been shown to reduce viral replication in its host cells, can be employed [21]. Also, viscosity modulators, antihistamines, Celastrol and/or immunosuppressors can be used in the COVID-19 disease for minimizing the cytokine storm that potentially is dangerous to some patients [18,21,22].

A carrier system as above, wherein the synthetic nanoparticle resembling the SARS-CoV-2, influenza, rhinoviruses and other viruses causing respiratory infection is loaded with proteome inhibitors or new molecular entities developed in the future for efficiently deliver the compounds in the target tissues with minimal off-target effects.

A carrier system as above, wherein the synthetic nanoparticle is decorated with molecules that has high affinity towards the SARS-CoV-2 virus or influenza, rhinoviruses and viruses causing respiratory infection e.g., proteins resembling that of the ACE-2, sialic acid, histo-blood group

antigens, ICAM-1, IGF1R receptor or any other pathogen of interest in order to bind and immobilize the infectious agent preventing or minimizing the potential risk of host entry.

A carrier system as above, wherein the synthetic nanoparticle resembling the SARS-CoV-2 virus, or any other pathogen for example influenza, rhinoviruses and viruses causing respiratory infection is decorated with epitopes to be used as a vaccination at target cell populations.

A carrier system as above, wherein the carrier system is loaded, stored or dispersed in a device or vessel capable of on-demand release of the carrier to the end-user. Therefore, in some embodiments, the carrier can be administered to the anatomy by the subject or patient, making the carrier capable of self-administration.

A carrier system as above, wherein the carrier system is loaded inside a dispenser such as an inhalation device, tablet, injectable substance, cream or ointment.

A carrier system as above, wherein the man-made materials are used to immobilize specific pathogens by adding the synthetic material in sanitation products and disinfectants.

A carrier system as above for minimizing the spread of diverse pathogens by binding to the target molecule in the host body or binding to the infectious agent itself and potentially inhibit the spread of the disease. Furthermore, as a combination treatment listed in the preceding embodiments hindering the replication of the infectious agent together with giving the immune system in the host a gained advantage to fight the disease similar to vaccines or immunoregulating drugs.

In further embodiments, the present invention is thus directed to a method for preparing a synthetic nanomaterial comprising a core object, particle, sheet, film or spheroid, triangle, star shaped, said object also comprising a coating or functionalization of organic polymers, amino acids, proteins or molecules mimicking the surface of the pathogen, such as the coronavirus of interest, e.g., SARS-CoV-2 and future variants alternatively influenza, rhinoviruses and other viruses causing respiratory infection. FIG. 7 exemplifies how the SARS-CoV-2 spike protein, and variation thereof, may be produced using a vector for producing the specific protein construct to be conjugated to the virus-like nanoparticle or synthetic carrier.

Producing a man-made material that has the capability of mimicking the pathogen of interest that has the capability of competing with the pathogen of interest for the same host target molecule, receptor, amino acid or nucleotide. Alternatively, in some embodiments, producing a material that has the capability of binding and immobilizing the pathogen of interest minimizing the possible infection in its host. However, in other embodiments, such immobilization is not required or necessary.

Producing a man-made material that has the capability of mimicking the coronavirus of interest, e.g., SARS-CoV-2, influenza, rhinoviruses and other viruses causing respiratory infection that has the capability of competing with the virus for the same host target molecule, receptor, amino acid or nucleotide e.g., ACE2 and/or TMPRSS2, sialic acid, histo-blood group antigens, ICAM-1, IGF1R receptors.

One embodiment comprises the steps of:

- a) providing a core material, e.g., a nano- and/or micro-material including nanoparticles, microparticles or any other object as disclosed herein;
- b) coating or functionalizing the core material with molecules, polymers, amino acids, proteins, API, drugs or other material as disclosed herein;
- c) loading the object with compounds, molecules, drugs, API, DNA or RNA etc.;

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- d) coating a second protective or functional layer on top of the object in particular for increasing its resistance that could be important in extreme environments such as the acidic environment in the stomach; and
- e) providing a small device, medical device, inhalation device or aerosol, sanitation product or consumer product that on-demand will release the containing synthetic material, particle or object for administration.

FIG. 7 shows an example of a SARS-CoV-2 spike RBD (receptor-binding domain) expression construct. Different expression cassettes (shown as the XXX region) can be used for expressing the desired construct e.g., influenza H7 haemagglutinin (indicated as "Signal sequence"), Tag, spacer, and SARS-CoV-2 RBD (using amino acid region 319 to 541, depicted in SEQ ID No: 5; PUBMED 32015508 [26]; the full amino acid sequence of the surface glycoprotein of SARS-CoV-2 is shown in in SEQ ID No: 6). The spike protein construct can be further optimized for ACE2 receptor interaction using other known amino acid sequences from SARS-CoV-2 variants such as but not limited to Alpha (B.1.1.7), Beta (B.1.351), Gamma (P.1), Delta (B.1.617.2), Omicron (B.1.1.529) and/or predicted amino acids or amino acid substitutions e.g., V367F [27], W436R, and/or N354D/D364Y.

FIG. 7 exemplifies how a spike protein, and variation thereof, or other viral protein, may be produced using a vector for producing the specific protein construct to be conjugated to the virus-like nanoparticle or synthetic man-made carrier. In an embodiment, the synthetic carrier or synthetic nanoparticle comprises and/or is coated with a peptide, a polypeptide or a protein having an amino acid sequence comprising (or consisting of) a sequence as set forth in any one of SEQ ID Nos. 1, 2, 3, 4, 5 and/or 6. In a further embodiment, the synthetic carrier or synthetic nanoparticle comprises and/or is coated with a polypeptide or a protein having an amino acid sequence comprising (or consisting of) a sequence as set forth in SEQ ID No. 5 and/or 6, wherein the sequence optionally comprises one or more amino acid substitutions. The one or more amino acid substitutions may be selected from (but are not limited to) V367F, W436R, and/or N354D/D364Y or other amino acid substitutes consisting of new coronavirus variants of concern (VOC) having higher affinity for the target receptor (table 1). To be used against contracting the COVID-19 disease and to release and shorten the disease progression and duration.

The following represent non-limiting embodiments of the present technology. The following is a non-exclusive list of embodiments, and as such, should not be seen to limit, in any way, the various inventions disclosed herein.

1. A method of preventing or reducing pathogen binding to target areas of cell surfaces of a host selected from mammals, comprising providing administering to the mammal a carrier comprising biocompatible particles having a maximum size in at least one dimension in the nanometer or micrometer range, forming a core, and further having a functionalized surface capable of binding to said target areas of said cell surfaces to at least temporarily block said target areas to prevent or minimize pathogen binding and thus, reducing the risk of the host contracting a disease caused by said pathogen.

2A. The method according to embodiment 1, wherein the carrier has the capacity of functioning as a targeted vaccination minimizing the pathogens' ability to bind and entry to the host lowering the risk of contracting the specific infectious agent.

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2B. The method according to embodiment 1, wherein the carrier has the capacity of binding and encapsulating the pathogen, thus immobilizing the pathogens' ability to bind and entry to the host lowering the risk of contracting the specific infectious agent.

3. The method according to embodiment 1 or 2, wherein the core structure of the carrier is being obtained by 3D printing, microfluidics, sol-gel method or other bottom-up and/or top-down method of fabrication.

4. The method according to any of embodiments 1 to 3, wherein the core material comprises organic or inorganic components, lipid droplets, amino acids, proteins, salts and minerals or other molecules or wherein the core material comprises mesoporous silica nanoparticles, in particular mesoporous silica particles with ordered mesostructures of pores that preferably are capable of being loaded with drugs.

5A. The method according to any of embodiments 1 to 4, wherein the core material is functionalized with substance selected from the group consisting of amino acids, epitopes, peptides, proteins and/or protein fragments, chemical agents, active pharmaceutical ingredients (API), organic or inorganic polymers or molecules and combinations thereof.

5B. The method according to any of embodiments 1 to 4, wherein the core material is functionalized with substance selected from the group consisting of peptides, proteins such as antibodies, chemical agents, active pharmaceutical ingredients (API), organic or inorganic polymers or molecules and combinations thereof.

6. The method according to any of embodiments 1 to 5, wherein the carrier functionalized for specifically binding to receptors, proteins and macromolecules at the cellular level in order to prevent and minimize pathogen entry to the host target tissues by competitive inhibition.

7. The method according to any of embodiments 1 to 6, wherein the synthetic nanoparticle and/or microparticle is used for reducing the spread of SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof.

8. The method according to any of embodiments 1 to 7, wherein said synthetic nanoparticle has a 3D-configuration generally matching the characteristics of the SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof., in particular the particle is fabricated to a size of around 100 nm and coated with similar amino acids as the glycoprotein spikes or protruding proteins at the surface of the viral particle or similar molecules that mimic the surface of the viral envelope.

9. The method according to any of embodiments 1 to 8, wherein said synthetic nanoparticle resembles the SARS-CoV-2 virus, influenza viruses, rhinoviruses, common cold viruses and/or noroviruses or is optimized for competitive inhibition.

10. The method according to any of embodiments 1 to 9, wherein the synthetic carrier exhibits a modified particle morphology, size or surface properties to achieve increased affinity for the target receptor angiotensin converting enzyme 2 (ACE-2), compared with the SARS-CoV-2 virus, and/or other viruses that causes a respiratory infection, diarrhea, common cold, in particular for increasing the binding affinity for the specific receptor e.g., sialic acid, histo-blood group antigens, ICAM-1, IGF1R blocking the internalization of the viral envelope more efficiently and potentially prolonging the gained viral protection.

11. The method according to any of embodiments 1 to 10, wherein said synthetic nanoparticle resembling the SARS-

CoV-2 virus, influenza viruses, rhinoviruses, common cold viruses and/or noroviruses is adapted for personalized medicine.

12. The method according to any of embodiments 1 to 11, wherein said the synthetic nanoparticle resembling the SARS-CoV-2 virus, influenza viruses, rhinoviruses, common cold viruses and/or noroviruses is loaded into or onto the nanoparticle for further enhancing the anti-viral properties.

13. The method according to any of embodiments 1 to 12, wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus, influenza viruses, rhinoviruses, common cold viruses and/or noroviruses is loaded with vehicles or proteome inhibitors for efficiently delivering the compounds in the target tissues with minimal off-target effects.

14. The method according to any of embodiments 1 to 13, wherein the synthetic nanoparticle is decorated with molecules that have high affinity towards the SARS-CoV-2 virus or any other pathogen of interest such as influenza viruses, rhinoviruses, common cold viruses and/or noroviruses in order to bind and immobilize the infectious agent preventing or minimizing the potential risk of host entry.

15. The method according to any of embodiments 1 to 14, wherein the synthetic nanoparticle resembling the SARS-CoV-2 virus or any other pathogen such as influenza viruses, rhinoviruses, common cold viruses and/or noroviruses is coated or decorated with epitopes to be used as a vaccination at target cell populations.

16. The method according to any of embodiments 1 to 15, wherein the carrier is loaded, stored or dispersed in a device or vessel capable of on-demand release of the carrier to the end-user. In some embodiments, the carrier is configured to be administered to the anatomy by the subject or patient, making the carrier capable of self-administration.

17. The method according to any of embodiments 1 to 16, wherein the carrier system is loaded inside a dispenser such as an inhalation device, tablet, injectable substance, cream or ointment.

18. The method according to any of embodiments 1 to 17, wherein the man-made materials is used for immobilizing specific pathogens by adding the synthetic material in sanitation products and disinfectants.

19. The method according to any of embodiments 1 to 18 for preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said method comprises minimizing the spread of diverse pathogens by binding to the target molecule in the host body or binding to the infectious agent itself and potentially inhibit the spread of the disease.

The following embodiments are disclosed. The following is a non-exclusive list of embodiments, and as such, should not be seen to limit, in any way, the various inventions disclosed herein.

1. A synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell surfaces of a host, said carrier comprising biocompatible particles having a maximum size which, in at least one dimension, is in the nanometer or micrometer range, forming a core, and further having a functionalized surface capable of binding to said target areas of said cell surfaces so as to at least temporarily block said target areas to prevent or minimize pathogen binding and, thus, reducing the risk of the host contracting a disease caused by said pathogen.

2A. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to embodiment 1, said carrier

having the capacity of eliciting an protective immunological reaction against said pathogen thus at least partially hindering the ability of the pathogen to bind to and enter the host, thereby lowering the risk of contracting the specific infectious agent and the related disease.

2B. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to embodiment 1, said carrier having the capacity of binding and encapsulating the pathogen, thus immobilizing the pathogens' ability to bind and entry to the host lowering the risk of contracting the specific infectious agent.

3. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to embodiment 1 or 2, wherein said core structure of the carrier is being obtained by 3D printing, microfluidics, sol-gel method or other bottom-up and/or top-down method of fabrication.

4. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of embodiments 1 to 3, wherein the core material comprises organic or inorganic components, lipid droplets, amino acids, proteins, salts and minerals or other molecules.

5. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein the core material comprises inorganic silica nanoparticles, in particular mesoporous silica particles, such particles preferably having ordered mesostructures of pores that preferably are capable of being loaded with drugs.

6A. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said core material is functionalized with substance selected from the group consisting of amino acids, epitopes, peptides, proteins or fragments of proteins, chemical agents, active pharmaceutical ingredients (API), organic or inorganic polymers or molecules and combinations thereof.

6B. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said core material is functionalized with substance selected from the group consisting of peptides, proteins such as antibodies, chemical agents, active pharmaceutical ingredients (API), organic or inorganic polymers or molecules and combinations thereof.

7. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said the carrier with its functionalization is used for specifically binding to receptors, proteins and macromolecules at the cellular level in order to prevent and minimize pathogen entry to the host target tissues by competitive inhibition.

8. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, said method comprising loading drugs, API, molecules, peptides inside or onto the carrier system.

9. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, comprising a functionalized and drug loaded carrier, said carrier being used for targeted drug delivery of

anti-pathogenic, anti-viral or anti-microbial compounds in order to decrease the growth of the pathogen, such as infectious agent.

10. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle and/or microparticle is used for reducing the spread of SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof.

11. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle has a 3D-configuration generally matching the characteristics of the SARS-CoV-2 virus or influenza viruses, rhinoviruses, common cold viruses and/or noroviruses, in particular the particle is fabricated to a size of around 100 nm and coated with similar amino acids as the glycoprotein spikes or other protruding proteins at the surface of the viral particle or similar molecules that mimic the surface of the viral envelope and thus binds to the same target receptor as the virus.

12. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle resembles the SARS-CoV-2 virus or influenza viruses, rhinoviruses, common cold viruses and/or noroviruses or is optimized for competitive inhibition.

13. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to embodiment 12, wherein the synthetic carrier exhibits a modified particle morphology, size or surface properties to achieve increased affinity for the target receptor angiotensin converting enzyme 2 (ACE-2) compared with the SARS-CoV-2 virus or influenza viruses, rhinoviruses, common cold viruses and/or noroviruses, in particular for increasing the binding affinity for the specific receptor e.g., sialic acid, histo-blood group antigens, ICAM-1, IGF1R blocking the internalization of the viral envelope more efficiently and potentially prolonging the gained viral protection.

14. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus, or influenza viruses, rhinoviruses, common cold viruses and/or noroviruses is adapted for personalized medicine.

15. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said the synthetic nanoparticle resembling the SARS-CoV-2 virus or influenza viruses, rhinoviruses, common cold viruses and/or noroviruses is loaded into or onto the nanoparticle for further enhancing the anti-viral properties.

16. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus, or influenza viruses, rhinoviruses, common cold viruses and/or noroviruses is loaded with vehicles or proteome inhibitors for efficiently delivering the compounds in the target tissues with minimal off-target effects.

17A. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle is decorated with molecules that mimics naturally occurring protrusion of SARS-CoV-2 virus or any other pathogen of interest for example influenza viruses, rhinoviruses, common cold viruses and/or noroviruses in order to bind to target receptor of said host and thus preventing or minimizing the potential risk of the infectious agent entry to host.

17B. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle is decorated with molecules that have high affinity towards the SARS-CoV-2 virus or any other pathogen of interest for example influenza viruses, rhinoviruses, common cold viruses and/or noroviruses in order to bind and immobilize the infectious agent preventing or minimizing the potential risk of host entry.

18. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus or any other pathogen for example or influenza viruses, rhinoviruses, common cold viruses and/or noroviruses is coated or decorated with epitopes to be used as a vaccination at target cell populations making the administration potentially easier for the end user e.g., inhalation compared to intra muscular injection used in traditional vaccinations.

19. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said carrier is loaded, stored or dispersed in a device or vessel capable of on-demand release of the carrier to the end-user.

20. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said carrier system is loaded inside a dispenser such as an inhalation device, tablet, injectable substance, cream or ointment.

21. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said the man-made materials is used for immobilizing specific pathogens by adding the synthetic material in sanitation products and disinfectants.

22. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said method comprises minimizing the spread of diverse pathogens by binding to the target molecule in the host body or binding to the infectious agent itself and potentially inhibit the spread of the disease.

23. Method of producing a synthetic carrier according to any of embodiments 1 to 22, comprising the steps of

- a) providing a core material, e.g., a nano- and/or micro-material including nanoparticles, microparticles or any other object as disclosed herein;
- b) coating or functionalizing the core material with molecules, polymers, amino acids, proteins, API, drugs or other material as disclosed herein;
- c) loading the object with compounds, molecules, drugs, API, DNA or RNA etc.;

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d) coating a second protective and/or functional layer on top of the object in particular for increasing its resistance that could be important in extreme environments such as the acidic environment in the stomach; and providing a small device, medical device, inhalation device or aerosol, sanitation product or consumer product that on-demand will release the containing synthetic material, particle or object for administration.

The following embodiments are non-limiting representative configurations of the present technology. The following is a non-exclusive list of embodiments, and as such, should not be seen to limit, in any way, the various inventions disclosed herein.

1. A method of preventing or reducing pathogen binding, in particular of preventing or reducing binding of SARS-CoV-2 or influenza viruses, rhinoviruses, common cold viruses and/or noroviruses and viral strains thereof, to target areas of cell surfaces of a host selected from mammals, comprising providing administering to the mammal a carrier comprising biocompatible particles having a maximum size in at least one dimension in the nanometer or micrometer range, forming a core, and further having a functionalized surface capable of binding to said target areas of said cell surfaces to at least temporarily block said target areas to prevent or minimize pathogen binding and thus, reducing the risk of the host contracting a disease caused by said pathogen.

2. The method according to embodiment 1, wherein the carrier has the capacity of binding and encapsulating the pathogen, thus immobilizing the pathogens' ability to bind and entry to the host lowering the risk of contracting the specific infectious agent.

3A. The method according to embodiment 1 or 2, wherein the carrier has the capacity of at least partly hindering (e.g., preventing) pathogens from binding and entering host cells, wherein the carrier is capable of binding to several said target areas of cell surfaces to at least temporarily and/or partially block viral entry, thereby giving the carrier dual targeting strategies and at least partially (e.g., partially, significantly, etc.) hindering the ability of one or more target pathogens to bind to and/or enter the host. In some embodiments, such carriers lower the risk of contracting the specific infectious agent.

3B. The method according to embodiment 1 or 2, wherein the carrier has the capacity of binding and encapsulating the pathogen thus immobilizing the pathogens' ability to bind and enter the host and capable of binding to said target areas of said cell surfaces to at least temporarily block viral entry, thus having dual targeting strategies thus significantly hinder the pathogens' ability to bind and entry to the host lowering the risk of contracting the specific infectious agent.

4. The method according to embodiment 1 to 3, wherein the core structure of the carrier is being obtained by 3D printing, microfluidics, sol-gel method or other bottom-up and/or top-down method of fabrication.

5. The method according to any of embodiments 1 to 4, wherein the core material comprises organic or inorganic components, lipid droplets, amino acids, proteins, salts and minerals or other molecules or wherein the core material comprises mesoporous silica nanoparticles, in particular mesoporous silica particles with ordered mesostructures of pores that preferably are capable of being loaded with drugs.

6A. The method according to any of embodiments 1 to 5, wherein the core material is functionalized with substance selected from the group consisting of amino acid, epitopes, peptides, proteins and/or fragments of proteins, chemical

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agents, active pharmaceutical ingredients (API), organic or inorganic polymers or molecules and combinations thereof.

6B. The method according to any of embodiments 1 to 5, wherein the core material is functionalized with substance selected from the group consisting of peptides, proteins such as antibodies, chemical agents, active pharmaceutical ingredients (API), organic or inorganic polymers or molecules and combinations thereof.

7. The method according to any of embodiments 1 to 6, wherein the carrier functionalized for specifically binding to receptors, proteins and macromolecules at the cellular level in order to prevent and minimize pathogen entry to the host target tissues by competitive inhibition.

8. The method according to any of embodiments 1 to 7, wherein the synthetic nanoparticle and/or microparticle is used for reducing the spread of SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof.

9. The method according to any of embodiments 1 to 8, wherein said synthetic nanoparticle has a 3D-configuration generally matching the characteristics of the SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof, in particular the particle is fabricated to a size of around 100 nm and coated with similar amino acids as the glycoprotein spikes at the surface of the viral particle or similar molecules that mimic the surface of the viral envelope.

10. The method according to any of embodiments 1 to 9, wherein said synthetic nanoparticle resembles the SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof, or is optimized for competitive inhibition.

11. The method according to any of embodiments 1 to 10, wherein the synthetic carrier exhibits a modified particle morphology, size or surface properties to achieve increased affinity for the target receptor angiotensin converting enzyme 2 (ACE2) compared with the SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof, in particular for increasing the binding affinity for the specific receptor e.g., sialic acid, histo-blood group antigens, ICAM-1, IGF1R blocking the internalization of the viral envelope more efficiently and potentially prolonging the gained viral protection.

12. The method according to any of embodiments 1 to 11, wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof, is adapted for personalized medicine.

13. The method according to any of embodiments 1 to 12, wherein said the synthetic nanoparticle resembling the SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof, is loaded into or onto the nanoparticle for further enhancing the anti-viral properties.

14. The method according to any of embodiments 1 to 13, wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus or other viruses that causes a respiratory infection, diarrhea, common cold, influenzas or generally discomfort or a combination thereof is loaded with vehicles or proteome inhibitors for efficiently delivering the compounds in the target tissues with minimal off-target effects.

15. The method according to any of embodiments 1 to 14, wherein the synthetic nanoparticle is decorated with molecules that have high affinity towards the SARS-CoV-2 virus or any other pathogen of interest for example influenzas, rhinoviruses and viruses causing respiratory infection in order to bind and immobilize the infectious agent preventing or minimizing the potential risk of host entry.

16. The method according to any of embodiments 1 to 15, wherein the synthetic nanoparticle resembling the SARS-CoV-2 virus or any other pathogen for example influenzas, rhinoviruses and viruses causing respiratory infection is coated or decorated with epitopes to be used as a vaccination at target cell populations.

17. The method according to any of embodiments 1 to 16, wherein the carrier is loaded, stored or dispersed in a device or vessel capable of on-demand release of the carrier to the end-user.

18. The method according to any of embodiments 1 to 17, wherein the carrier system is loaded inside a dispenser such as an inhalation device, tablet, injectable substance, cream or ointment.

19. The method according to any of embodiments 1 to 18, wherein the man-made materials is used for immobilizing specific pathogens by adding the synthetic material in sanitation products and disinfectants.

20. The method according to any of embodiments 1 to 19 for preventing or reducing pathogen binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said method comprises minimizing the spread of diverse pathogens by binding to the target molecule in the host body or binding to the infectious agent itself and potentially inhibit the spread of the disease.

The following embodiments are disclosed. The following is a non-exclusive list of embodiments, and as such, should not be seen to limit, in any way, the various inventions disclosed herein.

1. A synthetic carrier for use in a method of preventing or reducing binding of a pathogen to target areas of cell structures of a host, said carrier comprising biocompatible particles having a maximum size in at least one dimension in the nanometer or micrometer range, forming a core, and further having a functionalized surface, which preferably mimics that of the pathogen capable of binding to said target areas of said cell surfaces to at least temporarily block said target areas to prevent or minimize pathogen binding and thus, reducing the risk of the host contracting a disease caused by said pathogen.

2. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to embodiment 1, wherein the pathogen is a coronavirus, in particular SARS-CoV-2 or viral strains derived thereof.

3. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to embodiment 1, wherein the cell structures are selected from ACE2 and TMPRSS2 receptors and combinations thereof.

4A. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to embodiment 1 or 2, said carrier having the capacity of binding to the pathogen's co-receptors (e.g., high-density lipoprotein (HDL) scavenger receptor B type 1 (SR-B1), thus at least partially hindering the ability of the pathogens to bind and enter the

host. Advantageously, this can lower the risk of contracting the specific infectious agent and the disease resulting therefrom.

4B. The synthetic carrier for use in a method of preventing or reducing pathogen binding to target areas of cell structures of a host according to embodiment 1 or 2, said carrier having the capacity of binding and encapsulating the pathogens co-receptors e.g., high-density lipoprotein (HDL) scavenger receptor B type 1 (SR-B1), thus immobilizing the pathogens' ability to bind and entry to the host lowering the risk of contracting the specific infectious agent.

5. The synthetic carrier for use in a method of preventing or reducing pathogen coronaviruses, such as SARS-CoV-2, binding to target areas of cell structures of a host according to any of embodiments 1 to 3, wherein said core structure of the carrier is being obtained by 3D printing, microfluidics, sol-gel method or other bottom-up and/or top-down method of fabrication, and wherein the core material comprises organic or inorganic components, lipid droplets, amino acids, proteins, salts and minerals or other molecules.

6. The synthetic carrier for use in a method of preventing or reducing pathogen, in particular coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein the core material comprises inorganic silica nanoparticles, in particular mesoporous silica particles, such particles preferably having ordered mesostructures of pores that preferably are capable of being loaded with drugs.

7A. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said core material is functionalized with substance selected from the group consisting of amino acids, epitopes, peptides, proteins and/or fragments of proteins, chemical agents, active pharmaceutical ingredients (API), organic or inorganic polymers or molecules and combinations thereof, and wherein said carrier with its functionalization is preferably used for specifically binding to receptors, proteins and macromolecules at the cellular level in order to prevent and minimize novel coronaviruses such as SARS-CoV-2 entry to the host target cells by competitive inhibition.

7B. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said core material is functionalized with substance selected from the group consisting of peptides, proteins such as antibodies, chemical agents, active pharmaceutical ingredients (API), organic or inorganic polymers or molecules and combinations thereof, and wherein said carrier with its functionalization is preferably used for specifically binding to receptors, proteins and macromolecules at the cellular level in order to prevent and minimize novel coronaviruses such as SARS-CoV-2 entry to the host target cells by competitive inhibition.

8. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, said method comprising loading drugs, API, molecules, peptides inside or onto the carrier system, wherein the carrier preferably comprises a functionalized and drug loaded carrier, said carrier being used for targeted drug delivery of anti-viral in order to decrease the replication of the virus inside the host cell.

9. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding

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embodiments, wherein said synthetic nanoparticle and/or microparticle is used for reducing the spread of SARS-CoV-2 or other coronaviruses strains and/or types derived from the SARS-CoV-2 that causes a respiratory infection, diarrhea, common cold, cytokine storm, death or generally discomfort or a combination thereof.

10. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle has a 3D-configuration generally matching the characteristics of the SARS-CoV-2 virus or future variants thereof, in particular the particle is fabricated to a size of around 100-120 nm and coated with similar amino acids as the glycoprotein spikes at the surface of the viral particle or similar molecules that mimic the surface of the viral envelope e.g., spike protein and thus binds to the same target receptor as the virus.

11. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle resembles the SARS-CoV-2 virus or is optimized for competitive inhibition, wherein preferably the synthetic carrier exhibits a modified particle morphology, size or surface properties to achieve increased affinity for the target receptor ACE2 and/or TMPRSS2, compared with the SARS-CoV-2 virus, in particular for increasing the binding affinity for the specific receptor blocking the internalization of the viral envelope more efficiently and potentially prolonging the gained viral protection.

12. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus is adapted for personalized medicine.

13. The synthetic carrier for use in a method of preventing or reducing coronavirus binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus is adapted for personalized medicine in the case of ACE2 receptor polymorphism or different animal host organisms for achieving receptor interaction.

14. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus is loaded into or onto the nanoparticle for further enhancing the anti-viral properties, or wherein said synthetic nanoparticle resembling the SARS-CoV-2 virus is loaded with vehicles or proteome inhibitors for efficiently delivering the compounds in the target cell and tissues with minimal off-target effects.

15. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said carrier is loaded, stored or dispersed in a device or vessel capable of on-demand release of the carrier to the end-user, wherein said carrier system is preferably loaded inside a dispenser such as an inhalation device, tablet, injectable substance, cream or ointment.

16. The synthetic carrier for use in a method of preventing or reducing coronaviruses binding to target areas of cell structures of a host according to any one of the preceding embodiments, wherein said synthetic nanoparticle is a self-

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assembling recombinant protein-based nanoparticle construct, such as a SpyTag/SpyCatcher system

17. Method of producing a synthetic carrier according to any of embodiments 1 to 16, comprising the steps of

- a) providing a core material, e.g., a nano- and/or micro-material including nanoparticles, microparticles or any other object as disclosed herein;
- b) coating or functionalizing the core material with molecules, polymers, amino acids, proteins, API, drugs or other material as disclosed herein;
- c) loading the object with compounds, molecules, drugs, API, DNA or RNA etc.;
- d) coating a second protective and/or functional layer on top of the object in particular for increasing its resistance that could be important in extreme environments such as the acidic environment in the stomach; and providing a small device, medical device, inhalation device or aerosol, sanitation product or consumer product that on-demand will release the containing synthetic material, particle or object for administration.

It will be obvious to a person skilled in the art that, as the technology advances, the inventive concept can be implemented in various ways. The invention and its embodiments are not limited to the examples described above but may vary within the scope of the claims.

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- Although several embodiments and examples are disclosed herein, the present application extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the inventions and modifications and equivalents thereof. It is also contemplated that various combinations or subcombinations of the specific features and aspects of the embodiments may be made and still fall within the scope of the inventions. Accordingly, it should be understood that various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the disclosed inventions. Thus, it is intended that the scope of the present inventions disclosed herein should not be limited

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by the particular disclosed embodiments described above, but should be determined only by a fair reading of the claims that follow.

While the inventions are susceptible to various modifications, and alternative forms, specific examples thereof have been shown in the drawings and are herein described in detail. It should be understood, however, that the inventions are not to be limited to the particular forms or methods disclosed, but, to the contrary, the inventions are to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the various embodiments described and the appended claims. Any methods disclosed herein need not be performed in the order recited. The methods summarized above and set forth in further detail below describe certain actions taken by a practitioner; however, it should be understood that they can also include the instruc-

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tion of those actions by another party. The methods summarized above and set forth in further detail below describe certain actions taken by a user (e.g., a professional in some instances); however, it should be understood that they can also include the instruction of those actions by another party. Thus, actions such as “delivering” include “instructing delivering.” The ranges disclosed herein also encompass any and all overlap, sub-ranges, and combinations thereof. Language such as “up to,” “at least,” “greater than,” “less than,” “between,” and the like includes the number recited. Numbers preceded by a term such as “about” or “approximately” include the recited numbers. For example, “about 10 mm” includes “10 mm.” Terms or phrases preceded by a term such as “substantially” include the recited term or phrase. For example, “substantially parallel” includes “parallel.”

SEQUENCE LISTING

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Sequence total quantity: 7
SEQ ID NO: 1      moltype = AA length = 6
FEATURE          Location/Qualifiers
source           1..6
                 mol_type = protein
                 organism = SARS-CoV-2

SEQUENCE: 1
YKYRYL                                     6

SEQ ID NO: 2      moltype = AA length = 6
FEATURE          Location/Qualifiers
REGION          1..6
                 note = Potential alternative synthetic hexapeptide variant
source          1..6
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 2
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SEQ ID NO: 3      moltype = AA length = 6
FEATURE          Location/Qualifiers
REGION          1..6
                 note = Potential alternative synthetic hexapeptide variant
source          1..6
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 3
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SEQ ID NO: 4      moltype = AA length = 28
FEATURE          Location/Qualifiers
source          1..28
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                 organism = SARS-CoV-2

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FEATURE          Location/Qualifiers
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                 organism = SARS-CoV-2

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CYGVSP TKLN DL CFTN VYAD SFVIRGDEVR QIAPGQTGKI ADYNYKLPPD FTGCVIAWNS 120
NNLDSKVGGN YNYLYRLFRK SNLKPFERDI STEIYQAGST PCNGVEGFNC YPPLQSYGFQ 180
PTNGVGYPY RVVLSFELL HAPATVCGPK KSTNLVKNKC VNF 223

SEQ ID NO: 6      moltype = AA length = 1273
FEATURE          Location/Qualifiers
source          1..1273
                 mol_type = protein
                 organism = SARS-CoV-2

SEQUENCE: 6
MFVFLVLLPL VSSQCVNLTT RTQLPPAYTN SFTRGVYYPD KVRSSVLHS TQDLFLPFFS 60
NVTWFHAIHV SGTNGTKRFD NPVLPFNDGV YFASTEKSNI IRGWIFGTTL DSKTQSL LIV 120
NNATNVVIKV CEFQFCNDPF LGVYYHKNNK SWMESEFRVY SSANNCTFEY VSQPFLMDLE 180

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-continued

GKQGNFKNLR	EFVFNKIDGY	FKIYSKHTPI	NLVRDLPGGF	SALEPLVDLP	IGINITRFQT	240
LLALHRSYLT	PGDSSSGWTA	GAAAYVGYL	QPRTFLLKYN	ENGTITDAVD	CALDPLSETK	300
CTLKSFTVEK	GIYQTSNFRV	QPTESIVRFP	NITNLCPPGE	VFNATRFASV	YAWNRRKRISN	360
CVADYSVLN	SASFSTFKCY	GVSPTKLNLD	CFTNVYDSF	VIRGDEVROI	APQGTGKIAD	420
YNYKLPPDDFT	GCVIAWNSNN	LDSKVGGNYN	YLYRLFRKSN	LKPFERDIST	EIYQAGSTPC	480
NGVEGFNCYF	PLQSYGFQPT	NGVGYPYRV	VVLSFELLHA	PATVCGPKKS	TNLVKNKCVN	540
FNFNGLTGTG	VLTESNKKFL	PFQQFGRDIA	DTDAVRDPQ	TLEILDITPC	SFGGVSVITP	600
GTNTSNQVAV	LYQDVNCTEV	PVAIHADQLT	PTWRVYSTGS	NVFPQTRAGCL	IGAHEVNNNSY	660
ECDIPIGAGI	CASYQTQNS	PRRARSVASQ	SIIAYTMSLG	AENSVAYSNN	SIAIPTNFTI	720
SVTTEILPVS	MTKTSVDCMT	YICGSTECS	NLLQYGSFC	TQLNRALTGI	AVEQDKNTQE	780
VFAQVKQIYK	TPPIKDGGF	NFSQILPDPS	KPSKRSFIED	LLFNKVTLAD	AGFIKQYGDC	840
LGDIAADLI	CAQKFNGLT	LPPLLTDEMI	AQYTSALLAG	TITSGWTFGA	GAALQIPFAM	900
QMAYRFNGIG	VTQNVLYENQ	KLIANQFNSA	IGKIQDSLSS	TASALGKLQD	VVNQNAQALN	960
TLVKQLSSNF	GAISSVLNDI	LSRLDKVEAE	VQIDRLITGR	LQSLQTYVTQ	QLIRAAEIRA	1020
SANLAATKMS	ECVLGQSKRV	DFCGGYHLM	SFPQSAPHGV	VFLHVTYVPA	QEKNTFTAPA	1080
ICHGKAHFP	REGVFSNGT	HWFVTQRNFY	EPQIITTDNT	FVSGNCDVVI	GIVNNTVYDP	1140
LQPELDSFKE	ELDKYFKNHT	SPDVLGDIS	GINASVVNIQ	KEIDRLNEVA	KNLNESLIDL	1200
QELGKYEYI	KWPYIWLGF	IAGLIAIVMV	TIMLCMTSC	CSCLKGCCSC	GSCKFDEDD	1260
SEPVLLKGVKL	HYT					1273

SEQ ID NO: 7 moltype = AA length = 272

FEATURE Location/Qualifiers

REGION 1..272

 note = Synthetic

VARIANT 1..49

 note = MISC_FEATURE - Different Expression Cassette -

 Signal Sequence - Tag - Spacer; X is any amino acid

source 1..272

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 7

XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	VQPTESIVRF	60
PNITNLCPPG	EVFNATRFAS	VYAWNRRKRIS	NCVADYSVLY	NSASFSTFKC	YGVSPTKLNLD	120
LCFTNVYADS	FVIRGDEVROI	IAPGQTGKIA	DYNYKLPPDDF	TGCVIAWNSN	NLDSKVGGNY	180
NYLYRLFRKS	NLKPFERDIS	TEIYQAGSTP	CNGVEGFNCY	FPLQSYGFQFP	TNGVGYPYR	240
VVLSFELLH	APATVCGEKK	STNLVKNKCV	NF			272

What is claimed is:

1. A method of delivery of at least one component to a host, the method comprising:
 - introducing the at least one component to the host using a carrier, the carrier comprising a maximum size in at least one dimension in a nanometer or a micrometer range;
 - wherein the carrier comprises a core and a functionalized surface configured to increase a binding affinity of the carrier to target cells of the host;
 - wherein the carrier is configured to attach to target cells of the host; and
 - wherein the at least one component is configured to be delivered and released to target cells of the host using the carrier; and
 - delivering the at least one component to target cells of the host to assist with a prevention or a treatment of a disease or other adverse condition.
2. The method of claim 1, wherein target areas comprise at least one of a receptor, a target molecule, an amino acid and a nucleotide.
3. The method of claim 1, wherein the functionalized surface comprises at least one of a targeting moiety, an amino acid, a protein and a drug.
4. The method of claim 3, wherein the at least one of a targeting moiety, an amino acid, a protein and a drug is added to the carrier covalently, allosterically, via conjugation or via coating.
5. The method of claim 1, wherein the functionalized surface comprises an amino acid sequence of a receptor binding motif configured to bind to target cells.
6. The method of claim 1, wherein the at least one component comprises at least one of the following: an API, a drug, a protein, an amino acid, RNA, DNA and a molecule.
7. The method of claim 1, wherein the carrier is used for personalized treatment or personalized diagnostics.
8. The method of claim 1, wherein the carrier is configured to be delivered to the host using an inhalation device.
9. The method of claim 1, wherein the carrier is configured to be delivered to the host using an aerosol, a spray, a drop, a tablet, a cream, an ointment, an injection or an intravenous administration.
10. The method of claim 1, wherein the carrier is configured to deliver genetic information to assist the host with a prevention or a treatment of the disease or other adverse condition.
11. A method of delivery of at least one component to a host, the method comprising:
 - introducing at least one component within the host using a carrier, the carrier comprising a maximum size in at least one dimension in a nanometer or a micrometer range;
 - wherein the carrier comprises a core and a functionalized surface configured to increase a binding affinity of the carrier to target tissue of the host;
 - wherein the functionalized surface is configured to specifically bind to target tissue of the host; and
 - wherein the at least one component is configured to be delivered to target tissue of the host using the carrier; and
 - delivering the at least one component to target tissue of the host.
12. The method of claim 11, wherein the functionalized surface comprises at least one of a targeting moiety, an amino acid, a protein and a drug.
13. The method of claim 12, wherein the at least one of a targeting moiety, an amino acid, a protein and a drug is added to the carrier covalently, allosterically, via conjugation or via coating.

14. The method of claim 11, wherein the functionalized surface comprises an amino acid sequence of a receptor binding motif configured to bind to target cells.

15. The method of claim 11, wherein target tissue comprises at least one of a receptor, a target molecule, an amino acid and a nucleotide. 5

16. The method of claim 11, wherein the at least one component comprises at least one of the following: an API, a drug, a protein, an amino acid, RNA, DNA and a molecule.

17. The method of claim 11, wherein the carrier is used for personalized treatment or personalized diagnostics. 10

18. The method of claim 11, wherein the carrier is configured to be delivered to the host using an inhalation device.

19. The method of claim 11, wherein the carrier is configured to be delivered to the host using an aerosol, a spray, a drop, a tablet, a cream, an ointment, an injection or an intravenous administration. 15

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